

Choosing Career and Family: The Multidimensional Returns to College Majors*

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September 2026

Abstract

This paper studies how college major choices shape earnings and family formation using administrative data that link students' preferences, admissions, graduation, earnings, and births. Beyond documenting heterogeneity in effects across fields, we characterize the structure of effect heterogeneity in these returns. Earning payoffs exhibit substantial preference-based heterogeneity but limited gender heterogeneity, while fertility effects exhibit limited idiosyncratic heterogeneity with more substantial gender heterogeneity. This contrast points to interpreting college majors as gender-specific career-family bundles, an interpretation supported by associations between fertility effects and measures of career-family compatibility that we document. We then show how these multidimensional returns relate to major choice and subsequent implications. Men's choices are more strongly associated with earnings effects, whereas women's choices are more strongly associated with family-formation effects; demand counterfactuals show that gender differences in payoff weights shape sorting and the earnings gap. Finally, we use our causal estimates to evaluate counterfactual assignment rules that target representation gaps in centralized assignment systems, illustrating how gender quotas could reduce assignment and earnings gaps with minimal aggregate earnings costs and small effects on aggregate fertility.

Keywords: preferences, returns to majors, gender gaps, centralized assignment
JEL Classification: I23, I26, J13, J16, J24

*We thank Marianne Bertrand, David Card, Claudia Goldin, Eleonora Guarnieri, Larry Katz, Camille Landais, Matt Notowidigdo, Alessandra Voena, and seminar participants at Duke, PUC-Chile, Universidad Adolfo Ibáñez, Universidad de los Andes, Universidad de Chile, West Point, SITE Gender 2024, and CEP Education Conference 2025 for their valuable comments and suggestions. This research uses information from the Registry of Social Information (RIS). We thank the Undersecretary of Social Evaluation, owner of the RIS, for the authorization to use the databases. The views expressed here are those of the authors and do not reflect the views of the Ministerio de Desarrollo Social. Camila Brown, Ryan Lee, and Ishira Shrivatsa provided outstanding research assistance. Christopher Campos gratefully acknowledges support from the Biehler Junior Faculty Fellowship. Dante Contreras acknowledges the financial support provided by the Centre for Social Conflict and Cohesion Studies (COES; ANID/-FONDAP/15130009) and Fondecyt Project 1210085. Bucarey: alonsobucareyc@gmail.com. Campos: Chicago Booth School of Business, Christopher.Campos@chicagobooth.edu. Contreras: Universidad de Chile, Department of Economics, dcontrer@fen.uchile.cl. Muñoz: Universidad de Chile, Department of Economics, pablonh@uchile.cl.

1 Introduction

A growing body of evidence establishes that college major choice sharply changes labor market trajectories (Altonji et al., 2016; Andrews et al., 2024; Hastings et al., 2013; Kirkebøen et al., 2016) and that there are stark gender differences in major choice (Sloane et al., 2021; Wiswall and Zafar, 2021; Zafar, 2009). Yet, the evidence is thin on the numerous other outcomes that college major may affect and that are central to inequality and life-cycle behavior. Multidimensional and gender-specific returns may be particularly important for understanding gender differences in major choice if men and women face different returns from choosing the same field or place different weight on these returns, with fertility potentially being one consequential margin. Different fields of study lead students into different occupations, work environments, marriage markets, and career trajectories, all of which may affect when individuals form families and how costly parenthood becomes (Blundell et al., 2018; Doepke et al., 2023; Goldin and Katz, 2008; Kirkebøen et al., 2025). Despite this, causal evidence on how college majors affect fertility—and whether these family-formation consequences help explain gender sorting across fields—remains limited.

Understanding how majors affect fertility is important for two reasons. First, fertility and its timing are meaningful outcomes. In many predominantly middle- and high-income countries, delayed childbearing has become a central feature of demographic change, and shifts in the timing of first births can depress period fertility and translate into lower completed fertility (Bongaarts and Feeney, 1998; Roustaei et al., 2019). Second, measurement of causal fertility effects provides a window into the non-pecuniary consequences of schooling (Oreopoulos and Salvanes, 2011). Prior work shows that students expect majors to affect not only earnings but also marriage and family outcomes (Kirkebøen et al., 2025, 2016), and descriptive evidence documents large fertility differences across fields of study (Hoem et al., 2006; Micheltore and Musick, 2014; Wiswall and Zafar, 2021). If some fields of study make it easier to combine career and family while others impose larger costs to parenthood, then major choice shapes not only earnings but also the life-cycle tradeoffs students face after college (Adda et al., 2017; Doepke et al., 2023; Goldin, 2021; Goldin and Katz, 2011). Field-specific fertility effects can therefore provide a summary measure of an important dimension of the family-formation consequences of choosing different educational paths. If students forecast these differences with some accuracy, they may also shape major choice and contribute to the gendered sorting we see across fields.

In this paper, we study the causal impact of fields of study on earnings and fertility—allowing for vertical and horizontal effects—and investigate how these multidimensional payoffs relate to gender differences in major choice and labor market outcomes. Our setting is Chile’s centralized college admissions system, where applicants submit rank-ordered lists over programs and are assigned through a deferred-acceptance mechanism. We combine administrative data on preferences, admissions, graduation, earnings, and births, allowing us to follow applicants into their thirties. We estimate causal effects using an approach that borrows insights from Kirkebøen et al. (2016) for fallback conditioning, the self-revelation logic emphasized by Dale and Krueger (2002), and the tractable framework for addressing selection in centralized assignment settings developed by Ab-

dulkadiroğlu et al. (2020); and we validate our approach using the experimental variation embedded within the centralized college admission system (Abdulkadiroğlu et al., 2017, 2022).

We structure our analysis in four steps. First, we estimate gender-specific causal returns to fields of study in terms of economic payoffs and fertility, distinguishing between average effects and preference-specific match effects. This allows us to assess if gender differences in effects of the same field or if differences in comparative advantage can partly explain gender differences in major choice. Second, we investigate the economic forces underlying heterogeneity in fertility effects. In particular, we assess whether fertility effect heterogeneity is associated with differences in career-family compatibility across fields using data on job characteristics, child penalties (Kleven et al., 2019a), and gender norms (Bertrand et al., 2015). Third, we assess the extent to which men and women differentially prioritize the different dimensions when ranking fields of study, an exercise we interpret as assessing gender differences in the balance of career and family. Finally, we use our estimated payoffs to study the effects of supply-side policies that could address broader policy-relevant gaps, while accounting for potential equity and efficiency impacts on both earnings and fertility outcomes.

We begin with a framework for estimating causal effects that borrows insights from Kirkebøen et al. (2016) in relation to fallback conditioning, the self-revelation principle of Dale and Krueger (2002), and the approach of Abdulkadiroğlu et al. (2020) that uses rich rank-ordered choice data to account for selection. Importantly, the framework captures average returns separately for men and women and empirically assesses the importance of preference-specific match effects. The model distinguishes between vertical returns, which represent the general benefits of completing a particular field, and horizontal returns, which capture preference-specific effects. This distinction allows us to later assess whether gender differences in major choice could reflect differences in average returns or potential differences in comparative advantage across fields. Making this distinction also clarifies the assumptions needed to identify each type of return using observational comparisons coupled with a conditional independence assumption that leverages the suite of information available in settings with centralized admissions, namely rank-ordered preferences and entrance exam scores that govern sharp admission cutoffs. A key feature of our single-offer, centralized admissions setting is that post-offer sorting—where students sort after receiving offers—is significantly costlier and thus less common. This reduces the concerns present in decentralized settings, where extensive post-admission sorting can undermine identification strategies that rely on similar self-revelation principles (Dale and Krueger, 2002; Hoxby, 2009).

Consistent with a growing body of evidence (Altonji et al., 2012, 2016; Andrews et al., 2024; Hastings et al., 2013; Kirkebøen et al., 2016), we document substantial heterogeneity in pecuniary payoffs across fields of study. All fields of study, except for Humanities, exhibit positive economic payoffs ranging from 17 to 80 log points, with Medicine, Business, and Engineering having the largest returns. This dispersion in returns mirrors the patterns in Altonji et al. (2012) in that heterogeneity in returns across majors rivals the college premium. We also document strong evidence of match effects in economic payoffs, indicating that students tend to sort in ways consistent with

Roy (1951)-like selection. Despite this substantial heterogeneity across majors and students, both vertical and horizontal returns are similar for men and women. These patterns remain even when we further condition on fallbacks (Kirkebøen et al., 2016). Gender differences in pecuniary returns are therefore unlikely to account for the stark gender differences in major choice.

These findings motivate us to examine nonpecuniary outcomes—such as family formation—which can shape sorting alongside earnings based comparative advantage or match effects (Taber and Vejlín, 2020). Consistent with existing evidence that college completion delays fertility (e.g., Currie and Moretti, 2003), we find that graduation reduces the probability of having a child by early adulthood, with average effects of 6.6 percentage points for women and 13 percentage points for men. These average effects mask substantial heterogeneity across fields of study, with a noise-adjusted standard deviation of fertility effects of 5 percentage points for women and 3 percentage points for men. The reductions are smallest in Teaching and largest in Law and Humanities. In contrast to earnings, however, fertility effects differ substantially by gender across fields with rather limited preference-specific match effects. Thus, the family formation consequences that we document are less tied to idiosyncratic comparative advantage and are more closely associated with gender-specific features and opportunities of career paths that each college major offers. This motivates us to study what economic forces potentially contribute to the fertility effect heterogeneity.

One natural interpretation of the fertility effect heterogeneity is that college majors bundle human capital investments with different career and family environments. As Goldin and Katz (2008) note, women in careers with greater predictability and smaller penalties for time away from work tend to have more children. More generally, fields of study may guide graduates towards careers that differ in their occupational flexibility, cost of career interruptions, and the overall ease of balancing career and family (Dahl and Loken, 2024; Doepke et al., 2023; Goldin, 2021; Olivetti et al., 2024), while also directly shaping marriage markets (Kirkebøen et al., 2025). Supporting this interpretation, we find positive correlations between fertility payoffs and measures of job security, progressive gender norms, and lower child penalties by field, while finding no association between wages and fertility payoffs.¹ Combined with the evidence of limited preference-specific match effects, these patterns suggest that fertility effects relate less to idiosyncratic fit than to shared opportunities of graduates who follow a field’s educational and career path, including the household production technologies associated with different career trajectories (Blundell et al., 2018).

Before turning to whether these estimated effects are reflected in major choice, we assess their credibility as causal by using the quasi-random variation embedded in Chile’s centralized admissions system. Following Abdulkadiroğlu et al. (2022), we benchmark the earnings and fertility effects from our preferred model against treatment effects identified from local variation in assignments around admission thresholds (Angrist et al., 2017). We find that both earnings and fertility effects are forecast unbiased. This evidence provides reassuring evidence that our framework adequately

¹We combine data from Chile’s largest household survey with complementary measures of field-specific gender norms and child penalties, which we estimate using administrative data following Bertrand et al. (2015) and Kleven et al. (2019b). Consistent with evidence that the relationship between income and fertility has weakened over time (Doepke et al., 2023), we find no systematic relationship between fertility payoffs and wages across fields of study.

characterizes selection and allows us to recover causal earnings and fertility effects, while accounting for both vertical and horizontal dimensions of gender-specific field payoffs.

Having established that fields generate substantial heterogeneity in both earnings and family-formation payoffs, we next ask whether these ex-post payoffs are reflected in the choices students make before entering college. If students anticipate that college majors differentially affect both earnings and fertility, preference heterogeneity may manifest in the implicit weights they place on these outcomes when choosing among fields. Similar to Wiswall and Zafar (2018) and Burbano et al. (2024) who document systematic gender differences in the valuation of job attributes and their role in shaping sorting across fields and occupations, we study whether and to what extent fertility-related considerations matter at the time of educational choice. Leveraging multivariate relationships between estimated mean utilities derived from rank-ordered choice data and our estimated fertility and earnings treatment effects, we recover gender-specific payoff weights. We document that men’s choices are more strongly associated with earnings effects, whereas women’s choices are more strongly associated with family-formation effects. Counterfactuals that equalize payoff weights narrows the absolute gender earnings gap by 21–25 percent. Separate exercises that equalize gender-specific earnings and family-formation payoff schedules show additional, but smaller, reference-dependent effects.

These counterfactuals highlight the role of gender differences in preferences in shaping sorting across fields and subsequent earnings gaps. However, shifting these preferences may require changing long-standing norms (e.g., Alesina et al., 2013; Carlana and Corno, 2024; Müller, 2023). We therefore study how much policymakers could achieve with supply-side policies that modify the allocation of college opportunities while taking preferences as given. Leveraging the centralized assignment structure, we assess how rule modifications can reshape the allocation of high-payoff majors—particularly Engineering, the field that contributes the most to earnings differences—and affect subsequent gender earnings gaps and family formation. We consider two policy levers: i) expanding Engineering seats uniformly and ii) reserving a share of these seats for women.² Our counterfactual results show that broad expansions of Engineering opportunities can improve overall access, but they inadvertently widen the gender earnings gap due to underlying differences in college major demand. By contrast, our estimates suggest that targeted seat reservations for women considerably reduce the representation gap and modestly improve earnings equity, all with minimal costs in terms of aggregate earnings and fertility relative to the baseline allocation.

Our work contributes to four connected literatures. The first relates to a large body of work focusing on estimating the returns to college and college selectivity (Card, 2001; Chetty et al., 2026; Dale and Krueger, 2002; Mountjoy, 2024; Mountjoy and Hickman, 2021; Taber, 2001; Zimmerman, 2014), and to a growing body of work that pivots focus to the returns to fields of study (Altonji et al., 2012; Andrews et al., 2024; Dahl et al., 2020; Hastings et al., 2013; Kirkebøen et al., 2025, 2016; Lovenheim and Smith, 2023). One reason for the more recent interest in returns to fields of study is that dispersion in field returns is as large as the dispersion in returns to college (Al-

²The latter affirmative-action-like policy receives broad support in Chile (Bursztyn et al., 2023).

tonji et al., 2012), a stylized fact that alludes to the potentially important role of major choice in explaining the gender earnings gap among college graduates (Aguirre et al., 2020; Sloane et al., 2021). We contribute to this literature by conceptualizing major returns as multidimensional and gender-specific, while allowing for both horizontal and vertical effects. In doing so, we provide new evidence on how college and college majors affect fertility—a dimension for which existing evidence is scant.³ Beyond documenting heterogeneity in average payoffs, we also characterize the structure of heterogeneity across the dimensions we study. Earnings effects exhibit substantial idiosyncratic heterogeneity with limited gender heterogeneity, while fertility effects exhibit limited match effects with more substantial gender heterogeneity. We also show that the relatively parsimonious framework we propose produces estimates that are statistically comparable to those based on local random assignment (Abdulkadiroğlu et al., 2022), while also allowing us to characterize treatment effect heterogeneity that tends to be difficult to recover from local comparisons alone.

Another body of work in higher education has documented vast gender differences in major choice (Ahimbisibwe et al., 2024; Bordon et al., 2020; Neilson et al., 2021; Wiswall and Zafar, 2018, 2021; Zafar, 2009). An important question in this literature is what drives gender differences in preferences (Patnaik et al., 2021). Wiswall and Zafar (2021) find that college students’ expectations about how their education will impact their future career and family significantly influence their choice of major and degree completion, with notable differences between men and women. We contribute to this literature by linking revealed major preference data to ex-post causal earnings and family-formation effects. This allows us to distinguish between two competing forces that potentially generate gendered sorting into majors: men and women may face different payoff schedules, or they may place different value on the different dimensions of field- and gender-specific returns. We quantify the importance of both channels for sorting across majors, and as a consequence, their association with downstream gender earnings gaps. Overall, our analysis connects gender differences in major choice to both the multidimensional nature of major payoffs and the preferences students reveal well before entering the labor market.

Our third contribution is to the literature on career-family tradeoffs, fertility, and educational choice (Adda et al., 2017; Doepke et al., 2023; Goldin, 2021; Goldin and Katz, 2008, 2011; Kleven et al., 2019a; Sauer and Taber, 2021; Wiswall and Zafar, 2018, 2021). We complement previous evidence by documenting substantial heterogeneity in the causal effects of fields of study on earnings and family formation, and by showing that heterogeneity in fertility effects is systematically related to measures of career-family compatibility, including job security, child penalties, and gender norms. Viewed through the lens of Blundell et al. (2018), fields of study may therefore partly shape household production technologies and career environments that subsequently reshape the distribution of work, leisure, and childcare. Combining this with our revealed preference analysis, our results connect the ex-ante role of career and family considerations in educational choice to the ex-post causal consequences of those choices.

³Kirkeboen et al. (2025) provide evidence showing that access to elite college programs causes women (but not men) to delay first births and reduce completed fertility in Norway. Most prior research finds that attending college tends to delay childbearing and reduce fertility among women (see Bharati et al. (2023) for a recent review).

Finally, our paper contributes to a growing body of evidence studying the potential effects of affirmative action in both decentralized admissions systems (Arcidiacono, 2005; Arcidiacono et al., 2016; Bertrand et al., 2010; Bleemer, 2023) and centralized assignment systems (Barahona et al., 2026; Carlana et al., 2024). Most existing work on affirmative action in higher education focuses on race or socioeconomic status. In contrast, we focus on the potential implications of gender-based affirmative action policies to address assignment gaps in high-earning fields of study. Our findings also offer novel insights into the broader literature linking education and delayed child-bearing (Baudin et al., 2015; Currie and Moretti, 2003; Goldin, 2021) by showing that gender quotas can reshape field assignments and reduce earnings gaps while having little effect on aggregate fertility.

2 Institutional Background, Data, and Descriptive Statistics

This section provides background information on Chile and its centralized college admission system. We discuss the data generated by the college admission process and its linkages with other administrative records. We then characterize gender differences in sorting across fields of study and document key descriptive differences in earnings and fertility outcomes across majors that motivate our causal framework and demand analysis that follow.

College Admissions in Chile: Higher education admissions are based on high-school GPA and students’ test scores from nationwide entrance exams. High school graduates can register for the national admission test (Prueba Selección Universitaria, PSU), an SAT-type exam with four sub-tests: Mathematics, Language, and a choice between Science or History. After receiving their scores, students can apply to major-institution combinations (e.g., civil engineering at the University of Chile) by submitting a rank-ordered list of up to ten preferences. Each major-institution (a program) also ranks applicants by setting weights on the subject test scores and high school GPA. Given students’ rankings, program-specific weights, and capacities, the system offers each student a seat in at most one program, ensuring that no student-program pair would prefer to be matched over their actual assignment. This is done using an assignment mechanism built on Gale and Shapley (1962)’s student-proposing deferred acceptance algorithm. This process creates an admission cutoff for each program, corresponding to the score of the lowest program-weighted score among admitted students.

Data: Our analysis combines administrative data on students’ college applications, graduation, fertility, and labor market outcomes. Data on college applications and admissions come from the agency responsible for administering the national college entrance exam (Departamento de Evaluación, Medición y Registro Educacional, DEMRE). For exam-takers, we observe gender, high school identifier and high school location, GPA, test scores, and rank-ordered preferences for major-institution pairs. To focus on the returns of college graduates, we merge these data with graduation

records from the Higher Education Information Service (Servicio de Información de la Educación Superior, SIES).

We classify students into fields of study based on the major from which they graduate. Degree programs are classified primarily based on the OECD Handbook for Internationally Comparative Education Statistics (OECD, 2004). There are eight broad categories: “*Agriculture*”, “*Science*”, “*Social Sciences, Business and Law*”, “*Teaching*”, “*Humanities and Arts*”, “*Engineering, Manufacturing and Construction*”, “*Health and Welfare*”, and “*Services*”. To ease the exposition, we reclassify “*Social Sciences, Business and Law*” into three separate fields: “*Social Sciences*”, “*Business*”, and “*Law*”. We also separate “*Medicine*” from “*Health and Welfare*” and exclude “*Agriculture*” and “*Services*” as they represent small (<5% of graduates) and non-homogeneous college programs in Chile. We end up with nine fields of study (see Appendix A.1 for details).

Fertility data come from the Civil Registration System (Registro Civil). We observe the date of birth of all the children of every student who applied to college. Labor market data come from the Chilean pension system (Superintendencia de Pensiones, SP), which manages social contributions for all formal public and private employment.⁴ We complement the SP data with administrative records from the Unemployment Insurance Administrator Agency (Administradora de Fondos de Cesantía, AFC). These data, limited to formal private sector employment, allow us to replicate our main results among a smaller set of individuals but with a higher wage censoring limit.

For our analysis, we focus on students who applied and were accepted into a college major through the centralized admission system between 2004 and 2007. Focusing on accepted applicants ensures that all students we consider have an offer, which is an important detail that is highlighted in our conceptual framework. The chosen period allows us to study labor market and fertility outcomes at least 13 years after college application.⁵ For earnings, this provides a sufficiently long window to evaluate divergences in the trajectories of individuals, while for fertility, it allows us to assess delays that can meaningfully translate into fertility reductions (Bongaarts and Feeney, 1998; Roustaei et al., 2019).

Table 1 presents descriptive statistics for individuals in our main analysis sample. The table shows substantial variation in the number of students, their average admission test scores, and proxies for socio-economic status—public secondary school enrollment, public health insurance enrollment, and parental education—across genders and fields of study. Accepted students tend to score above average on the college entrance exam, underscoring the competitive nature of college admissions in Chile. Medicine and Law programs are the most competitive, although important elite programs are also contained in Business and Engineering (Zimmerman, 2014). As for gender differences in admissions, engineering has the highest number of male graduates, whereas Teaching

⁴While this dataset is comprehensive and of high quality, it has two limitations. First, it does not include the hours in the workers’ contract. However, part-time employment is not that common among college graduate workers in Chile. According to the household survey CASEN 2017, only 8.6% of prime-age college graduates report having worked part-time. Second, wages are censored at the Social Security contribution limit. We address this issue by imputing wages above the limit as in Dustmann et al. (2009). See Appendix A.2 for details.

⁵Access to more recent data would allow us to extend the analysis to at least 17 years after college application. While our current access is limited, our data agreement permits extending the analysis in the event of a revision.

Table 1: Descriptive Statistics

	Non-Grad (1)	Law (2)	Science (3)	Social Science (4)	Teaching (5)	Humanities (6)	Engineering (7)	Health (8)	Medicine (9)	Business (10)
PSU Score	0.03	1.00	0.57	0.41	-0.02	0.69	0.76	0.57	1.53	0.40
Father is College Grad	0.20	0.46	0.29	0.34	0.17	0.47	0.38	0.26	0.52	0.28
Mother is College Grad	0.15	0.41	0.24	0.29	0.13	0.39	0.30	0.22	0.46	0.23
Has Public Health Insurance	0.59	0.37	0.52	0.47	0.62	0.36	0.45	0.53	0.32	0.50
Attended Public K-12 School	0.29	0.14	0.24	0.21	0.33	0.12	0.15	0.25	0.10	0.20
Observations	10,477	1,697	1,307	4,833	12,184	1,280	4,365	8,085	3,374	4,961
Panel A: Female										
PSU Score	0.24	1.18	0.63	0.59	0.18	0.64	0.84	0.65	1.72	0.65
Father is College Grad	0.22	0.50	0.29	0.34	0.17	0.35	0.37	0.23	0.50	0.36
Mother is College Grad	0.17	0.44	0.23	0.30	0.14	0.29	0.28	0.19	0.44	0.27
Has Public Health Insurance	0.59	0.32	0.51	0.47	0.65	0.46	0.45	0.58	0.33	0.45
Attended Public K-12 School	0.29	0.16	0.26	0.25	0.35	0.22	0.20	0.33	0.18	0.20
Observations	15,494	1,584	3,146	2,223	4,861	797	11,772	2,493	3,030	4,634
Panel B: Male										

Notes: This table presents average standardized test scores (PSU scores) and available background characteristics for all students in our main analysis sample, by gender and field of study. PSU scores are standardized using the mean and standard deviation of the full applicant population. Student characteristics come from the forms that students must complete before taking the college admission exam. Our main analysis sample includes all students who applied and were accepted into a program through the centralized admission system between 2004 and 2007 and who were employed in 2019.

has the highest number of female graduates. Medicine is the most selective field, while Teaching is the least selective. Compared to Medicine graduates, Teaching graduates come from a more disadvantaged background (e.g., less than 20% of their fathers have a college degree versus 50% of the fathers of graduates from Medicine). Overall, our primary sample is positively selected on incoming achievement due to the competitive nature of the centralized assignment system, but there is still substantial earnings inequality among graduates in the system.

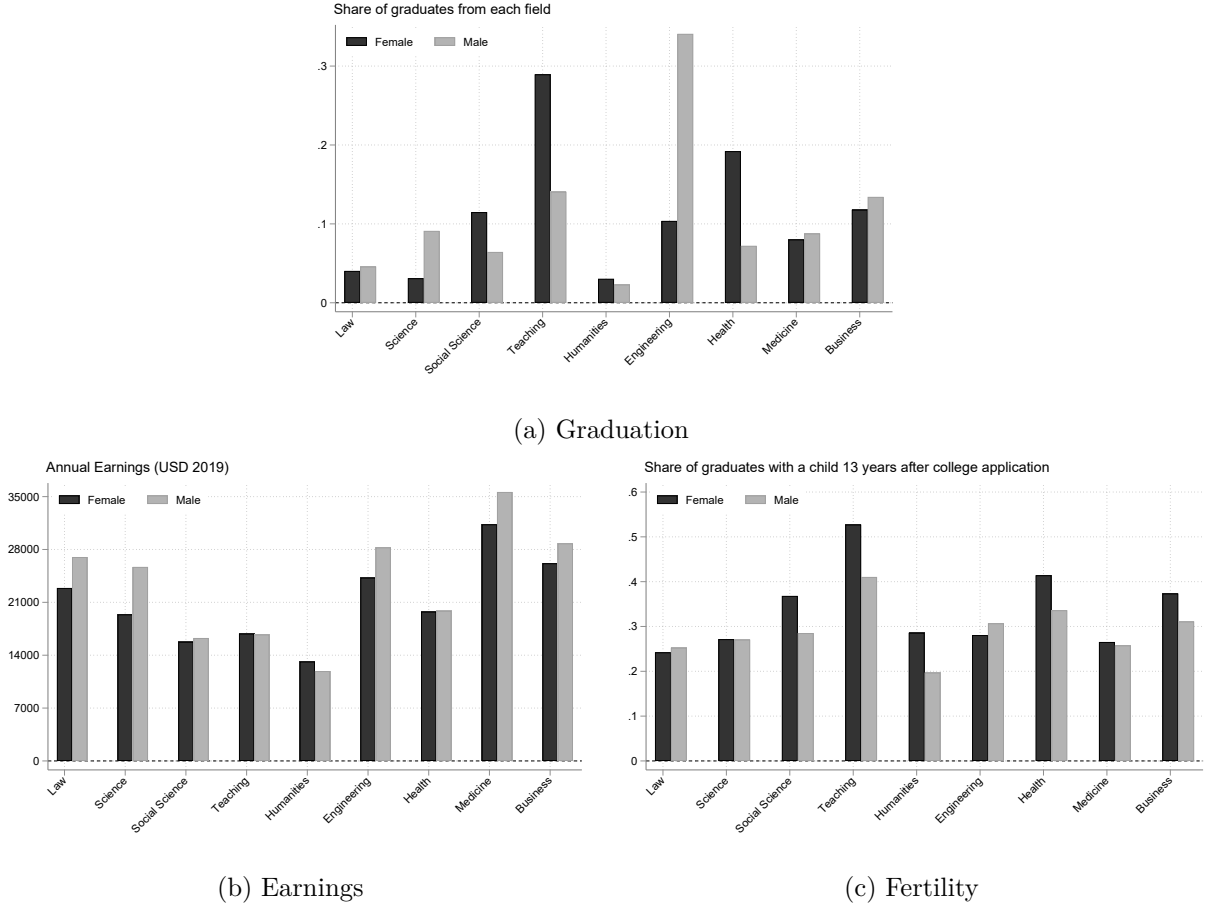
2.1 Major Choice, Earnings, and Fertility in Chile

There are stark differences in major choice by gender and major choice is strongly related to both earnings and fertility in Chile. Panels (a) to (c) of Figure 1 show graduation shares, average annual earnings, and the share of graduates with a child thirteen years after college application by gender and field of study. Three facts stand out. First, men and women sort into different fields: men are disproportionately represented in Engineering and Science, while women are disproportionately represented in Health and Teaching. Second, these fields differ sharply in graduates' eventual earnings. Engineering and Science are among the higher-paying fields, whereas Teaching and Health are associated with lower earnings. Third, there are stark fertility differences across fields of study. The fields of study with the highest fertility are Teaching and Health, and those with the lowest are Humanities, Medicine, and Law. The overall low fertility across fields is consistent with Chile's national context, where the 2022 fertility rate was approximately 1.5 children per woman of reproductive age, among the lowest in Latin America (World-Bank, 2025).⁶ These patterns suggest that major choice may shape gender disparities not only by sorting men and women into fields with different earnings prospects, but also by exposing them to different career-family tradeoffs.

These descriptive patterns also indicate that gender differences in field sorting are quantitatively important. A simple decomposition demonstrates that differences in representation account for between 46% and 74% of the gender earnings gap among college graduates, depending on whether female or male field-specific earnings are held fixed (reported later in Figure 7). This fact motivates our later focus on gender differences in major demand and on assignment policies that alter access to high-return fields. At the same time, the patterns in Figure 1 are only descriptive. Differences in earnings and fertility across fields may reflect selection, differences in student ability, preferences, family background, or other unobserved determinants of later outcomes. The next section therefore develops a framework for estimating the causal effects of fields of study on earnings and fertility while accounting for selection into majors using applicants' rank-ordered choices and admissions scores.

⁶Drawing on data from the World Bank World Development Indicators database, Appendix Figure A.4 shows that Chile's recent fertility trends align with patterns observed in most high-income economies (Doepke et al., 2023).

Figure 1: Graduation, Earnings, and Fertility



Notes: Panel (a) shows the share of male and female graduates in each field of study. Panel (b) shows the annual earnings (as of 2019) of male and female graduates in each field of study. Both panels consider students who were employed in 2019. Panel (c) shows the share of students with at least one child thirteen years after their college application. All panels consider students who applied through the centralized admission system between 2004 and 2007 and graduated between 2007 and 2019.

3 Conceptual Framework

To understand how college major choice contributes to gender gaps among college graduates, we begin with a framework to estimate causal effects of college majors on both earnings and fertility. Given our focus on major choice and the potential impact of reallocation policies, we adopt a framework that accounts for both vertical and horizontal dimensions of major effects. We begin with a largely non-parametric discussion of major effects, following the approach of Mountjoy and Hickman (2021), which transparently analyzes how student sorting affects observational comparisons and delineates how different sorting margins interact with centralized admission systems. The general intuition behind our approach is that there are two sorting margins of concern: pre- and post-offer sorting. Dale and Krueger (2002)-like logic takes care of the first, while institu-

tional details in settings with centralized assignment make the latter plausibly less important. Our framework therefore draws on approaches developed to study the returns to college selectivity (Dale and Krueger, 2002), and connects to more recent approaches which use rank-ordered choice data (Abdulkadiroğlu et al., 2020) and the fallback-conditioning approach of Kirkebøen et al. (2016).

3.1 Environment and Assumptions

We index the population of accepted college applicants by $i \in \mathcal{I}$ and majors by $j \in \mathcal{J}$. Define $\mathcal{J}_0 \equiv \{0\} \cup \mathcal{J}$, where 0 denotes non-graduation. The potential outcome of student i completing major j is:⁷

$$Y_{ij} = Y_{i0} + \mu_{ij} \quad (1)$$

where Y_{i0} is their potential outcome if they do not graduate from college, and μ_{ij} is the idiosyncratic return of major j . For expositional purposes, we denote the non-graduation potential outcome as $a_i = Y_{i0}$.⁸

In a setting with centralized matches, applicants submit a rank-ordered list of majors, $R_i \in \mathcal{R}$, and have a vector of program-specific admission scores, $S_i \in \mathcal{S}$.⁹ A student type is defined as $\theta_i = (R_i, S_i) \in \Theta$. A centralized system allocates students to majors, producing an offer $Z_i \in \mathcal{J}$. Offers are assumed to be unrelated to potential outcomes conditional on type: $\{Y_{ij}\}_{j \in \mathcal{J}_0} \perp\!\!\!\perp Z_i \mid \theta_i$.

After receiving an offer, students may comply or not. Denote $F_i \in \mathcal{J}_0$ as student i 's completed field, where $F_i=0$ denotes non-graduation. The completed field depends on whether the student complies with the assignment Z_i . Let $D_i = \mathbf{1}\{F_i = Z_i\}$ be an indicator for compliance with the assignment. We can then write

$$F_i = \begin{cases} Z_i & \text{if } D_i = 1 \\ F'_i & \text{if } D_i = 0 \end{cases},$$

where $F'_i \in \mathcal{J}_0$ denotes the field completed by student i when they do not comply with their offer.

Previous work argues that information in college application and admissions portfolios reveals information about student ability (Dale and Krueger, 2002). In settings with centralized assignments, where access to majors is strongly governed by entrance exam scores, the intuition behind the self-revelation principle also applies. We assume that student type adequately summarizes the student's ability and residual ability is mean independent of offers.¹⁰

Assumption 1. *Student types characterize non-graduation potential outcomes,*

$$a_i = \phi(\theta_i) + u_i,$$

⁷We suppress additional notation for different outcomes for expositional purposes.

⁸As our emphasis is on returns to college majors $j \in \mathcal{J}$, we do not consider program-level heterogeneity. We do consider field-specific fallback heterogeneity and that is discussed further below.

⁹Applicants formally rank programs. Throughout the analysis, we map each ranked program to its corresponding field and order the distinct fields according to the position of the highest-ranked program in each field. Therefore, our notation throughout focuses on fields.

¹⁰We suppress dependence on additional covariates for expositional ease but introduce them in the empirical framework that follows in a subsequent section.

with $E[u_i | Z_i, \theta_i] = E[u_i | \theta_i] = 0$.

Assumption 1 clarifies the differences that govern selection on levels and pre-offer sorting, namely that they operate through differences in type. A separate concern is whether decisions made after offers are assigned reveal additional outcome-relevant heterogeneity within student type. In general, post-offer sorting in decentralized systems opens up the scope for additional sources of bias that plague the identification of treatment effects (Hoxby, 2009). The single-offer structure of the centralized system limits one important source of outcome-relevant residual sorting, motivating the following assumption.

Assumption 2. *Compliance Uninformativeness.* For every $j \in \mathcal{J}_0$,

$$E[Y_{ij} | F_i, Z_i, \theta_i] = E[Y_{ij} | Z_i, \theta_i].$$

Assumption 2 states that, conditional on type and realized offers, compliance or non-compliance provides no additional information about major returns or student ability. This mirrors Dale and Krueger (2002)-like logic that revealed preferences and offers account for pre-offer sorting, while the institutional details in centralized environments limit the extent of post-offer sorting justifying Assumption 2. As currently stated, however, the identification strategy suggests the need to condition on both student type and offers. Following the logic from Angrist et al. (2024), we can show that Assumption 2 implies the need to only condition on type.

Lemma 1. Suppose offers are assigned via a centralized assignment system, are conditionally ignorable, and Assumption 1 and Assumption 2 hold. Then, for every $j \in \mathcal{J}_0$,

$$E[Y_{ij} | F_i, \theta_i] = E[Y_{ij} | \theta_i].$$

Consequently, for every $j \in \mathcal{J}$, $E[\mu_{ij} | F_i, \theta_i] = E[\mu_{ij} | \theta_i]$, and $E[u_i | F_i, \theta_i] = 0$.

Proof. See Appendix B □

Combining Lemma 1 with the assignment property in the centralized assignment system eliminates the need to condition on realized offers, providing a tractable approach that we further develop in Section 3.3. Importantly, the conclusions implied by Lemma 1 do not rule out selection on gains, and instead restrict selection on gains to be summarized by student type with residual post-offer sorting limited by the centralized assignment system. In the next section, we more clearly outline the treatment effect heterogeneity this framework allows.

3.2 Sources of Treatment Effect Heterogeneity

With access to rank-ordered choice data and entrance exam scores, we can now assess how Lemma 1 affects type-specific observational comparisons. With observed outcomes Y_i , and fixing $\theta_i = \theta$,

we can compare the mean outcomes of major j graduates to those of non-graduates ($j = 0$):

$$E[Y_i|F_i = j, \theta] - E[Y_i|F_i = 0, \theta] = \underbrace{E[\mu_{ij}|F_i = j, \theta]}_{ATT_j(\theta)} + \underbrace{E[a_i|F_i = j, \theta] - E[a_i|F_i = 0, \theta]}_{\text{Selection bias} = 0, \text{ by Lemma 1}}. \quad (2)$$

Equation (2) shows that observational comparisons within types θ recover θ -specific average treatment effects on the treated, as Lemma 1 takes care of selection bias. While empirically interesting and important to some extent, this quantity masks different sources of treatment effect heterogeneity that may be important to policymakers. To see this concretely, we can write the $ATT_j(\theta)$ as:

$$ATT_j(\theta) = \underbrace{E[\mu_{ij} | \theta]}_{ATE_j(\theta)} + \underbrace{(E[\mu_{ij} | F_i = j, \theta] - E[\mu_{ij} | \theta])}_{\text{Post Offer Sorting} = 0, \text{ by Lemma 1}}.$$

The first component measures the average treatment effect for all students of type θ . The second component isolates differences in treatment effects arising from post-offer sorting: it captures the extent to which the treatment effect for type θ students who complete j differs from the average effect for type θ students. Lemma 1 rules out this kind of post-offer sorting and institutional features of centralized assignment largely limit this form of sorting on gains.¹¹

Lemma 1 therefore addresses selection bias and post-offer sorting effects that raised concerns in prior work. Our empirical focus is now on distinguishing between the vertical and horizontal components of payoffs. We can write the type θ -specific conditional average treatment effect as:

$$ATE_j(\theta) = \underbrace{E[\mu_{ij} | \mathcal{I}]}_{ATE_j(\mathcal{I})} + \underbrace{(E[\mu_{ij} | \theta] - E[\mu_{ij} | \mathcal{I}])}_{M_j(\theta)}, \quad (3)$$

where $ATE_j(\mathcal{I})$ represents the payoff to the typical student in the pool of accepted applicants—the vertical component—and $M_j(\theta)$ captures the horizontal component, i.e., if those of type θ experience systematically different gains than the typical admitted student, then $M_j(\theta)$ will be nonzero.¹² This sorting is intimately linked to the formation of the rank-ordered list, an element of θ , which is itself governed by inherent preferences. In this respect, the type of sorting considered by our approach is analogous to that considered by Kirkeboen et al. (2016), where preference heterogeneity is associated with potential outcomes.

¹¹This type of sorting has raised concerns in studies estimating the returns to college selectivity (Dale and Krueger, 2002). As Hoxby (2009) points out, “the [Dale and Krueger] strategy generates estimates that rely entirely on the small share of students who make what is a very odd choice. These are students who know that they could choose a much more-selective college and who have already expressed interest in a much more-selective college (they applied), yet, they choose differently than 9 out of 10 students. Almost certainly, these odd students are characterized by omitted variables that affect both their college decision and their later life outcomes.” That observation underscores the empirical relevance of this kind of sorting in decentralized markets. In settings with centralized assignment, however, this kind of concern is much attenuated.

¹²For simplicity, Equation 3 is defined over the full accepted population. Below, we generalize this to allow for gender-specific heterogeneity in both vertical and horizontal dimensions of treatment effect heterogeneity.

3.3 Empirical Approach

We now specify a choice model that helps translate complex student types, $\theta = (R_i, S_i)$, into a more manageable, lower-dimensional characterization of ability and preferences. We further characterize observable heterogeneity in terms of observed applicant characteristics X_i , gender G_i , and a field-specific distance vector d_i .¹³ Then through a combination of distributional assumptions on unobserved preference heterogeneity in the choice model and by embedding elements of preferences into potential outcomes, this approach also enables us to recover both the vertical component, $ATE_j(\mathcal{I})$, and the horizontal component. We begin by detailing the structure of the choice model.

Major Choice: We model demand for majors, abstracting away from demand for particular programs or institutions relative to a left-out field of study.¹⁴ For student i , the indirect utility from choosing major j is specified as:

$$U_{ij} = \delta_{c(X_i)j} - \kappa_{c(X_i)} d_{ij} + \eta_{ij}, \quad (4)$$

where $\delta_{c(X_i)j}$ is the mean utility for applicants in a cell $c(X_i)$ defined by their observed covariates (region, school type, PSU score group range, and gender). A given d_{ij} in the vector of distances $d_i = (d_{i1}, \dots, d_{iJ})$ measures the distance between the student’s high school and the nearest program corresponding to field j , and η_{ij} is an unobserved preference shock.

Equation (4) imposes several assumptions that enable us to summarize demand in a parsimonious manner. First, we assume that students form rational expectations about mean potential outcomes in each major. Second, we parameterize preference heterogeneity as covariate-specific and aggregate it into distinct cells $c(X_i)$. Third, and related to financial aid and net price, students in Chile face different financial aid schedules governed by rigid discontinuities in the PSU distribution. We account for this in our choice of cell strata, which include different non-overlapping PSU score groups. This allows us to assume that students’ net price is constant within a cell. These three assumptions allow us to summarize demand with cell-specific mean utilities and a distance shifter. Last, we assume that η_{ij} are independent and identically distributed with an Extreme Value Type 1 distribution conditional on $\delta_{c(X_i)j}$ and d_{ij} .

Identification of Payoffs: Our empirical approach next formalizes how selection into majors maps into potential outcomes. Following the logic of schooling selection models, we allow unobserved determinants of educational choices to affect both baseline outcomes and the gains from completing a particular field.

¹³Note that G_i is contained in X_i but we distinguish it to clarify the gender heterogeneity in what follows.

¹⁴Collapsing demand estimation is common in the literature. Abdulkadiroğlu et al. (2020) estimate demand for schools as opposed to programs, similar to us estimating demand for majors instead of programs. Laverde (2022) uses similar aggregations in the context of pre-K in Boston.

Assumption 3. Let η_{ij} correspond to the unobserved preference heterogeneity governing choices in Equation (4). We assume student ability is summarized by:

$$a_i = \gamma(X_i, d_i) + \sum_{l \in \mathcal{J}} (\psi_l + \psi_l^G G_i) (\eta_{il} - \bar{\eta}) + \tilde{u}_i,$$

and treatment effects are parameterized as:

$$\mu_{ij} = \alpha_j + \alpha_j^G G_i + (\psi_j^* + \psi_j^{*G} G_i) (\eta_{ij} - \bar{\eta}).$$

These parametric assumptions imply the following outcome equation:

$$\begin{aligned} Y_{ij} = & \alpha_j + \alpha_j^G G_i + \gamma(X_i, d_i) \\ & + \sum_{l \in \mathcal{J}} (\psi_l + \psi_l^G G_i) (\eta_{il} - \bar{\eta}) \\ & + (\psi_j^* + \psi_j^{*G} G_i) (\eta_{ij} - \bar{\eta}) + \tilde{u}_i, \quad j = 1, \dots, J. \end{aligned} \tag{5}$$

Equation (5) in Assumption 3 adds three layers of structure to the potential-outcome model. First, selection on levels is fully captured by observed characteristics X_i (which include entrance-exam scores) together with the vectors summarizing distance and preference heterogeneity.¹⁵ Two students who appear identical on paper may therefore experience different outcomes if their underlying preferences differ. The coefficient ψ_l translates this selection on levels for men: a positive ψ_l means applicants who especially favor field l tend to perform better everywhere (i.e., no matter which major they finish). For women, selection on levels governed by preferences is summarized by $\psi_l + \psi_l^G$. Second, the vertical dimension of major quality is allowed to vary by gender through the parameters α_j and α_j^G . Third, the horizontal dimension of major quality captures the preference-specific match quality, is linear in the unobserved preference heterogeneity, and may also differ by gender. The ψ_j^* captures the match effects for men while $\psi_j^* + \psi_j^{*G}$ summarizes it for women, both commonly referred to as *selection on gains* in these types of models (Abdulkadiroğlu et al., 2020; Bruhn et al., 2023; Campos et al., 2025). In the case that $\psi_j^* > 0$, men with a higher preference for major j and who complete major j experience an additional earnings gain, evidence of positive Roy (1951)-like selection. If $\psi_j^{*G} \neq 0$, then that captures the differential horizontal return that women experience or differential selection on gains.

¹⁵Note that $\tilde{u}_i$ inherits the same properties as u_i defined in Assumption 1, Assumption 2, and Lemma 1, but is now the empirical counterpart to the conceptual u_i .

The final step is to arrive at an empirical specification for the observed outcomes. Following Abdulkadiroğlu et al. (2020), the conditional mean of observed outcomes is:

$$\begin{aligned} E[Y_i \mid R_i, S_i, X_i, G_i, d_i, F_i = j] &= \alpha_j + \alpha_j^G G_i + \gamma(X_i, d_i) \\ &+ \sum_{l \in \mathcal{J}} (\psi_l + \psi_l^G G_i) \lambda_{il} \\ &+ (\psi_j^* + \psi_j^{*G} G_i) \lambda_{ij}. \end{aligned} \quad (6)$$

where $\lambda_{ij} \equiv E[\eta_{ij} - \bar{\eta} \mid R_i, S_i, X_i, G_i, d_i]$ is the control function for student i 's unobserved preference for field j . As shown by Abdulkadiroğlu et al. (2020) in the school-choice setting, these control functions can be derived from the rank-ordered choice data in R_i under an Extreme Value Type 1 (EVT1) assumption on unobserved preference heterogeneity. This distributional assumption yields analytical expressions for the control functions, thereby avoiding the need to compute the multi-dimensional integrals implied by the students' choice problem. Finally, the random variation in assignments generated by the centralized admissions process induces variation in completed fields conditional on students' preferences and observed characteristics.

Equation (6) highlights how Assumption 3 connects to Assumption 1 and Assumption 2, allowing us to summarize the conditional average treatment effect of a given major j as:

$$ATE_j(R_i, S_i, X_i, G_i, d_i) = \underbrace{\alpha_j + \alpha_j^G G_i}_{ATE_j(G_i)} + \underbrace{(\psi_j^* + \psi_j^{*G} G_i) \lambda_{ij}(R_i, S_i, X_i, G_i, d_i)}_{M_{ij}(G_i)}. \quad (7)$$

Equation (7) illustrates how treatment effects vary with respect to applicant type—via the composition of their rank-ordered list, R_i , the characteristics embedded in X_i (including test scores), gender, G_i , and distance, d_i . This approach bears similarity to fallback conditioning as treatment effects vary according to the composition of a student's rank-ordered list, R_i . For example, the return to a given field includes a match effect governed by the submitted rank-ordered list, including most-preferred field of study and any fallback options the student ranks. This means that treatment effects for major j differ for individuals with different fallbacks, and even among individuals with similar fallbacks, additional differences in the composition of the rank-ordered list produce additional differences in treatment effects. Therefore, our assumptions governing selection into majors play a dual role. They allow us to estimate causal effects by accounting for preference heterogeneity flexibly while simultaneously allowing us to overcome power limitations from other data-demanding approaches that emphasize the importance of fallback conditioning (Dahl et al., 2020; Kirkebøen et al., 2016). Accounting for this heterogeneity is also important for policy evaluation because students whose assignments change may have systematically different gains from moving between fields (Kirkebøen et al., 2016).

For reallocative policies that are our focus in Section 5, students get shuffled from field k to field j . The conditional average treatment effect of such a move is:

$$\begin{aligned}
ATE_{jk}(R_i, S_i, X_i, G_i, d_i) &\equiv E[Y_{ij} - Y_{ik} \mid R_i, S_i, X_i, G_i, d_i] \\
&= \underbrace{(\alpha_j - \alpha_k) + (\alpha_j^G - \alpha_k^G)G_i}_{ATE_{jk}(G_i)} \\
&\quad + \underbrace{\left(\psi_j^* + \psi_j^{*G}G_i\right)\lambda_{ij}(R_i, S_i, X_i, G_i, d_i) - \left(\psi_k^* + \psi_k^{*G}G_i\right)\lambda_{ik}(R_i, S_i, X_i, G_i, d_i)}_{M_{ijk}(R_i, S_i, X_i, G_i, d_i)}.
\end{aligned} \tag{8}$$

When the fallback is non-graduation, Equation (8) reduces to Equation (7), with the vertical and match components associated with non-graduation normalized to zero.

Estimating the Choice Model: We leverage rich information contained in applicants' rank-ordered lists to estimate the parameters of the choice model that then allow us to construct the control function $\lambda_{ij}(R_i, X_i, G_i, d_i)$. We rely on the fact that truthful reporting is a weakly dominant strategy in the Chilean centralized assignment system—which uses a variant of a student-proposing deferred acceptance algorithm (Gale and Shapley, 1962). A student reveals field j as their most preferred if they list it at the top of their list, the next-stated field is their second preferred, and so on. Preferences are summarized by a vector $R_i = (R_{i1}, R_{i2}, \dots, R_{iK_i})$, where R_{i1} is the top-listed field and the length of the list, K_i , is allowed to vary across students.¹⁶ Thus, under truthful reporting, students' top-ranked field satisfies:

$$R_{1i} = \arg \max_{s \in \mathcal{J}} U_{is}$$

while the remaining options k satisfy:

$$R_{ki} = \arg \max_{s \in \mathcal{J} \setminus \{R_{i1}, \dots, R_{i,k-1}\}} U_{is}, \quad k > 1.$$

If we let $c_i \equiv c(X_i)$, the EVT1 assumption allows us to express the likelihood of observing the major-specific rank-ordered list as follows:

$$\mathcal{L}(R_i \mid X_i, d_i) = \prod_{k=1}^{K_i} \frac{\exp(\delta_{c_i, R_{ik}} - \kappa_{c_i} d_{i, R_{ik}})}{\sum_{r \in \mathcal{J} \setminus \{R_{i1}, \dots, R_{i,k-1}\}} \exp(\delta_{c_i, r} - \kappa_{c_i} d_{i, r})}.$$

We estimate preference models separately for 36 covariate cells (i.e., 2 school types, 3 macro-regions, 3 financial aid-relevant PSU ranges, and 2 genders) via maximum likelihood to obtain a list of field-specific mean utilities.

One potential concern with our estimation approach is potential deviations away from truthful reporting. As Fack et al. (2019) point out, some students may skip the impossible, or constrained

¹⁶For example, some students rank a single field across their list, while others rank multiple fields.

list lengths may introduce other empirical challenges (Haeringer and Klijn, 2009). Estimating preferences under an *ex-post stability* assumption is an increasingly popular approach that circumvents these issues. We choose not to use this approach in our main analysis because it requires implicit extrapolations that impose strong restrictions on preference heterogeneity across the entrance exam score distribution. Instead, we adopt a strategy that uses all of the information contained in each individual’s rank-ordered list and validate it by comparing our estimates to those obtained from an alternative approach that leverages local random assignment to majors (Angrist et al., 2017).

Empirical Outcome Model Using the sample of students admitted through the centralized admission system, we estimate gender-specific average major effects and match effects, while allowing selection on outcome levels to differ by gender. For a given outcome, we estimate the empirical analog of Equation (6) that incorporates the estimated control functions $\hat{\lambda}_{ij}$:

$$\begin{aligned} Y_i = & \sum_j \alpha_j F_{ij} + \sum_j \alpha_j^G F_{ij} G_i + \sum_j \psi_j \hat{\lambda}_{ij} + \sum_j \psi_j^G \hat{\lambda}_{ij} G_i \\ & + \sum_j \psi_j^* \hat{\lambda}_{ij} F_{ij} + \sum_j \psi_j^{*G} \hat{\lambda}_{ij} F_{ij} G_i + \beta \mathbf{W}_i + \xi_a + \zeta_{c(X_i)} + \varepsilon_i, \end{aligned} \quad (9)$$

where the outcomes of interest Y_i are log earnings and a binary indicator for having at least one child thirteen years post-college application. The variable $F_{ij} \equiv \mathbb{1}(F_i = j)$ is an indicator that equals 1 if individual i completed field j , and 0 otherwise; and G_i is another indicator variable that equals 1 if an individual i is female. $\hat{\lambda}_{ij}$ is the control function proxying for individual i ’s preferences for field j and $\mathbf{W}_i$ are quadratic polynomials of the college admission scores in mathematics and language and a distance vector. We include cohort (i.e., year of application) ξ_a and cell $\zeta_{c(X_i)}$ fixed effects, where a cell is defined by 2 school types (public and private), 3 macro-regions (north, center, and south), 3 financial aid-relevant PSU ranges (<450 , $\in (450, 550)$, >550), and 2 genders. To ease interpretation, we standardize $\hat{\lambda}_{ij}$ within each cell.

For inference, we use a parametric bootstrapping procedure to account for estimation errors in the estimated control functions $\hat{\lambda}_{ij}$. Concretely, for each cell c , we have an estimate of the asymptotic distribution of (δ_{cj}, κ_c) , which we can draw from in each bootstrap iteration b to construct different control functions. Thus, we estimate model (9) using ordinary least squares and the bootstrapped control function $\hat{\lambda}_{ij}^b$ one hundred times. For each of the model’s parameters, we report the mean estimate across bootstrap iterations and the mean of the standard errors.

4 Causal Effects of College Majors, Fertility, and Gender Gaps

We now use our framework to study how major choice shapes the tradeoff between career and family by affecting both earnings and fertility trajectories. We proceed in five steps. First, we estimate gender-specific earnings effects and ask whether differences in average returns or preference-based match effects have enough variation to potentially explain gendered sorting into majors. Second,

we turn to fertility and document substantial heterogeneity across fields and gender but limited preference-linked heterogeneity. Third, having established that fertility effects are largely homogeneous within gender, we ask whether the cross-field dispersion in fertility effects is systematically related to characteristics of the careers linked to each field—such as job security, child penalties, and gender norms—that may make work and family more or less compatible. Fourth, we validate our empirical approach by comparing its predictions with quasi-experimental estimates based on locally random assignment to programs. Finally, we assess whether students’ choices are associated with the estimated ex-post earnings and fertility payoffs and conduct counterfactual analysis to gauge how gender differences in causal effects and the relative importance assigned to these causal effects relate to gendered sorting into majors.

4.1 Do differences in earnings effects or comparative advantage explain gendered major sorting?

We begin by reporting causal earnings effects by gender to assess if there is sufficient gender heterogeneity—both vertical and horizontal—to potentially explain the gendered sorting into majors we observe. One possibility is that the gendered sorting may be due to differences in the pecuniary payoffs men and women can expect to have in the labor market, or in other words, if observationally similar women and men obtain different average returns from completing the same field, they may rationally rank fields differently. But even if average effects are similar, horizontal effect differences—driven by preference heterogeneity—could add an important wrinkle: women and men may differ in their earnings-based comparative advantage and therefore expect different earnings gains from similar field allocations. Our earnings-based analysis therefore explores these possibilities. A lack of strong gender differences in monetary payoffs suggestively points to other factors governing gendered sorting patterns, of which we focus on fertility in the next subsection.

Table 2 reports how obtaining degrees from different fields of study affects log earnings among accepted college applicants. Our first main finding concerns our general estimates to the returns to college graduation: the unweighted average return is 39.5 log points for women and 40.5 log points for men, both relative to gender-specific average earnings among non-graduate accepted college applicants. Weighting by the size of the graduate pool for each gender yields an average return for women of 43 log points and 51 log points for men, underscoring the fact that the allocation across majors matters for the observed gender earnings gap. Our second finding, which is consistent with a growing body of work, is that payoffs vary substantially across majors. Except for Humanities, all fields generate positive payoffs ranging from 17 to 80 log points. Medicine, Business, and Engineering stand out with the largest earnings premiums, ranging from 55 to 80 log points. The noise-adjusted standard deviation of these causal returns is 22 log points for women and 27 log points for men.¹⁷ This dispersion is similar to patterns found by Altonji et al. (2012), who find that the dispersion in returns rivals the college premium.

¹⁷We calculate the noise-adjusted variance using the formula: $\frac{1}{J} \sum_{j=1}^J ((\hat{\alpha}_j - \mu_\alpha)^2 - (SE(\hat{\alpha}_j))^2)$, where μ_α is the overall mean across field returns $\hat{\alpha}_j$ and $SE(\hat{\alpha}_j)$ is the estimation error.

Although we find evidence of substantial cross-field variation in causal earnings returns, we find limited gender differences in these effects. Men have statistically significantly higher effects in Science and Engineering, by approximately 20 and 7 log points, respectively, whereas women have an approximately 9 log points higher effect in Teaching. We cannot reject earnings effects are the same for the six remaining fields of study. Taken together, average earnings effects are largely similar across gender, and of the differences we do find, the evidence does not point uniformly toward an advantage for one group. This latter finding is notable given evidence that gender earnings gaps can persist even in relatively homogeneous populations with similar degrees (Graddy and Pistaferri, 2000).

Before turning to the horizontal dimension, it is worth emphasizing that the magnitude of the average effects we document is in line with credible estimates in the broader literature. Card (1999) documents that an additional year of schooling raises earnings by roughly 8-10 percent, so completing a four- to five-year degree—as is done in Chile—amounts to gains of 32-50 percent. Quasi-experimental estimates of marginal college admission in the United States imply earnings gains of roughly 8–22 percent (Hoekstra, 2009; Mountjoy, 2024; Zimmerman, 2014), although these estimates isolate admissions effects instead of degree completion. A closer setting-specific comparison is Hastings et al. (2013), who also document substantial heterogeneity in returns across more granular degree programs in Chile using admission cutoffs. Although their threshold-crossing approach similarly combines admission, enrollment, completion and a host of other factors, under reasonable assumptions about the completion first stage, their estimates imply returns of a similar magnitude as our estimates.

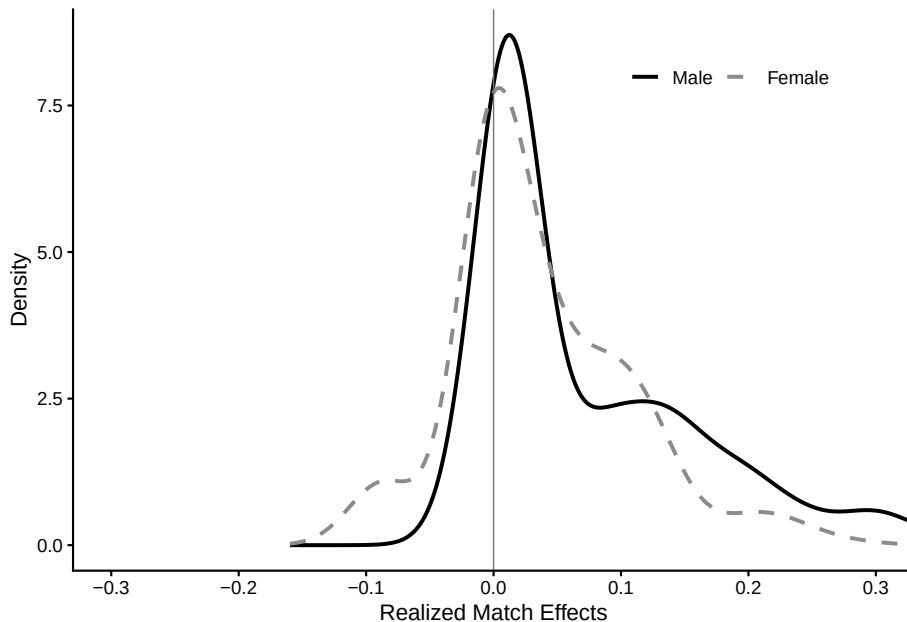
We have shown that there are minimal gender differences in field-specific returns, but that does not rule out differences in horizontal payoffs. Table 2 shows that students with stronger tastes for the majors they ultimately enroll in experience larger earnings returns. This evidence is consistent with positive Roy (1951)-like selection on gains documented in other settings (Kirkebøen et al., 2016). Our estimates suggest that a female student with a one-standard deviation larger preference intensity for Law is predicted to have an additional 6 log point boost in earnings, while a similar male student experiences a 9 log point boost to earnings. The positive selection on gains is present in nearly all fields for both genders. Across fields, however, we find minimal evidence that match-effect coefficients are different: we reject equality only in Teaching and Business. Figure 2 reports the distribution of realized match effects, which correspond to $\hat{\psi}_j^* \hat{\lambda}_{ij}$, for men and women. We find a mean of 0.04 for women and 0.07 for men. These are realized match effects relative to a student with typical preferences for the field, for whom $\hat{\lambda}_{ij} = 0$. Although the realized match effects are moderately different, the coefficients are largely the same indicating minimal gender differences in preference-based comparative advantage. Taken together, the evidence shows that students sort into majors on comparative advantage, as in Kirkebøen et al. (2016), but much less evidence that the strength of selection on gains differs systematically by gender.

Table 2: Gender Differences in Pecuniary Returns: Log Earnings in 2019

	Law (1)	Science (2)	Social Science (3)	Teaching (4)	Humanities (5)	Engineering (6)	Health (7)	Medicine (8)	Business (9)
Panel A: Female									
Returns	0.364 (0.069)	0.333 (0.053)	0.214 (0.026)	0.415 (0.010)	-0.056 (0.054)	0.554 (0.027)	0.347 (0.022)	0.779 (0.036)	0.606 (0.018)
Selection on Gains	0.062 (0.023)	0.029 (0.026)	0.038 (0.014)	-0.004 (0.008)	0.081 (0.022)	0.011 (0.015)	0.059 (0.011)	-0.035 (0.014)	0.017 (0.012)
Panel B: Male									
Returns	0.437 (0.081)	0.536 (0.028)	0.171 (0.043)	0.321 (0.018)	-0.184 (0.070)	0.623 (0.015)	0.338 (0.036)	0.801 (0.042)	0.598 (0.022)
Selection on Gains	0.088 (0.027)	0.039 (0.016)	0.068 (0.019)	0.058 (0.010)	0.095 (0.025)	0.013 (0.012)	0.076 (0.017)	0.003 (0.019)	0.050 (0.010)
Est. Females = Est. Males									
Δ Returns (p-val)	0.491	0.001	0.376	0.000	0.145	0.020	0.818	0.689	0.751
Δ Match Effects (p-val)	0.458	0.730	0.193	0.000	0.674	0.926	0.401	0.100	0.028

Notes: This table presents the estimates obtained from our main regression model, presented in Equation (9), using log earnings in 2019 as the outcome. We display the averages—of coefficients, robust standard errors, and p-values—across 100 regressions, each of which uses a different set of control functions obtained after drawing from the asymptotic distribution of the demand model estimates. The sample includes 102,597 observations of students who applied and were accepted into a program through the centralized admission system between 2004 and 2007 and were employed in 2019. Panel (a) focuses on parameter estimates for Female students, and Panel (b) focuses on Male students. In each panel, we report return estimates and selection on gains parameters, also referred to as match effects in the text. Underneath each estimate in parentheses is the bootstrapped standard error. The final two rows of the table report p-values from two additional statistical tests. The Δ Returns p-values correspond to the null hypothesis that the Female and Male returns are equivalent for a given column. The Δ Match Effects p-values correspond to the null hypothesis that the Female and Male match effects are equivalent for a given column. In addition to main effects for field of study and control functions and gender interactions, all specifications include quadratic polynomials of the college admission scores in mathematics and language, a distance vector, cohort fixed effects and cell fixed effects, where a cell is defined by school type, macro-region, financial-aid relevant score ranges, and gender. For ease of interpretation, $\hat{\lambda}_{ij}$ is standardized within each cell. The R^2 of this regression is 0.19.

Figure 2: Distribution of Realized Match Effect Estimates



Notes: This figure shows the density of students' realized match effect estimates relative to students with typical preferences for their completed major and similar gender (i.e., $\hat{\psi}_j^* \times (\hat{\lambda}_{ij} - 0)$). For each student, we consider the $\hat{\psi}_j^*$ —reported in Table 2—corresponding to their completed field and interact it with the student's control function estimate, $\hat{\lambda}_{ij}$, for the same field. The figure uses the 2004–2007 graduate cohorts composition in the estimation-sample extract used for the gender-gap decomposition. Individual control functions within each gender–field cell are recentered to the corresponding estimation-sample cell mean and the densities are weighted to reproduce that cell's share. The mean (standard deviation) of the realized match effect is 0.07 (0.09) for men and 0.04 (0.07) for women.

Our evidence thus far points to surprisingly limited gender-specific heterogeneity in vertical and horizontal earnings effects within majors. This evidence suggests that the scope for gender heterogeneity driving gendered sorting patterns is limited. One remaining concern that goes against this interpretation is that field-specific averages may average out important fallback-specific heterogeneity (Kirkeboen et al., 2016). There could be stark gender differences in returns after we isolate fallback- and gender-specific returns. Appendix Table C.1 documents substantial heterogeneity in fallback-specific payoffs, consistent with Kirkeboen et al. (2016). Figure 3 then asks whether the fallback heterogeneity varies substantially across gender. For each of the ordered first-ranked and second-ranked field pairs observed for both genders,¹⁸ the figure plots the estimated male relative payoff against the corresponding female relative payoff. It also plots the nine main field effects from Table 2 in maroon. Although the fallback-specific estimates vary considerably across pairs, the male and female estimates lie close to the 45-degree line: the fitted slopes are 1.05 for the main field effects and 1.23 for the fallback-specific effects, and neither is statistically distinguishable from one.¹⁹ We also selectively color options where Humanities is the focal option in purple and pairs

¹⁸Four field-gender combinations cannot be estimated: Humanities–Medicine for both women and men, Medicine–Humanities for women, and Law–Medicine for men.

¹⁹The corresponding p -values are 0.15 and 0.20, respectively. These are approximate coefficient-level delta-method tests that use estimated coefficient variances while ignoring covariance terms. Because fallback-specific effects share underlying coefficient estimates, the tests should be interpreted with this qualification in mind.

where Business, Engineering, and Medicine are the fallback option in brown, orange, and green. We do this to emphasize that a majority of the negative relative payoffs are due to diversions away from high paying fields or into low-paying fields. Overall, the evidence in Figure 3 shows that the similarity of returns by gender is not sensitive to fallback conditioning and we conclude that there is surprisingly limited gender heterogeneity in field-specific returns to fields of study.

Appendix C presents additional evidence on economic payoffs. Appendix Table C.2 reports estimates for total rather than log earnings to account for the extensive margin, and Appendix Table C.3 uses unemployment-insurance records from the AFC, which have a higher top-coding threshold but cover only private-sector earnings. The results remain qualitatively similar.²⁰

Taken together, the estimates reveal substantial heterogeneity in pecuniary payoffs across fields and economically meaningful selection on gains, but limited evidence of systematic gender differences in the underlying vertical or horizontal payoff schedules. Pecuniary payoffs remain important for the earnings consequences of where students sort, but gender-specific pecuniary payoffs do not appear to be a primary culprit governing why women and men sort into such different fields. This motivates turning to another dimension along which majors differ: their effects on fertility and the career–family bundles they provide.

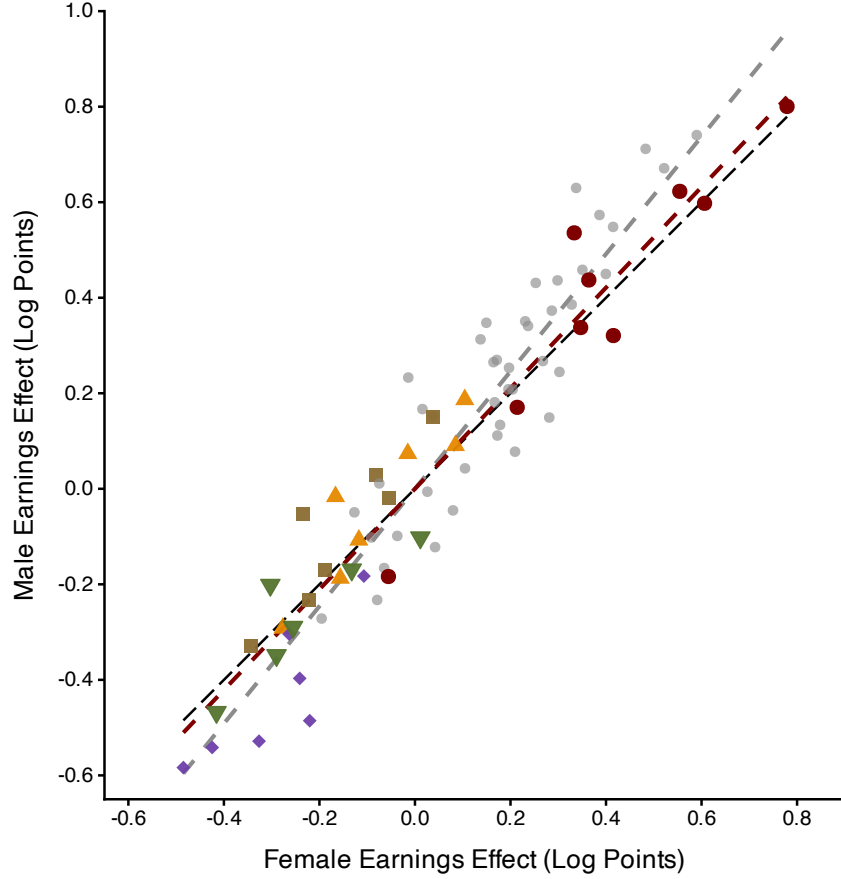
4.2 Majors as Career-Family Bundles

Majors shape more than labor-market payoffs. We now turn to fertility and ask whether fields of study operate as career-family bundles, combining human-capital investments with distinct career and family trajectories. As we will see, this margin offers a sharp contrast with the earnings results. Earnings effects display economically meaningful heterogeneity associated with students’ preferences for their chosen fields but comparatively modest gender differences in the payoff schedules. As we show next, fertility effects exhibit the reverse pattern: they display modest heterogeneity associated with students’ preferences for their chosen fields but economically meaningful gender differences in the payoff schedules.

Table 3 reports major completion effects on the probability of having at least one child thirteen years after college application. Because the outcome is measured before completed fertility is fully realized, we interpret the estimates as effects on parenthood by this horizon rather than necessarily on lifetime fertility. We emphasize that the timing is economically meaningful as postponing the transition to parenthood compresses the fertility window and is associated with lower completed fertility (Bongaarts and Feeney, 1998; Roustaei et al., 2019). Averaging the nine field coefficients equally, completing a field lowers the probability of having become a parent by 6.6 percentage

²⁰One exception is Medicine, for which the estimated payoff becomes smaller in the AFC data. One explanation is the substantial role of the public sector as an employer of doctors. Public hospitals account for an average of 72% of all hospital beds across OECD countries, and Chile is close to this average (Muñoz and Otero, 2025). Because many doctors practice across public and private institutions, estimates derived solely from AFC data may therefore capture only part of their total annual earnings.

Figure 3: Male-Female Earnings Effects Association



Notes: This figure compares male and female estimates of the effects of fields of study on log earnings. Female estimates are plotted on the horizontal axis and male estimates on the vertical axis. Maroon circles report the nine field-specific effects from Table 2, measured relative to the corresponding gender-specific earnings of non-graduates. The remaining points report fallback-specific relative payoffs. For each gender and ordered pair of first-ranked field j and second-ranked field k , these estimates compare the predicted payoff to j with the predicted payoff to k , evaluated at the mean control functions among students who submit that pair. Other fallback-specific estimates are shown as gray circles. Purple diamonds denote pairs with Humanities as the first-ranked field; brown squares, orange upward triangles, and green downward triangles denote pairs with Business, Engineering, and Medicine, respectively, as the fallback field. The black dashed line is the 45-degree line. The maroon and gray dashed lines are separate unweighted regressions of the male estimates on the female estimates for the main and fallback-specific effects, respectively. The corresponding slopes are 1.05 and 1.23. Approximate coefficient-level diagonal delta-method tests of a unit slope yield p -values of 0.15 and 0.20. The calculation fixes cell mean control functions, sets off-diagonal coefficient covariances to zero, and retains the dependence induced by shared coefficients across fallback cells.

points for women and 13.2 percentage points for men.²¹ Appendix Table C.4 shows that the broad pattern also appears when fertility is measured by the number of children.

²¹The more negative effect for men at this horizon is consistent with evidence that men tend to have children later in life (Wang et al., 2023).

The effects differ meaningfully across fields. For women, the estimates range from a 4.5 percentage point increase among Teaching graduates to a 15.6 percentage-point reduction among Law graduates, while the estimate for Health is close to zero. For men, all nine fertility timing effects are negative, ranging from a 7.0 percentage point reduction among Teaching graduates to a 21.0 percentage point reduction among Humanities graduates. Adjusting for estimation error, the standard deviation across fields is 5.2 percentage points for women and 3.4 percentage points for men. Taken together, we find strong evidence that fields of study strongly influence fertility.

A sharper pattern emerges when we focus on gender differences in fertility effects. The male effect is substantially more negative in eight of the nine fields, by an average of 6.6 percentage points. We reject that effects are equal across gender at conventional levels in seven fields, in contrast to just three for earnings effects. Figure 4 visualizes the different patterns across earnings and fertility effects. Although the nine main fertility effects are positively correlated across gender, with a correlation of 0.685, we can decisively reject they are equal on average. Following the same exercise as in Figure 3, Figure 4 reports the relationship between fallback-specific relative effects across gender. In contrast to the earnings effects, these estimates are less closely aligned across gender and we reject they are the same on average. Overall, the evidence in Figure 4 shows much less gender alignment in fertility payoff schedules than in earnings payoff schedules.

Another stark contrast between earnings and fertility effects is related to the horizontal dimension: fertility preference-based match effects are small. Their mean absolute magnitude is approximately 1.0 percentage point for women and 0.9 percentage points for men, showing no consistent pattern across fields or gender. Only 4 of 18 parameters are distinguishable from statistical noise. Overall, the match effects provide limited evidence that fertility effects vary systematically with applicants’ preference intensity.

The evidence therefore suggests that a field’s fertility effect is less a return to idiosyncratic fit than an attribute shared by students of the same gender who follow a common educational and career path. Consistent with this viewpoint, choosing a major partly selects a “household production technology” where fields guide graduates toward careers that differ in job security, predictability, the penalties attached to work interruptions, and the ease of combining paid work with childcare (Adda et al., 2017; Blundell et al., 2018; Goldin, 2021). At the same time, fields of study may also affect family formation through peer composition and the marriage market (Kirkebøen et al., 2025). These are all plausible mechanisms that may contribute to the variation in fertility effects we document. Regardless of the mechanism, the relatively homogeneous gender-specific structure of the effects motivates asking whether the cross-field variation is systematically related to measures of career–family compatibility, including job security, child penalties, and gender norms. We formally explore this connection next.

4.3 Career–Family Correlates of Fertility Effects

What accounts for the substantial heterogeneity in fertility effects across fields? As documented above, fertility effects vary sharply across fields and gender but only weakly with applicants’ revealed

Table 3: Gender Differences in Non-Pecuniary Returns: Any child 13 years after college application

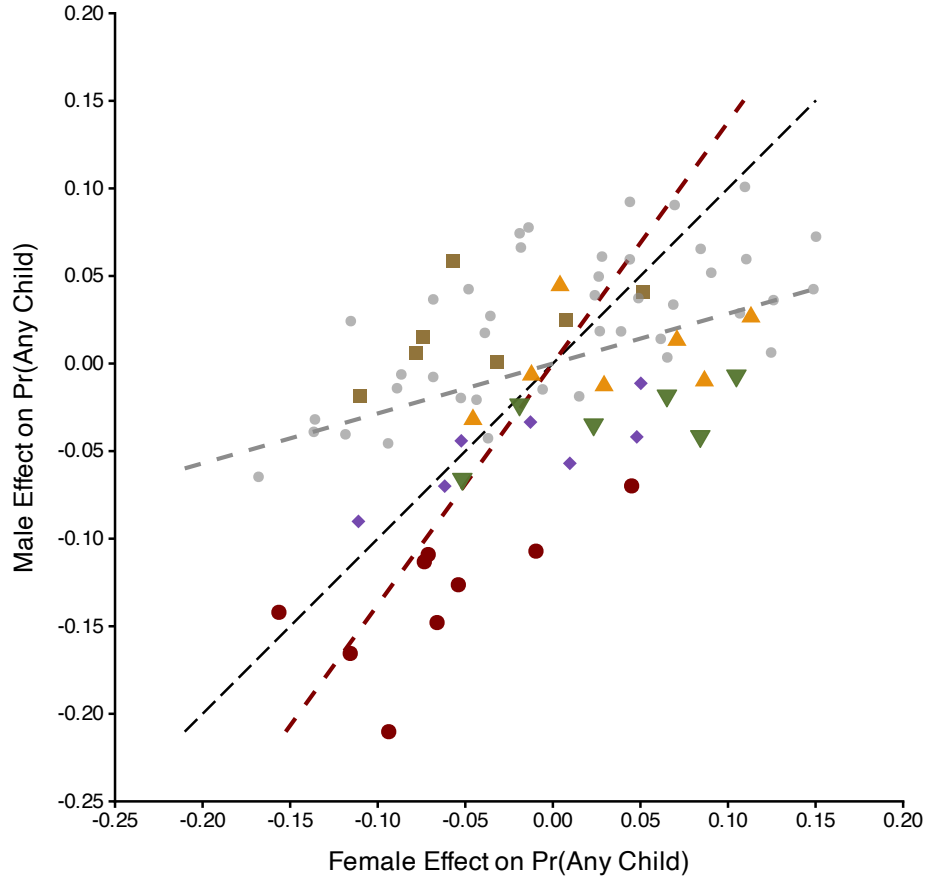
	Law (1)	Science (2)	Social Science (3)	Teaching (4)	Humanities (5)	Engineering (6)	Health (7)	Medicine (8)	Business (9)
Panel A: Female									
Returns	-0.156 (0.025)	-0.116 (0.023)	-0.066 (0.012)	0.045 (0.008)	-0.094 (0.020)	-0.071 (0.015)	-0.010 (0.011)	-0.073 (0.018)	-0.054 (0.010)
Selection on Gains	0.019 (0.009)	-0.006 (0.011)	0.006 (0.007)	-0.011 (0.005)	0.008 (0.009)	-0.013 (0.008)	-0.001 (0.006)	-0.002 (0.007)	0.022 (0.007)
Panel B: Male									
Returns	-0.142 (0.026)	-0.166 (0.012)	-0.148 (0.017)	-0.070 (0.012)	-0.210 (0.021)	-0.109 (0.008)	-0.107 (0.016)	-0.113 (0.017)	-0.126 (0.011)
Selection on Gains	0.009 (0.009)	0.013 (0.007)	0.011 (0.008)	-0.006 (0.007)	0.016 (0.008)	0.001 (0.007)	0.005 (0.008)	0.015 (0.008)	0.003 (0.005)
Est. Females = Est. Males									
Δ Returns (p-val)	0.686	0.052	0.000	0.000	0.000	0.018	0.000	0.102	0.000
Δ Match Effects (p-val)	0.410	0.151	0.629	0.576	0.528	0.176	0.520	0.109	0.021

Notes: This table presents estimates from our main regression model in Equation (9), using an indicator for having any child 13 years after college application as the outcome. We display the averages—of coefficients, robust standard errors, and p-values—across 100 regressions, each of which uses a different set of control functions obtained after drawing from the asymptotic distribution of the demand model estimates. The sample includes 137,339 observations of students who applied and were accepted into a program through the centralized admission system between 2004 and 2007. Panel (a) focuses on parameter estimates for Female students and Panel (b) focuses on Male students. In each panel, we report return estimates and selection on gains parameters, also referred to as match effects in the text. Underneath each estimate in parentheses is the bootstrapped standard error. The final two rows of the table report p-values from two additional statistical tests. The Δ Returns p-values correspond to the null hypothesis that the Female and Male returns are equivalent for a given column. The Δ Match Effects p-values correspond to the null hypothesis that the Female and Male match effects are equivalent for a given column. In addition to main effects for field of study and control functions and gender interactions, all specifications include quadratic polynomials of the college admission scores in mathematics and language, a distance vector, cohort fixed effects and cell fixed effects, where a cell is defined by school type, macro-region, financial-aid relevant score ranges, and gender. For ease of interpretation, $\hat{\lambda}_{ij}$ is standardized within each cell. The R^2 of this regression is 0.06

preference intensity. This pattern points toward features shared by graduates who follow a field’s educational and career path. A growing literature emphasizes the interdependence of fertility decisions and economic outcomes (Blundell et al., 2018; Jakobsen et al., 2025), including how child-related interruptions in work affect women’s human capital accumulation and wage growth (Sauer and Taber, 2021) and the role of career and occupational considerations in fertility decisions, which includes both the timing of childbearing and the tradeoffs between career paths and family formation (Adda et al., 2017; Doepke et al., 2023). Building on this literature, we hypothesize that fields of study affect fertility through the types of careers they enable, particularly through differences in how easily those careers allow individuals to balance work and family life (Goldin, 2021).

To assess this mechanism, we use data from Chile’s largest household survey to characterize the job attributes typically faced by graduates of each major and we complement these survey-based measures with field-level proxies for gender norms and child penalties (Bertrand et al., 2015; Kleven et al., 2019a), constructed from administrative data (see Appendix D for details on how we estimate child penalties and gender norms). We then examine whether these measures of career-family compatibility are systematically related to the estimated fertility effects across fields.

Figure 4: Male–Female Fertility Effects Association



Notes: This figure compares male and female estimates of the effects of fields of study on the probability of having any child 13 years after college application. Female estimates are plotted on the horizontal axis and male estimates on the vertical axis. Maroon circles report the nine field-specific effects from Table 3, measured relative to the corresponding gender-specific fertility outcome among non-graduates. The remaining points report fallback-specific relative payoffs. For each gender and ordered pair of first-ranked field j and second-ranked field k , these estimates compare the predicted payoff to j with the predicted payoff to k , evaluated at the mean control functions among students who submit that pair. Other fallback-specific estimates are shown as gray circles. Purple diamonds denote pairs with Humanities as the first-ranked field; brown squares, orange upward triangles, and green downward triangles denote pairs with Business, Engineering, and Medicine, respectively, as the fallback field. The black dashed line is the 45-degree line. The maroon and gray dashed lines are separate unweighted regressions of the male estimates on the female estimates for the main and fallback-specific effects, respectively.

Figure 5 presents the point estimates and confidence intervals obtained from separate regressions of fertility payoffs on jobs and field-specific attributes.²² We find no significant association between the average wages of graduates in each field of study and our estimated fertility payoffs. However, we observe a positive correlation between fertility impacts and factors that facilitate balancing career and family. Majors with greater job security, proxied by the percentage of graduates who hold a job in the public sector, are associated with higher fertility. Fertility payoffs are also positively correlated with the progressiveness of gender norms in each field of study—measured by the size

²²We use the estimated fertility payoffs by field of study and gender as the dependent variable. All independent variables are standardized to have mean zero and standard deviation one, and all regressions include gender fixed effects.

of the cliffs at the point where the mother starts to earn more than the father.²³ The percentage of graduates who are married or cohabitating, a key feature of the “child production function” (Albanesi et al., 2023), is also positively correlated with fertility. Conversely, fertility impacts are more negative in fields where unemployment is high and where the career costs of children are higher, as proxied by our estimated child penalties. A regression of fertility payoffs on a career-and-family compatibility index (calculated as the average of the standardized variables mentioned above, adjusted such that higher values indicate greater compatibility) yields a coefficient of 0.080 with a standard error of 0.019. Including logged wages as a control variable in this regression—in the spirit of Oreopoulos and Salvanes (2011)—leaves the coefficient unchanged and only slightly increases the standard error to 0.020. This result aligns with the fact that the income-fertility relationship has flattened over time and with modern fertility models that shift the focus from opportunity costs to the influence that policies, social norms, and workplace flexibility have in shaping fertility decisions (Dahl and Loken, 2024; Doepke et al., 2023).

These results suggest that the heterogeneous fertility payoffs documented above are systematically related to differences in career-family compatibility rather than average earnings. Our results also resonate with Goldin and Katz (2008) in that “It is, perhaps, not surprising that women who pursue different career paths and have earned degrees in different fields have different numbers of children. It appears that women in careers with the greatest predictability and the smallest financial penalty for time out have the most children.” Of course, there are other ways a college major can influence fertility, such as its impact on the marriage market (Angrist, 2002; Kirkeboen et al., 2025).

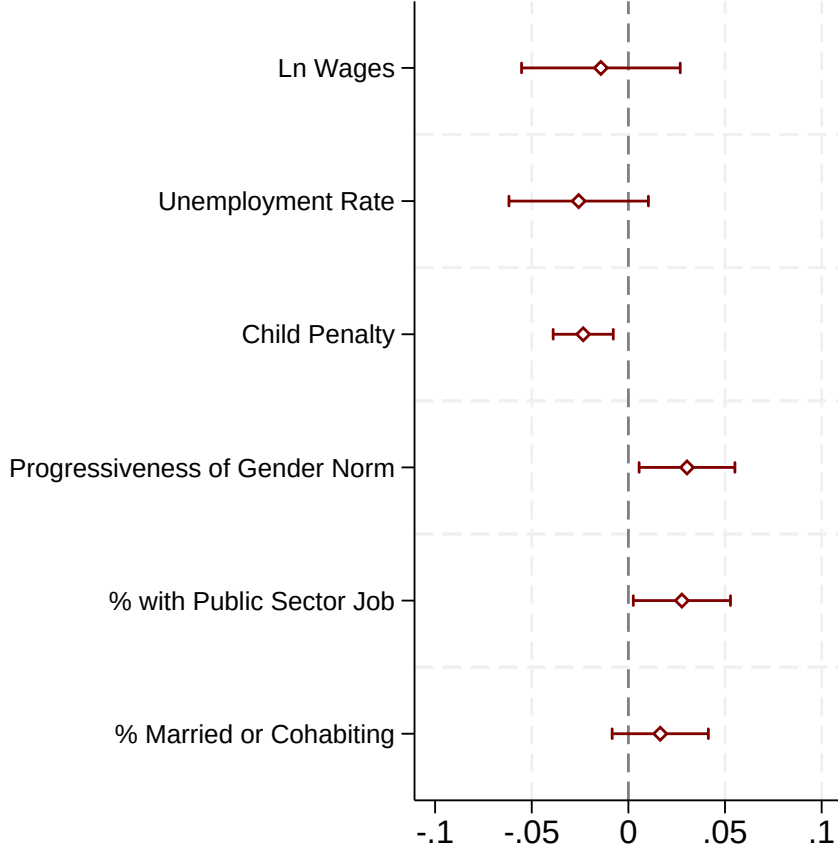
While other channels may also be important, data limitations prevent us from directly examining them. We therefore do not take a stand on which mechanism is most prominent. Instead, we interpret the heterogeneous fertility effects as summary measures of the family-formation payoffs associated with each field. Whether students anticipate these gender-specific payoffs—and whether doing so helps explain gendered sorting across majors and, ultimately, the gender earnings gap—is therefore an important empirical question. Before turning to this question, however, we first validate the underlying payoff estimates against quasi-experimental variation in assignment to college majors.

4.4 Validation Exercise

Our empirical strategy—which allows us to distinguish between vertical and horizontal returns—relies on accounting for preference heterogeneity captured by the partial ordering observed in rank-ordered lists. Alternative approaches leverage capacity constraints to identify the numerous quasi-experiments embedded within centralized assignment mechanisms (Abdulkadiroğlu et al., 2017, 2022). The richness of our data and the institutional setting allow us to validate our approach

²³This finding may be interpreted as highlighting the importance of a more equitable distribution of child care responsibilities between mothers and fathers (Doepke and Kindermann, 2019).

Figure 5: Fertility Correlations



Notes: This figure reports the point estimates and confidence intervals obtained from separate regressions of fertility payoffs by field of study and gender on job and field-specific attributes. All regressions include gender fixed effects, and all independent variables are standardized to have mean zero and standard deviation one. Confidence intervals are constructed using robust standard errors. For details on the construction of Child Penalties and Progressiveness of Gender Norm, see Appendix D.

by directly comparing its predicted payoffs with causal estimates obtained from these local quasi-experiments.

The key ingredients of the quasi-experimental approach are applicant types, θ_i , the vector of program cutoffs, τ , and a bandwidth, δ , defining neighborhoods around the relevant admission cutoffs. Abdulkadiroğlu et al. (2022) show that the many quasi-experiments embedded in centralized assignment systems can be summarized by local propensity scores that capture the probability of assignment induced jointly by applicants' preferences, program capacities, and their position relative to the relevant cutoffs. Let $p_{ij}(\theta_i, \tau, \delta)$ denote applicant i 's local propensity score for an

offer in field j , obtained by aggregating the corresponding program-level assignment probabilities.²⁴ With non-random tie-breakers, identification is local to the program-specific admission cutoffs and additionally requires flexible controls for the corresponding running variables. Let r_i collect these program-specific running variables and let $g_s(\theta_i, \tau, r_i; \delta)$, for $s = 1, 2$, denote the associated local polynomial controls.

Because applicants do not apply to every program, let a_{im} indicate whether applicant i applied to program m , and let a_i collect these application indicators. Let $p_i = (p_{i1}, \dots, p_{iJ})$ denote the vector of field-specific local propensity scores. For expositional purposes, we summarize the controls entering the first- and second-stage equations by

$$f_s(a_i, p_i, g_s) = \sum_m \gamma_{ms}^a a_{im} + \sum_j \gamma_{js}^p p_{ij}(\theta_i, \tau, \delta) + g_s(\theta_i, \tau, r_i; \delta), \quad s = 1, 2.$$

Conditional on the local propensity scores and the corresponding running-variable controls, variation in program offers near the admission cutoffs is locally random (Abdulkadiroğlu et al., 2022). Aggregating these program-level offers to fields of study therefore provides an independent source of quasi-experimental variation with which to validate our estimated payoffs.

We use this locally random variation to test the predictive validity of the payoff estimates generated by our preferred model, following Angrist et al. (2017). As shown above, our model estimates the earnings and fertility payoffs associated with completing each field, while the centralized assignment mechanism provides independent quasi-experimental variation in offers across fields. If these estimated payoffs correctly capture causal effects, offer-induced variation in model-predicted outcomes should translate one-for-one into variation in realized outcomes.

Formally, let Y_i denote the observed outcome for individual i and let $\hat{Y}_i$ denote the outcome predicted by our model. We estimate the following system by two-stage least squares:

$$\begin{aligned} \hat{Y}_i &= \sum_{j \in \mathcal{J}} \pi_j Z_{ij} + f_1(a_i, p_i, g_1) + e_{1i} \\ Y_i &= \phi \hat{Y}_i + f_2(a_i, p_i, g_2) + e_{2i}, \end{aligned} \tag{10}$$

where Z_{ij} indicates an offer in field j . The first stage isolates variation in the model's predicted outcomes generated by locally random offers, while the second stage asks whether this variation translates one-for-one into realized outcomes. If the model's predictions align, on average, with the causal effects implied by the quasi-experimental variation, then $\hat{\phi} = 1$ and the model is forecast unbiased (Angrist et al., 2017). Because offers to multiple fields provide more instruments than are required to identify the scalar forecast coefficient ϕ , the model is overidentified. This allows us to additionally test whether the model's predictions are jointly consistent with the separate quasi-experimental contrasts generated by offers to different fields.

²⁴Because the assignment mechanism generates at most one offer per applicant, propensity scores can be aggregated across programs belonging to the same field.

We conduct the validation exercise for both fertility and earnings models separately. Table 4 reports test results for three different models of interest. Column 1 reports test results for an uncontrolled model; Column 2 augments the model with quadratic polynomials in college admission scores, cell fixed effects, and cohort fixed effects; Column 3 adds the control functions. Column 1 is a useful starting point to demonstrate the prevalence of bias in models that do not adjust for observable differences that correlate with major choice. The second model contains observables available in a host of papers and serves as a useful reference point to gauge the potential bias present in those papers (Altonji et al., 2012, 2016; Andrews et al., 2024). Column 3 is our preferred model corresponding to the specification used to obtain the estimates in Table 2 and Table 3.

Panel (a) shows that uncontrolled earnings models exhibit substantial bias, with a forecast coefficient of 0.72. Models that include cell and cohort fixed effects and polynomials of PSU scores, analogous to SAT scores in other papers, show an improvement with a forecast coefficient of 0.86. The forecast coefficient in Column 2 is marginally significantly different from 1. Our preferred model produces a forecast coefficient of 0.98, underscoring the importance of accounting for preference heterogeneity to improve the average predictive validity of earnings impacts. This evidence bodes well for other approaches and settings that use preference data to account for selection bias (Dale and Krueger, 2002; Mountjoy and Hickman, 2021) but may ultimately depend on the presence of a centralized assignment system.

Panel (b) reports the corresponding validation results for fertility effects. As with earnings, the uncontrolled model exhibits substantial forecast bias, and adding rich controls attenuates this bias, although to a lesser extent than in the earnings model. Adding control functions that characterize selection into majors produces a forecast coefficient of 0.95, and we cannot reject that it equals one. We also fail to reject the overidentifying restrictions for our preferred specification at the 5 percent level in either panel, providing no strong evidence that the model’s predictions are inconsistent with the separate quasi-experimental contrasts across fields.

Taken together, these validation results support the field-specific earnings and fertility effects previously reported: estimates from our preferred model closely align with quasi-experimental benchmarks, while allowing us to characterize match effects and gender differences in payoffs. With this validation in hand, we now return to whether these ex-post payoffs are reflected in students’ major choices.

4.5 Implications for Sorting and the Earnings Gap

Limited preference-linked heterogeneity in fertility effects does not imply that fertility is irrelevant for choice: applicants may respond to payoffs common within a gender-field pair even when those payoffs vary little with individual preference intensity. We therefore ask whether the realized (ex-post) earnings and fertility payoffs estimated with our preferred model are reflected in individuals’ major choice decisions. Wiswall and Zafar (2021) showed that students’ ex-ante beliefs about both fertility and earnings outcomes affect their choices. Whereas they elicit subjective expectations when decisions are still pending, we examine whether the objective, causal payoffs that graduates

Table 4: Testing for Bias

	Panel (a): Earnings		
	Model 1	Model 2	Model 3
	(1)	(2)	(3)
Forecast Coefficient ϕ	0.717 (0.066)	0.856 (0.078)	0.984 (0.091)
Covariates in the model:			
Controls	No	Yes	Yes
Control function	No	No	Yes
Tests:			
First-stage F-statistic:	251	213	172
p-values			
Forecast bias ($\phi = 1$)	0.000	0.065	0.861
Overidentification	0.068	0.212	0.087
Observations	44,684	44,684	44,684
	Panel (b): Fertility		
	Model 1	Model 2	Model 3
	(1)	(2)	(3)
Forecast Coefficient ϕ	0.555 (0.109)	0.768 (0.144)	0.946 (0.176)
Covariates in the model:			
Controls	No	Yes	Yes
Control function	No	No	Yes
Tests:			
First-stage F-statistic	471	320	239
p-values			
Forecast Bias ($\phi = 1$)	0.000	0.107	0.758
Overidentification	0.047	0.110	0.116
Observations	59,859	59,859	59,859

Notes: This table reports the results of tests for bias following Equation (10). Column 1 reports test results for an uncontrolled model; Column 2 augments the model with quadratic polynomials in PSU scores, cell fixed effects, and cohort fixed effects; Column 3 adds the control functions. The forecast bias test checks whether the forecast coefficient equals 1, and the overidentification test checks the IV model's overidentifying restrictions. Panel (a) considers logged earnings in 2019 as the outcome, and Panel (b) considers fertility (any child 13 years after college application) as the outcome.

ultimately experience are reflected in aggregate choice patterns. In particular, we relate the cell-specific mean utilities of each field of study to the cell-specific payoffs of each field. Building on our choice model, we estimate:

$$\hat{\delta}_{jc} = \delta_c + \rho^E \bar{Y}_{jc}^E + \rho^F \bar{Y}_{jc}^F + \beta X_j + e_{jc}, \quad (11)$$

where the parameters of interest are ρ^E and ρ^F , which gauge the relative importance of earnings and fertility payoffs in shaping major choice decisions. To measure $\bar{Y}_{jc}^E$ and $\bar{Y}_{jc}^F$ we consider the causal impact of graduation from major j on earnings and fertility, as captured by our main model (9) estimated at the cell level c . The vector of controls X_j includes the average cost of the programs within each field of study, as well as field-specific attributes related to career and family previously used. The inclusion of cell effects, δ_c , ensures all comparisons are within-cell.

This specification assesses whether students’ revealed choices place implicit weight on earnings and fertility—factors that strongly influence the career-family trade-off. This trade-off emerges from the different sets of job opportunities available to students upon graduation, which vary by field of study. Students are assumed to anticipate these opportunities, with heterogeneous preferences generating gender differences in sorting across college majors. In this regard, our approach aligns closely with Adda et al. (2017), who examine the interplay between fertility decisions, labor supply, and occupational choice. However, our framework focuses on an earlier stage—the choice of college major—which subsequently shapes the occupational decisions available to graduates.

Table 5 reports estimates of Equation (11). In Panel (a), earnings and fertility payoffs each predict mean utility when entered separately and remain predictive when entered jointly. In Column (3), a one-standard-deviation increase in the earnings payoff is associated with a 0.23-standard-deviation increase in mean utility, while the corresponding fertility coefficient is 0.16. In the spirit of Oreopoulos and Salvanes (2011), Column (4) controls for field-specific attributes that may mediate the relationship between major choice and fertility payoffs. These controls include the unemployment rate, the share of graduates employed in the public sector, the share of graduates who are married or cohabiting, and our estimates of child penalties and breadwinner gender norms by field of study. Because earnings risk may also shape educational choices (Andrews et al., 2024; Kirkebøen, 2012), we also include the variance of wages within each field. Adding these controls leaves the earnings coefficient similar and attenuates the fertility coefficient, consistent with fertility payoffs partly capturing differences in work-family compatibility across fields.

Panel (b) allows the payoff loadings to differ by gender. In Column (3), the earnings coefficient is 0.36 for men and 0.14 for women, whereas the fertility coefficient is 0.03 for men and 0.27 for women. This pattern is consistent with men’s choices being more strongly associated with monetary payoffs and women’s choices with family-formation payoffs (Goldin, 2021). With the career-family controls in Column (4), the corresponding coefficients are 0.34 and 0.19 for earnings and 0.03 and 0.13 for fertility.

These estimates point to two distinct sources of gendered sorting: gender differences in the weights placed on earnings and fertility payoffs, and gender differences in the payoff schedules

Table 5: Preferences for Pecuniary and Non-Pecuniary Returns

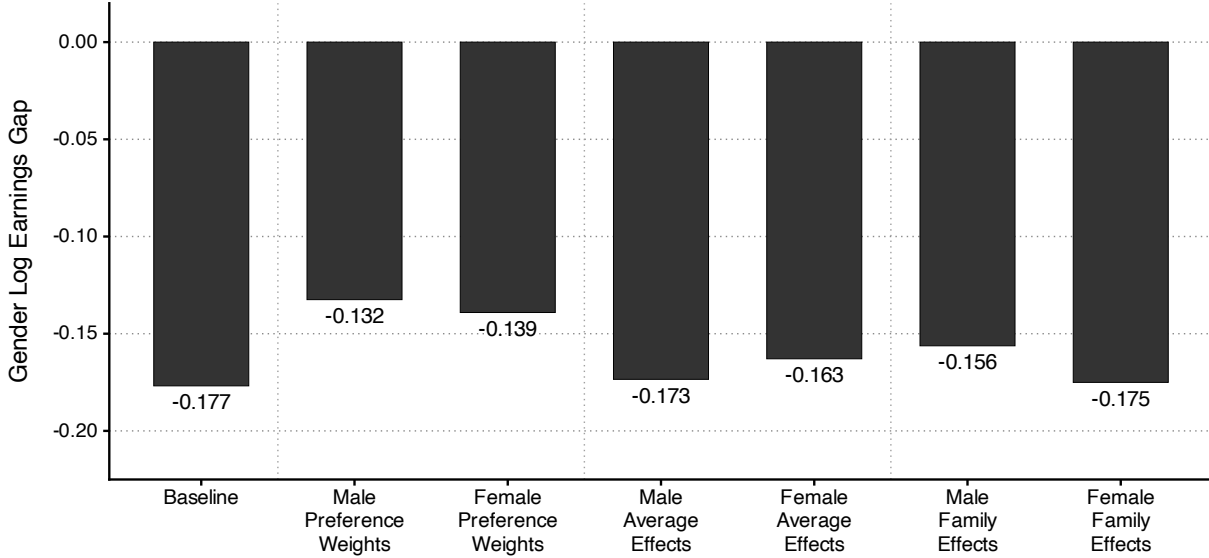
	Mean Utility			
	(1)	(2)	(3)	(4)
Panel (a): Pooled Estimates				
Earnings	0.240 (0.055)		0.232 (0.055)	0.260 (0.046)
Fertility		0.166 (0.054)	0.157 (0.053)	0.088 (0.043)
Panel (b): Estimates by Gender				
Earnings	0.359 (0.080)		0.355 (0.079)	0.344 (0.070)
× Female	-0.234 (0.115)		-0.217 (0.111)	-0.152 (0.094)
Fertility		0.069 (0.076)	0.027 (0.073)	0.035 (0.059)
× Female		0.192 (0.107)	0.240 (0.103)	0.095 (0.081)
R-squared:				
Panel (a)	0.22	0.20	0.24	0.53
Panel (b)	0.23	0.21	0.27	0.54
Career-Family Controls	No	No	No	Yes
Observations	324	324	324	324

Notes: This table reports regressions of standardized cell-by-field mean utilities on standardized earnings and fertility effects. The sample contains nine fields in each of 36 cells defined by macro-region, test-score range, high-school type, and gender. All specifications absorb cell fixed effects and control for average field cost; Panel (b) allows the payoff and cost coefficients to differ by gender. Column (4) additionally includes field-level unemployment, public-sector employment, marriage or cohabitation, wage dispersion, child penalties, and breadwinner gender norms. Robust standard errors are in parentheses. When a cell-by-field effect cannot be estimated, we impute the corresponding within-cell mean before standardization.

themselves. To gauge their relative importance for sorting and the earnings gap, we conduct three counterfactual exercises. We separately equalize (i) the preference weights on both average earnings and family-formation effects, (ii) the average-earnings effect schedule, and (iii) the family-formation effect schedule. We then recalculate demand under these changes and project downstream consequences on earnings gaps based on counterfactual choices.

Figure 6 shows that the preference-weight exercise has the largest and most consistent effect. Assigning men’s earnings and family weights to women reduces the absolute log earnings gap

Figure 6: Counterfactual Earnings Gaps: Preferences and Payoff Schedules



Notes: This figure reports the female-minus-male log earnings gap at the observed baseline and under six one-at-a-time counterfactuals. Preference Weights jointly equalizes the coefficients on average earnings and family-formation effects in Equation (11); Average Effects equalizes the gender-specific cell-by-field average-earnings effect schedule; and Family Effects equalizes the corresponding family-formation effect schedule. “Male” assigns the men’s weights or schedule to women, leaving men unchanged; “Female” does the reverse. Counterfactual shares are obtained from calibrated cell-specific mean utilities, so the common baseline reproduces observed first-ranked application shares. The earnings gap is valued using fixed observed gender-by-field mean log earnings. The demand regressions use all 36 cells and exclude the career-family controls.

by 25.1 percent; assigning women’s weights to men reduces the two measures by 21.3 and 18.7 percent, respectively. Equalizing average-earnings effect schedules reduces the absolute earnings gap by 1.9 percent under the male returns and 7.9 percent under the female returns. Equalizing family-formation effect schedules reduces the gap by 11.7 and 1.0 percent and sorting by 5.4 and 2.0 percent.

Taken together, these exercises highlight that gender differences in how earnings and family-formation payoffs enter major choice are consequential for gendered sorting and, in turn, the earnings gap. At the same time, these differences may be difficult to shift directly, particularly if they reflect deeply rooted social norms (e.g., Alesina et al., 2013). This motivates considering policy interventions that take observed preferences as given and instead modify the set of college opportunities available to students.

5 Supply-Side Policies for Gender Gaps in College Majors

In this final section, we assess the potential effects of supply-side policies that seek to reduce gender differences in representation across fields of study while taking students’ preferences as given. Because fields differ in both earnings and fertility payoffs, reallocative policies may affect not only

gender gaps in labor market outcomes but also aggregate family formation. These counterfactuals therefore allow us to investigate whether policies that close representation gaps and narrow gender earnings gaps can be pursued without meaningful economic or demographic sacrifice. We motivate our analysis with a descriptive decomposition of the broader gender earnings gap, demonstrating that differences in representation in Engineering explain a sizable share. We then use our gender-specific fertility and earnings payoffs by field of study and the structure of college admissions to evaluate the impact of different policies on female representation, the gender earnings gap, aggregate earnings, and aggregate fertility.

As a proof of concept of what can be done in systems with centralized assignment, we assess two policy scenarios: expanding capacity in all Engineering programs by 10%, 30%, or 50%, and reserving 10%, 30%, or 50% of Engineering seats for female students while keeping total program capacities fixed. Capacity expansions are gender-neutral policies, with their distributional impacts shaped by preferences within the applicant pool. Quotas, in contrast, explicitly target female applicants and may displace male applicants who would otherwise receive those seats.²⁵ The emphasis on Engineering as opposed to an omnibus STEM oriented policy is motivated by a decomposition of the gender earnings gap that we turn to next.

Motivating Decomposition We motivate and inform our counterfactual analysis with a descriptive decomposition of the gender earnings gap. However, a conventional decomposition into representation and within-major differences combines different causal and selection components and is generally uninformative about how reallocative policies would affect outcomes. Our framework instead allows for a decomposition that separates gender differences in average field-specific returns, preference-based match effects, and selection; components that need not disappear when students are reallocated across fields. This decomposition therefore clarifies which parts of the observed gap can potentially be altered through reallocation and which may persist under alternative assignments.

Let $\bar{y}_j^g$ stand for the average earnings of individuals of gender $g \in \{F, M\}$ who graduate from major j , and let s_j^g stand for the share of individuals who graduate from field j among those of gender g . Using this notation, we can write the average gender earnings gap among college graduates as:

$$\mathbb{E}[\text{Earnings} \mid \text{Female}] - \mathbb{E}[\text{Earnings} \mid \text{Male}] = \sum_j s_j^F \bar{y}_j^F - \sum_j s_j^M \bar{y}_j^M.$$

²⁵In the most recent admissions processes, exclusive seats were reserved for women in STEM-related fields, aiming to reduce the gender gap in Chile. The country ranks fourth lowest among OECD nations in the proportion of women graduating in STEM (MINEDUC, 2023; La Tercera, 2023). Bursztyn et al. (2023) find that 74% of Chilean citizens support affirmative action policies.

This expression has two algebraically equivalent decompositions:

$$\begin{aligned}
\sum_j s_j^F \bar{y}_j^F - \sum_j s_j^M \bar{y}_j^M &= \underbrace{\sum_j (s_j^F - s_j^M) \bar{y}_j^F}_{\Delta \text{Representation}} + \underbrace{\sum_j s_j^M (\bar{y}_j^F - \bar{y}_j^M)}_{\Delta \text{Within-major}} \\
&= \underbrace{\sum_j (s_j^F - s_j^M) \bar{y}_j^M}_{\Delta \text{Representation}} + \underbrace{\sum_j s_j^F (\bar{y}_j^F - \bar{y}_j^M)}_{\Delta \text{Within-major}}.
\end{aligned} \tag{12}$$

Both lines of Equation (12) separate differences in sorting patterns from within-major earnings differences. The first line evaluates representation differences using female field-specific earnings and weights within-major gaps using male shares. The second line evaluates representation differences using male major-specific earnings and weights within-major gaps using female shares. The gender- and major-specific mean earnings $\bar{y}_j^g$ combine causal and selection components. We can therefore further decompose the within-major contribution to the overall gap.

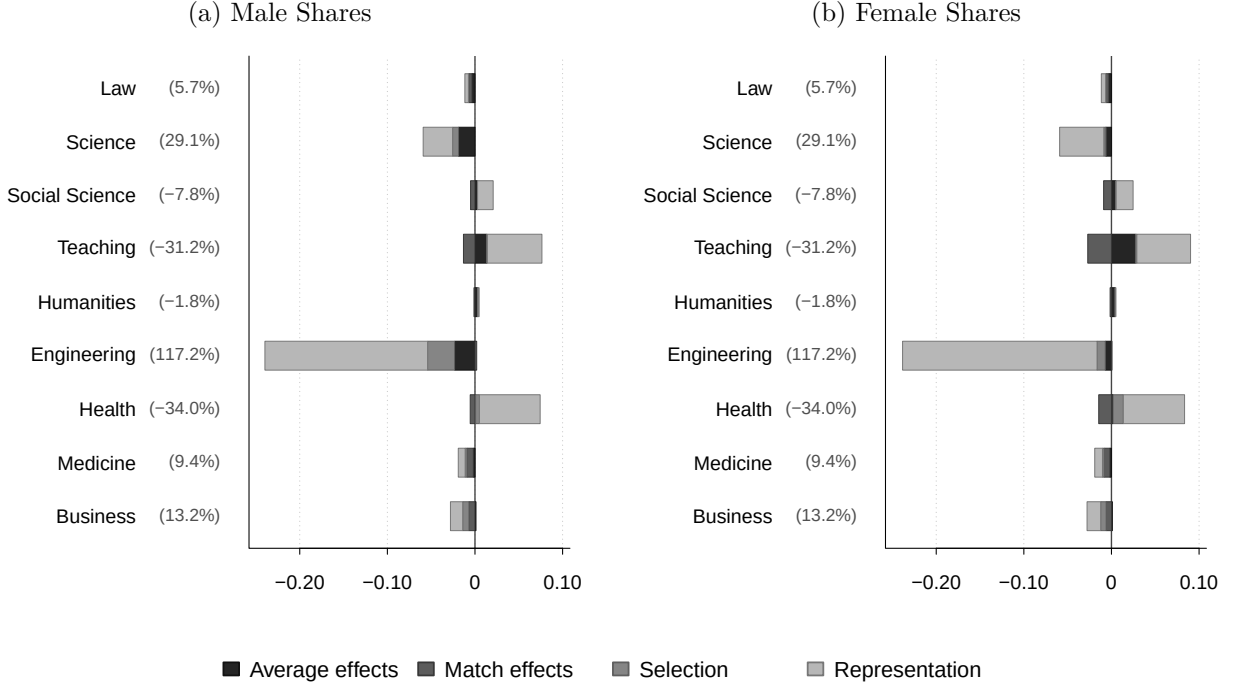
Define $\Delta_j = \bar{y}_j^F - \bar{y}_j^M$. The empirical specification in Equation (9) implies that the within-field gender gap can be decomposed into differences in average causal effects, match effects, and remaining selection components. For compactness, define $\psi_j^{*M} \equiv \psi_j^*$ and $\psi_j^{*F} \equiv \psi_j^* + \psi_j^{*G}$. Then,

$$\Delta_j = \underbrace{\alpha_j^G}_{\text{Average Effect Diff}} + \underbrace{(\psi_j^{*F} \bar{\lambda}_{Fj} - \psi_j^{*M} \bar{\lambda}_{Mj})}_{\text{Match Effect Diff}} + \underbrace{(\bar{S}_{Fj} - \bar{S}_{Mj})}_{\text{Selection Diff}}, \tag{13}$$

where $\bar{\lambda}_{gj}$ denotes the average control function for field j among gender- g graduates of that field, and $\bar{S}_{gj}$ collects the remaining components of predicted earnings associated with observed characteristics and selection on levels. Equation (13) therefore separates gender differences in the causal payoff to graduating from field j from differences in realized match effects and other selection components.

Figure 7 reports the decomposition in Equation (12), which builds on the decomposition in Equation (13). Panel (a) reports decomposition results which use the male shares to value the within-major differences, while Panel (b) uses female shares. The two weighting schemes yield qualitatively similar conclusions. Although the share of the overall gender earnings gap attributed to the within-major component varies somewhat with the choice of weights—representation accounts for 46.5 percent of the signed gap when valued at female earnings and weighted by male shares, versus 74.2 percent when valued at male earnings and weighted by female shares—the prominent role of representation is evident in both panels. Engineering is especially important, accounting for 117 percent of the overall gender earnings gap. By contrast, Teaching and Health partially offset the aggregate gender earnings gap, reflecting their relatively high female representation and earnings patterns that are more favorable to women. The causal contribution to within-major earnings differences varies across fields but is generally modest relative to representation-based differences. Taken together, the decomposition points to gender differences in representation—particularly the underrepresentation of women in Engineering—as the largest contributor to the aggregate gender earnings gap. We therefore turn next to policies that can directly affect this margin.

Figure 7: Earnings Gap: Representation versus Within-Major Differences



Notes: This figure reports the two algebraically equivalent decompositions in Equation (12) for the female-minus-male gap in mean log earnings. Panel (a) values representation differences using female field-specific mean log earnings and weights within-major differences using male major shares. Panel (b) values representation differences using male field-specific mean log earnings and weights within-major differences using female major shares. Following Equation (13), the within-major contribution is separated into gender differences in average effects, match effects, and selection. Representation contributions use the female non-graduate mean log earnings as a common anchor in both panels. Bars are signed field-level contributions to the observed gap. The percentage shown beside each major is its signed total contribution divided by the signed observed gap. These percentages are identical across panels because the two decompositions are algebraically equivalent within each major; the different reference shares change the composition of each bar, but not its total contribution. The contributions across majors sum to the observed gap of -0.203 log points. We consider the 2019 earnings of students who applied through the centralized admission system between 2004 and 2007, graduated between 2007 and 2019, and were employed in 2019.

5.1 Counterfactual Framework

We begin with the population of college applicants $\mathcal{I}^{all} \supseteq \mathcal{I}$ and simulate student assignments under alternative capacity and quota policies.²⁶ The inputs of the counterfactual analysis are a vector of student types $\theta = (\theta_i)_{i \in \mathcal{I}^{all}}$,²⁷ programs' capacities $\mathbf{q}$, reserves ω , and a matching algorithm, i.e., a function that takes as inputs the previous elements and outputs a vector of student assignments $\mathcal{A}$. Changes in the policies we consider correspond to a change in $\mathbf{q}$ for gender-neutral seat expansions

²⁶We use the population of accepted applicants $\mathcal{I}$ to estimate the outcome model. Equipped with outcome model estimates, we can calculate treatment effects for the entire pool of applicants.

²⁷Each applicant's type is $\theta_i = (R_i, S_i)$, where R_i is the submitted rank-ordered list and S_i is the vector of program-specific admission scores.

or to a change in ω for gender quotas, both of which produce a change in the allocation of students $\mathcal{A}$.²⁸

The assignment mechanism operates at the program level, whereas our payoff estimates are defined at the field level. To harmonize this, we let $m \in \mathcal{M}$ index programs, where a program is a major-institution pair, and let $h(m) \in \mathcal{J}$ denote the broad field associated with program m . We denote the unassigned outside option by U . Thus, an assignment rule produces a program-level assignment $\mathcal{A}_i \in \mathcal{M} \cup \{U\}$, while expected outcomes are assigned using the induced field or outside-option state. For compactness, write $j_i(\mathcal{A}) \equiv h(\mathcal{A}_i)$ when $\mathcal{A}_i \in \mathcal{M}$, and $j_i(\mathcal{A}) = U$ when $\mathcal{A}_i = U$.

For each outcome $r \in \{e, f\}$, where e denotes earnings and f denotes fertility, let $\mathcal{C}_i \equiv (R_i, S_i, X_i, G_i, d_i)$ denote the applicant-level information set entering the outcome model, and let $\pi_{ij}^{\text{grad}} \equiv (\pi_{ijk}^{\text{grad}})_{k \in \mathcal{J}_0}$ denote the vector of assignment-specific graduation probabilities for majors k , where $\pi_{ijk}^{\text{grad}} \equiv P(F_i(j) = k \mid \mathcal{C}_i)$. We define the relevant conditional outcome components as:

$$\begin{aligned} m_{ij,1}^r &\equiv E[Y_{ij}^r \mid \mathcal{C}_i], \quad j \in \mathcal{J}, \\ m_{i0}^r &\equiv E[Y_{i0}^r \mid \mathcal{C}_i], \\ m_{iU}^r &\equiv E[Y_{iU}^r \mid \mathcal{C}_i], \end{aligned}$$

Here, $m_{ij,1}^r$ is the conditional mean outcome if applicant i is assigned to j and graduates from j , whereas m_{i0}^r is the conditional mean outcome if applicant i is assigned to a field but does not graduate. The object m_{iU}^r is the corresponding conditional mean outcome in the unassigned outside option. Importantly, the model we use to predict m_{iU}^r uses the same conditioning variables as the other conditional mean models we consider for the pool of accepted applicants.

Equipped with the baseline causal model for assigned students and an auxiliary model for earnings among the unassigned,²⁹ we combine these conditional mean outcomes with assignment-specific graduation probabilities. We define applicant i 's graduation-risk-adjusted conditional expected outcome under assignment state j , for outcome $r \in \{e, f\}$, as follows:

$$\bar{Y}_{ij}^r = \begin{cases} \sum_{k \in \mathcal{J}_0} \pi_{ijk} m_{ik}^r, & j \in \mathcal{J}, \\ m_{iU}^r, & j = U. \end{cases} \quad (14)$$

The implied expected outcome $\bar{Y}_{ij}^r$ combines the relevant conditional mean outcomes under graduation and non-graduation using the graduation probability associated with assignment state j . Note that Equation (14) requires offers (or assignments) to be independent of potential completion paths, $\{F_i(j)\} \perp\!\!\!\perp Z_i \mid \theta_i$, which is natural given the centralized assignment environment.

²⁸To not over-complicate notation, we suppress the implicit dependence of $\mathcal{A}$ on $\theta, \mathbf{q}$, and ω .

²⁹We elaborate on the details of the auxiliary model in the next section.

For a given allocation $\mathcal{A}$, applicant i 's allocation-selected expected outcome is:

$$\bar{Y}_i^r(\mathcal{A}) \equiv \bar{Y}_{i,j_i(\mathcal{A})}^r.$$

We define the individual counterfactual effect of changing the allocation from $\mathcal{A}$ to $\mathcal{A}'$ as:

$$\Xi_i^r(\mathcal{A}, \mathcal{A}') \equiv \bar{Y}_i^r(\mathcal{A}') - \bar{Y}_i^r(\mathcal{A}).$$

This general definition applies to all transitions, including those into and out of the unassigned outside option U . For an assigned field $j \in \mathcal{J}$, define the conditional mean graduation payoff as

$$ATE_j^r(\mathcal{C}_i) \equiv m_{ij,1}^r - m_{i0}^r$$

where $ATE_j^r(\mathcal{C}_i)$ is the outcome r -specific average effect in Equation (7) in Section 3. Equation (14) can then be written as $\bar{Y}_{ij}^r = m_{i0}^r + \sum_{k \in \mathcal{J}} \pi_{ijk} ATE_k^r(\mathcal{C}_i)$ for assigned students. For applicants assigned to fields under both $\mathcal{A}$ and $\mathcal{A}'$, if we define $p_{ij}^{\text{grad}} \equiv \sum_{k \in \mathcal{J}} \pi_{ijk} = 1 - \pi_{ij0}$, then we can arrive at the following decomposition:

$$\begin{aligned} \Xi_i^r(\mathcal{A}, \mathcal{A}') &= \underbrace{\left(p_{i,j_i(\mathcal{A}')}^{\text{grad}} - p_{i,j_i(\mathcal{A})}^{\text{grad}} \right)}_{\Delta \text{ Graduation Prob.}} ATE_{j_i(\mathcal{A}')}^r(\mathcal{C}_i) \\ &\quad + p_{i,j_i(\mathcal{A})}^{\text{grad}} \underbrace{ATE_{j_i(\mathcal{A}')}^r(\mathcal{C}_i)}_{\Delta \text{ Payoffs}} \\ &\quad + \underbrace{\sum_{k \in \mathcal{J} \setminus \{j_i(\mathcal{A}')\}} \pi_{i,j_i(\mathcal{A}'),k} ATE_{k,j_i(\mathcal{A}')}^r(\mathcal{C}_i) - \sum_{k \in \mathcal{J} \setminus \{j_i(\mathcal{A})\}} \pi_{i,j_i(\mathcal{A}),k} ATE_{k,j_i(\mathcal{A})}^r(\mathcal{C}_i)}_{\text{Adjustment for completion in other fields}}. \end{aligned} \tag{15}$$

where the first term values the change in graduation probability at the focal fields, while the second term captures the difference in graduation payoffs where $ATE_{jk}^r(\mathcal{C}_i)$ is defined in Equation (8) in Section 3. These first two terms treat completion as occurring in the assigned field. The last term aggregates potential non-compliance effects. For assigned students in each scenario, this decomposition sheds light on the various mechanisms at play with reassignment policies. For example, if reassignment lowers the probability of graduation—due, for example, to mismatch effects (Arcidiacono and Lovenheim, 2016; Arcidiacono et al., 2016)—the first term for earnings is negative whenever the destination-field earnings payoff is positive.³⁰ If a student is reallocated

³⁰The empirical evidence on mismatch effects is mixed. In the University of California system, Arcidiacono et al. (2016) find that less prepared minority students at top-ranked campuses would have higher graduation rates had they attended lower-ranked campuses, while Bleemer (2023) finds that minority students experienced worse outcomes as they transitioned into lower tier universities following the ban of California's Proposition 209. Also, in the US, Black et al. (2023) find that introducing the Texas Top Ten Percent rule improved enrollment and graduation of highly ranked students at more disadvantaged high schools. In Latin America, Barahona et al. (2026) do not find strong

from a major where their returns are higher to one where they are lower, then the second term may also be negative. All three reported mechanisms are at play in our counterfactual analysis.

Using these allocation-selected expected outcomes, we define four counterfactual statistics of interest: the Engineering assignment rate gap, the gender earnings gap, aggregate predicted earnings, and aggregate expected fertility.

Assignment rate gap in Engineering: Let N_M and N_F correspond to the total number of male and female applicants, and let $j^E \in \mathcal{J}$ correspond to Engineering. The assignment rate gap—in Engineering—under allocation $\mathcal{A}$ is defined as:

$$\Delta_{j^E} S(\mathcal{A}) = \frac{\sum_{i:G_i=0} \mathbb{1}\{j_i(\mathcal{A}) = j^E\}}{N_M} - \frac{\sum_{i:G_i=1} \mathbb{1}\{j_i(\mathcal{A}) = j^E\}}{N_F} \quad (16)$$

where $\Delta_{j^E} S(\mathcal{A})$ measures male overrepresentation in Engineering, defined as the difference in the within-gender shares of Engineering assignments between men and women under allocation $\mathcal{A}$. Across counterfactual scenarios, our object of interest is the percentage change in the Engineering assignment rate gap, i.e., $100 \times (\Delta_{j^E} S(\mathcal{A}') - \Delta_{j^E} S(\mathcal{A})) / \Delta_{j^E} S(\mathcal{A})$.

Earnings gap: Applicant i 's expected earnings under allocation $\mathcal{A}$ are $\bar{Y}_i^e(\mathcal{A}) = \bar{Y}_{i,j_i(\mathcal{A})}^e$. We define the female-minus-male gender earnings gap under allocation $\mathcal{A}$, as follows:

$$\Delta(\mathcal{A}) = \frac{\sum_{i:G_i=1} \bar{Y}_i^e(\mathcal{A})}{N_F} - \frac{\sum_{i:G_i=0} \bar{Y}_i^e(\mathcal{A})}{N_M}.$$

Thus, when the allocation changes from $\mathcal{A}$ to $\mathcal{A}'$, the corresponding percentage change in the gender earnings gap is given by $100 \times (\Delta(\mathcal{A}') - \Delta(\mathcal{A})) / \Delta(\mathcal{A})$.

Total Earnings: To summarize aggregate implications of each counterfactual assignment $\mathcal{A}'$, we consider the percentage change in aggregate predicted earnings relative to the baseline assignment $\mathcal{A}$, defined as:

$$\mathcal{E}(\mathcal{A}, \mathcal{A}') = 100 \times \frac{\sum_i \Xi_i^e(\mathcal{A}, \mathcal{A}')}{\sum_i \bar{Y}_i^e(\mathcal{A})}. \quad (17)$$

The numerator sums applicant-level changes in expected earnings, and the denominator is the already-defined aggregate expected earnings under the baseline allocation. A positive value of $\mathcal{E}(\mathcal{A}, \mathcal{A}')$ indicates that $\mathcal{A}'$ leads to higher aggregate expected earnings than $\mathcal{A}$, while a negative value implies the opposite.

Parenthood: Finally, to account for the impacts on parenthood by roughly age 31, we define the percentage change in expected fertility relative to the baseline allocation:

$$\mathcal{F}(\mathcal{A}, \mathcal{A}') = 100 \times \frac{\sum_i \Xi_i^f(\mathcal{A}, \mathcal{A}')}{\sum_i \bar{Y}_i^f(\mathcal{A})}.$$

evidence of mismatch effects in Brazil, while Carlana et al. (2024) find evidence of mismatch effects for marginal students but not for the average students in Chile.

Expected fertility is the model-implied probability of having at least one child 13 years after college application. The numerator sums applicant-level changes in that probability, and the denominator is aggregate expected fertility under the baseline allocation. A positive value of $\mathcal{F}(\mathcal{A}, \mathcal{A}')$ indicates that $\mathcal{A}'$ leads to higher aggregate expected fertility than $\mathcal{A}$, while a negative value implies the opposite.

5.2 Empirical Methods

To construct the counterfactual statistics, we estimate the conditional outcome components and graduation probabilities required for every applicant and assignment state.³¹ For each outcome $r \in \{e, f\}$, the baseline causal model described in Section 3 provides a framework to obtain $\hat{m}_{ij,1}^r$ and $\hat{m}_{i0}^r$ for assigned fields $j \in \mathcal{J}$. Specifically, estimates of Equation (9) provide the conditional means $\hat{m}_{ij,1}^r$ and $\hat{m}_{i0}^r$ that we use to construct model-implied predicted outcomes for assigned fields $j \in \mathcal{J}$. For $\hat{m}_{iU}^r$ we use an auxiliary model to predict outcomes of the unassigned as a function of the same covariates in the causal model. Allocation $\mathcal{A}$ then selects the expected outcome associated with applicant i 's induced field or outside-option state. We hold treatment effects fixed across counterfactual policies, abstracting from peer effects, behavioral responses to changes in student composition, and equilibrium adjustments in skill prices, a common assumption in the literature (Barahona et al., 2026), but limiting the interpretation of our results to partial equilibrium effects.³²

To estimate graduation probabilities as a function of assignments, we use a variant of our control function model. In particular, for each major k we can model the conditional graduation probability following Equation (6) where the outcome is graduating from k as $P(F_i(j) \mid \mathcal{C}_i)$. This allows us to recover a matrix of causal effects of assignments on graduation in different fields of study that we incorporate in our construction of expected earnings in Equation (14). As noted above, our approach that combines our baseline outcome model with the graduation model further requires that potential field completion paths be ignorable of assignments conditional on type. Appendix Figure E.1 assesses the forecast bias among held-out applicants in 2007. The estimated slope is 1.011 (robust standard error 0.029), and we do not reject a unit slope ($p = 0.708$), providing suggestive evidence that the model adequately predicts graduation probabilities in addition to estimating causal effects of assignment to different fields on graduation. We now turn to evaluating the policies of interest.

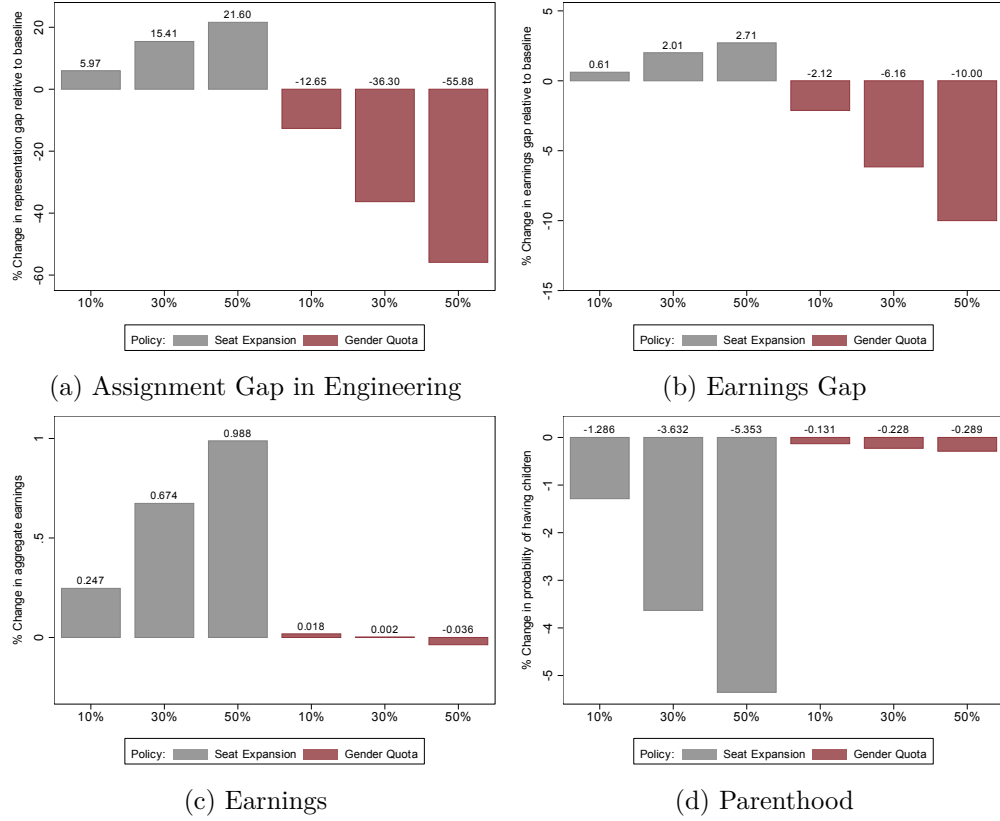
5.3 Results

Figure 8 summarizes the impacts of several counterfactual policies targeting Engineering admissions. The first three gray bars in Panel (a) show that expanding Engineering seats without gender-specific intervention worsens the assignment gap. Because men demand Engineering seats more

³¹We use estimates reported in Table 3 and Appendix Table C.2 for fertility and earnings outcomes, respectively. The choice of Appendix Table C.2 for earnings effects stems from using earnings levels for counterfactual exercises as opposed to log earnings.

³²Reallocation policies may affect outcomes through changes in peer composition (Altonji et al., 2015), and through changes in the supply of graduates that alter equilibrium wages and educational choices (Heckman et al., 1998).

Figure 8: Counterfactual Policies



Notes: These figures report the changes (relative to baseline) implied by different counterfactual scenarios. We consider 10, 30, and 50% seat expansions in Engineering programs (gray bars) and a 10, 30, and 50% quota for women in Engineering (red bars). Panel (a) shows the impacts on the Engineering assignment gap. Panel (b) reports the impacts on the total gender earnings gap. Panels (c) and (d) show the total impacts on earnings and fertility. The sample includes students who applied through the centralized admission system in 2007.

strongly, they secure most of the newly available places, further widening the gender imbalance: the assignment gap grows by 6-22 percent relative to its 16 percentage point baseline. In contrast, the next three maroon bars illustrate that gender quotas of 10%, 30%, and 50% mechanically lower the assignment gap.³³ However, even these quotas do not fully eliminate the baseline disparity, reflecting persistent differences in application rates between men and women.

The allocative changes may dampen or widen baseline earnings gaps as students are shuffled across fields with varying idiosyncratic graduation probabilities and idiosyncratic returns, with potentially contrasting effects. Panel (b) reports how each policy affects the gender earnings gap among the pool of applicants. Gender-neutral Engineering seat expansions not only widen the assignment gap, but also increase the overall pay gap modestly by 0.6 to 2.7 percent. In contrast, quota policies reduce the gender earnings gap by a more sizable 2 to 10 percent, depending on the

³³When introducing quotas, we modify the matching algorithm to consider reserved seats by processing open seats first and then applying the quotas. See Dur et al. (2018) for the significance of precedence in allocating reserved seats.

quota size. Panel (c) shows relatively modest aggregate impacts partly due to the relatively small share of students affected by the policies of interest: Appendix Figure E.2 shows that even under the sizable 50% seat expansion or quota policies, at most 10 and 6 percent of male and female applicants are reallocated.

The differences in policy effects primarily stem from variation in where students come from and where they ultimately go. Under the seat expansion policy, although some students enter Engineering from Science and Business, the majority of men and women are drawn from the outside option, thereby boosting overall enrollment (see Appendix Table E.1). Importantly, the proportion of men affected by seat expansions is nearly twice that of women, accruing a larger proportion of the Engineering gains to men. Because assignments are inherently linked, some men and women are diverted away from Engineering and into other fields they ranked more highly than Engineering. The gender quota policy, by contrast, affects men and women in relatively similar proportions but triggers more extensive reshuffling: a large share of women *enter from* the outside option—yet roughly 30% are diverted away from Science and Business—while about 50% of men are *displaced into* the outside option and just under 40% into Science or Business (see Appendix Table E.2).

Table 6 examines the local effects underlying the evidence in Figure 8, with an emphasis on students moved in and out of Engineering programs. Panel (a) shows that across all seat expansion policies, 55–62 percent of affected women enter Engineering programs and 75–76 percent of affected men similarly enter Engineering programs (Column 1). Because of knock-on effects in the centralized match, some students exit Engineering programs as slots into more preferred programs are available, but this tends to consistently correspond to about 6 percent or less of affected students (Column 6). Across the three seat expansion scenarios, women entering Engineering are predicted to have an 18% increase in earnings, while men are predicted to have a 12–13% increase. Column 4 shows that although both groups are expected to experience an increase in earnings, their conditional causal return to Engineering is slightly below that of unaffected Engineering students. Entering students’ mean percentile ranks range from 43 to 48 in the unaffected Engineering return distribution, modestly below the reference group’s mean rank of 50.

Panel (b) of Table 6 examines the local effects of the gender quota policies. In this class of policies, 82–88% of affected women are entering Engineering programs, and 85–87% of affected men are exiting Engineering programs. The gains accrued to women tend to exceed the losses accrued to men in percentage terms, although the baseline earnings differ across genders. The women entering Engineering programs via a quota tend to be similar to those entering via seat expansions in terms of their Engineering returns, but the men entering Engineering programs via knock-on effects induced by the quotas have relatively lower Engineering returns, with mean percentile ranks of 20–29 in the unaffected reference distribution. Men exiting Engineering programs due to the quotas experience lower earnings but also had returns that would put them near the middle of the incumbent return distribution. Taken together, these results provide little indication that the policies we consider generate substantial allocative losses via poorer matches into or out of Engineering programs. This evidence, however, focused on focal Engineering programs of interest and as Table 6 demonstrates,

other students were also affected. We therefore next examine whether the distributional changes induced by these policies have aggregate effects on earnings and fertility timing.

One may be concerned that reallocative policies could generate aggregate earnings losses or further reduce fertility, a particularly salient issue in Chile, where the fertility rate has fallen to 1.5, well below the replacement level. We measure these possibilities by aggregating the outcomes of interest implied by our counterfactuals, considering all applicants in the centralized admission system. Panel (c) of Figure 8 reports small impacts on aggregate earnings ($< 1\%$ absolute change relative to the baseline). Due to the seat expansion, more students gain college access, leading to a small increase in aggregate earnings, while gender quotas lead to very small changes in aggregate earnings. Finally, Panel (d) of Figure 8 reports impacts on overall fertility, measured as the probability of having at least one child 13 years after college application. In this case, we find that both policies would reduce predicted aggregate fertility, but quotas would lead to very small declines ($< 0.3\%$).

All in all, the results highlight the importance of how access to high-return fields is allocated and how different policy objectives affect subsequent policy-relevant outcomes. A gender-neutral expansion of access may increase earnings for many, but may have downstream effects on gender earnings gaps if underlying demand differs strongly across groups. In contrast, quota policies have limited impacts on affecting aggregate earnings, but do attenuate gaps that policymakers may be interested in narrowing. Our framework that captures both vertical and horizontal dimensions of returns allows us to further show that women entering Engineering through quotas have predicted returns close to those of incumbent women, suggesting that the local effects of such policies do not generate strong allocative losses in the horizontal dimension while unlocking substantial vertical returns for those benefiting and preserving modest aggregate impacts. In other words, our estimates suggest that policies that close assignment gaps and narrow gender earnings gaps can be pursued without meaningful economic and demographic sacrifice.³⁴

³⁴The latter finding contributes to the growing evidence that policies tend to have a modest impact on fertility outcomes (Dahl and Loken, 2024).

Table 6: Engineering Returns and Earnings Changes of Entering and Exiting Students

Policy size	Gender	Entering Engineering					Exiting Engineering				
		Share of affected (%)	Change in mean earnings (%)	Match effect difference (log points)	Relative return (%)	Mean percentile	Share of affected (%)	Change in mean earnings (%)	Match effect difference (log points)	Relative return (%)	Mean percentile
		(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Panel (a): Seat Expansions											
10%	Men	74.93	11.88	-0.0016	-0.26	48.1	4.96	-1.27	-0.0080	-1.26	22.0
	Women	55.46	18.04	-0.0015	-0.27	46.3	6.00	-2.55	-0.0071	-1.24	25.1
30%	Men	75.78	12.87	-0.0019	-0.29	48.4	4.93	-0.55	-0.0099	-1.56	17.1
	Women	61.22	17.78	-0.0018	-0.31	44.2	4.27	-3.39	-0.0066	-1.16	25.3
50%	Men	76.12	12.76	-0.0021	-0.33	47.1	4.30	-0.29	-0.0105	-1.65	15.2
	Women	62.02	17.58	-0.0020	-0.35	43.4	3.99	-2.96	-0.0092	-1.61	17.4
Panel (b): Gender Quotas											
10%	Men	2.34	2.36	-0.0069	-1.09	28.9	85.45	-9.49	-0.0012	-0.19	48.4
	Women	81.64	15.83	-0.0012	-0.21	46.8	1.37	-1.50	-0.0036	-0.63	43.3
30%	Men	1.78	-0.24	-0.0082	-1.28	20.2	87.43	-10.34	-0.0017	-0.26	45.4
	Women	87.34	16.02	-0.0016	-0.28	44.8	0.96	-2.09	-0.0059	-1.02	26.2
50%	Men	2.38	-0.87	-0.0086	-1.35	21.0	85.24	-10.33	-0.0019	-0.29	43.6
	Women	88.31	16.92	-0.0018	-0.32	44.2	0.57	-4.97	-0.0034	-0.59	37.1

Notes: Columns (1) and (6) report the shares of affected same-gender applicants entering or exiting Engineering in each policy-specific paired sample. An applicant is affected if the broad-field assignment changes relative to baseline, including the outside option; the shares need not sum to 100 because some applicants move between other fields. The reference group s comprises same-gender applicants assigned to Engineering under both allocations. For entering or exiting group h , columns (2) and (7) report $100(Y_h^{\text{policy}} - \bar{Y}_h^{\text{baseline}})/Y_h^{\text{baseline}}$, using the expected earnings in Equation (14) and the linear control-function completion model in Figure 8. Columns (3) and (8) report $\bar{M}_h - \bar{M}_s$ in log points, where $M_{iE} = \psi_{E, g_i} \lambda_{iE}$ is the Engineering match component. Columns (4) and (9) report $100(\bar{\tau}_h/\bar{\tau}_s - 1)$, where $\tau_{iE} = \alpha_{E, g_i} + M_{iE}$ is the Engineering completion return relative to non-graduation. Columns (5) and (10) average individual return percentiles in the same-gender, same-policy reference distribution, with half weight for ties. Program changes within Engineering count as unaffected.

6 Conclusion

This paper examines how college majors shape multidimensional life-cycle outcomes and how these causal effects relate to gender differences in major choice and earnings. Our focus on estimating effects on earnings and fertility timing shows that major choice is not simply an investment with pecuniary returns. Instead, major choice also places students on career paths with distinct family-formation consequences. Characterizing these multidimensional payoffs, together with gender differences in preferences over them, provides a lens for understanding how the trajectories of men and women begin to diverge well before they enter the labor market.

Using administrative data that link higher education rank-ordered choices to earnings and fertility, we document substantial heterogeneity across majors but with distinct patterns across outcomes. Earnings effects vary widely across fields and exhibit economically meaningful idiosyncratic heterogeneity with similar vertical and horizontal returns for men and women. In contrast, fertility effects vary meaningfully across majors and gender, but with limited idiosyncratic heterogeneity. Fertility effects are systematically related to measures of career-family compatibility, such as job security, gender norms, and child penalties, but not to average wages, allowing us to interpret them as summary measures of an important family-formation dimension of major choice.

To connect back to the motivating puzzle regarding gendered sorting patterns, we use our causal effects to investigate how men and women weigh these ex-post payoffs when choosing fields of study. Men’s choices are more strongly associated with earnings effects, while women’s choices are more strongly associated with family-formation effects. Jointly equalizing these estimated payoff weights narrows the absolute log earnings gap by 21–25 percent. In contrast, equalizing gender-specific earnings or family-formation effect schedules is associated with modest reductions in subsequent gender earnings gaps.

Finally, taking existing preferences and our causal estimates as given, we evaluate supply-side policies that modify the allocation of college opportunities. We find that expanding Engineering capacity largely benefits men and widens the wage gap, but reserving 30–50 percent of seats for women narrows the gap by up to 10 percent, with virtually no loss in aggregate earnings and less than a half-percentage point reduction in aggregate fertility. Our results underscore two policy lessons: (i) admissions rules can move equity outcomes quickly when preferences are slow to change, and (ii) alleviating the career-family trade-off through workplace flexibility or more egalitarian norms may offer a longer-run path to convergence that does not rely on quotas (Goldin, 2021).

While we have documented systematic associations between the career-family balance embodied in majors and students’ choices, a comprehensive understanding of the factors driving gender differences in these relationships remains elusive. Understanding how preference-shifting forces take root, and whether similar dynamics arise in decentralized admissions systems—where students make major choice decisions at older ages—remain promising avenues for future work.

7 References

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Online Appendix for:
Choosing Career and Family:
The Multidimensional Returns to College Majors

Christopher Campos Pablo Muñoz Alonso Bucarey Dante Contreras

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A Data Appendix

This section discusses additional details regarding the data used in our analysis. We begin by providing details about our categorization of majors that are crucial in our analysis. Then, since earnings records are top-coded, we discuss our imputation procedure. We also report additional descriptive evidence related to trends in gender representation, preferences, and standardized test scores mentioned in the paper.

A.1 Major Classification

We classify students into fields of study based on the major from which they graduate. Degree programs are classified by field of study mostly based on the OECD Handbook for Internationally Comparative Education Statistics (OECD, 2004). There are eight broad categories: “*Agriculture*”, “*Science*”, “*Social Sciences, Business and Law*”, “*Teaching*”, “*Humanities and Arts*”, “*Engineering, Manufacturing and Construction*”, “*Health and Welfare*”, and “*Services*”.

We reclassify the category “*Social Sciences, Business and Law*” into three separate fields of study “*Social Sciences*”, “*Business*”, and “*Law*”. The field of “*Social Sciences*” includes the degrees of anthropology, library science, political sciences, social communication, geography, journalism, psychology, sociology, and social work. The field of “*Business*” includes the degrees: commercial engineering, accounting, commerce engineering, business administration, marketing engineering, logistics engineering, foreign trade engineering, management control engineering, human resources engineering, finance engineering, public management, advertising, and public relations; while the field of “*Law*” includes law degrees.

We also separate “*Medicine*” from “*Health and Welfare*”. In particular, we include all degrees covered by the “Medical Law” in Chile (Ley N 19,664) into “*Medicine*”. These degrees are in medicine, dentistry, pharmaceutical chemistry, and biochemistry. Finally, we drop “*Agriculture*” and “*Services*” as they represent very few graduates (less than 5%) from non-homogeneous college programs in Chile.

A.2 Data Imputations

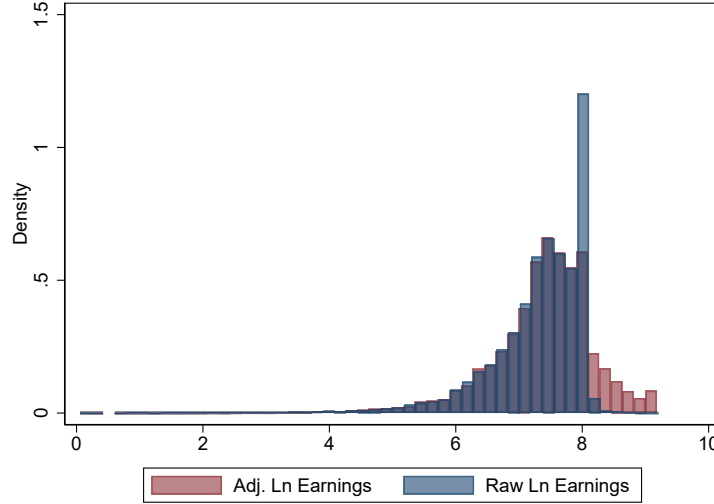
To impute wages above the social security contribution limit, we proceed as in Dustmann et al. (2009) and Card et al. (2013). First, we fit a series of Tobit models to log wages separately by gender, including the type of high school, test-score range, and region controls.

Then, we impute an uncensored value for each censored observation using the estimated parameters of these models and random drawings ϵ from a truncated distribution. Following Gartner et al. (2005):

$$\epsilon_i = \Phi^{-1} \left(u \times \left[1 - \Phi \left(\frac{c - X_i' \hat{\beta}}{\hat{\sigma}} \right) \right] + \Phi \left(\frac{c - X_i' \hat{\beta}}{\hat{\sigma}} \right) \right),$$

where $u \sim U[0,1]$, c is the social security contribution limit, $X'\hat{\beta}$ is the Tobit prediction, and $\hat{\sigma}$ is the standard deviation of the Tobit error. Figure A.1 below presents both the distribution of the original log earnings and the imputation-adjusted log earnings used in our analysis.

Figure A.1: Log Earnings Distribution



Notes: This figure presents the histogram of the original log earnings and the imputation-adjusted log earnings. We adjust earnings at the contribution limit following Dustmann et al. (2009) and Card et al. (2013). For details, see section A.2.

A.3 Descriptive Evidence

We present descriptive evidence that motivates and complements our work.

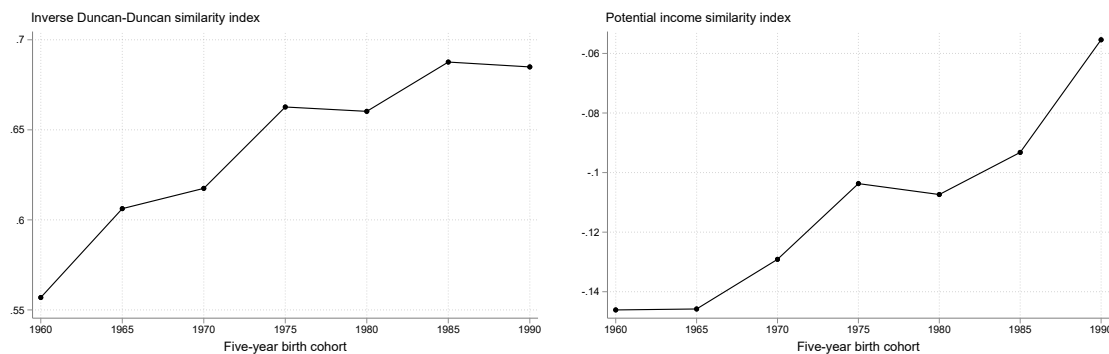
Historical Trends: Figure A.2 documents trends in gender representation across majors and its potential contribution to the overall gender gap. In the spirit of Sloane et al. (2021), and leveraging the largest household survey in Chile, the figure shows the major “similarity index” (as a re-normalization of the inverse Duncan-Duncan index) and a “potential wage index” (that assigns to everyone within a major the median wages of prime-age male workers in that major). This figure shows that majors in Chile have become less segregated over time.

Fields of Study: To characterize preferences for fields of study, Figure A.3 reports fallback major prevalence. Overall, we find sensible patterns. For example, Social Science is the most common fallback option among students who rank Law programs as their most-preferred field, while Science is the most common fallback major for individuals who rank Engineering as their most-preferred field.

Fertility: Figure A.4 uses data from the World Bank World Development Indicators database and replicates the Figure presented in Doepke et al. (2023), including Chile. The figure shows total

fertility rates since the seventies and highlights that Chile's fertility rates today are remarkably similar to those of high-income countries.

Figure A.2: Similarity and Potential Wage Indexes by Field of Study Across Cohorts

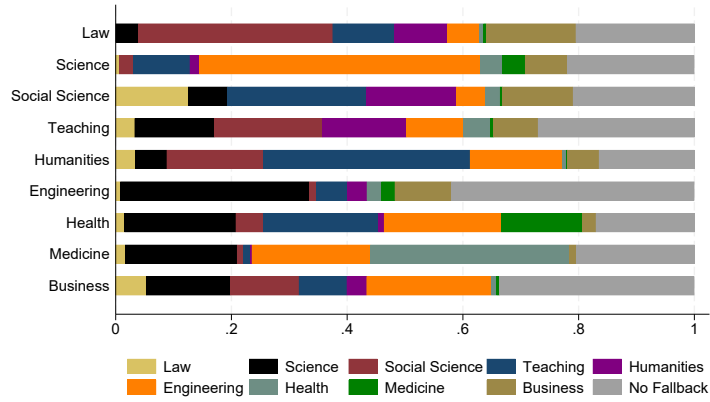


(a) Gender Similarity Index

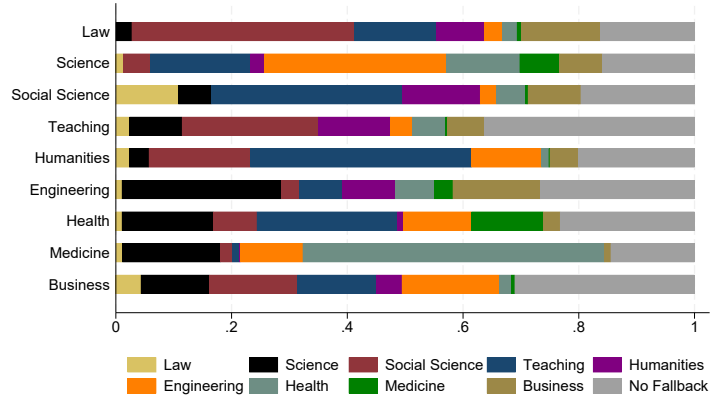
(b) Potential Wage Index

Notes: This figure presents trends in the similarity index and wage potential by field of study across cohorts, as in Sloane et al. (2021). Data comes from the largest household survey in Chile and is restricted to those with at least a bachelor's degree. Panel (a) plots the renormalized, inverse Duncan-Duncan index for different cohorts of Chilean college graduates. Panel (b) plots the potential wage index for different cohorts of Chilean college graduates.

Figure A.3: Applicants' Fallbacks



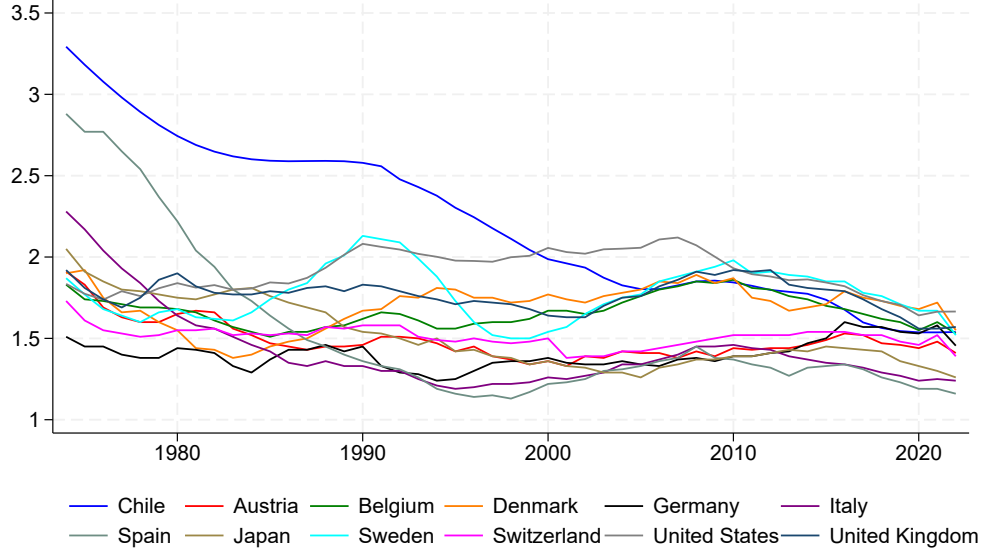
(a) Male



(b) Female

Notes: This figure shows students' fallback fields of study based on their ranked order lists. Among students ranking a given field as their first choice (y-axis), we compute the share of students ranking each of the remaining fields of study as a second option (i.e., fallback). Panels (a) and (b) show the relevance of each fallback for males and females, respectively. We consider all students who applied and were accepted through the centralized admission system between 2004 and 2007 and were employed in 2019.

Figure A.4: Total Fertility Rates Over Time



Notes: This figure shows the total fertility rates for the high-income countries considered in Doepke et al. (2023) and Chile (in blue). The data comes from the World Bank World Development Indicators database.

B Proofs

Lemma 1. Suppose offers are conditionally ignorable and Assumptions 1 and 2 hold. Then, for every $k \in \mathcal{J}_0$,

$$E[Y_{ik} \mid F_i, \theta_i] = E[Y_{ik} \mid \theta_i].$$

Consequently, for every $j \in \mathcal{J}$,

$$E[\mu_{ij} \mid F_i, \theta_i] = E[\mu_{ij} \mid \theta_i],$$

and

$$E[u_i \mid F_i, \theta_i] = 0.$$

Proof of Lemma 1. For every option $k \in \mathcal{J}_0$, using the law of iterated expectations we have

$$\begin{aligned} E[Y_{ik} \mid F_i, \theta_i] &= E[E[Y_{ik} \mid F_i, Z_i, \theta_i] \mid F_i, \theta_i] \\ &= E[E[Y_{ik} \mid Z_i, \theta_i] \mid F_i, \theta_i] \\ &= E[E[Y_{ik} \mid \theta_i] \mid F_i, \theta_i] \\ &= E[Y_{ik} \mid \theta_i], \end{aligned}$$

where the second equality follows from Assumption 2, and the third follows the assumption that offers are unrelated to potential outcomes conditional on type: $\{Y_{ij}\}_{j \in \mathcal{J}_0} \perp\!\!\!\perp Z_i \mid \theta_i$.

Next, it follows that for the idiosyncratic return μ_{ij} ,

$$\begin{aligned} E[\mu_{ij} \mid F_i, \theta_i] &= E[Y_{ij} \mid F_i, \theta_i] - E[Y_{i0} \mid F_i, \theta_i] \\ &= E[Y_{ij} \mid \theta_i] - E[Y_{i0} \mid \theta_i] \\ &= E[\mu_{ij} \mid \theta_i]. \end{aligned}$$

Finally, Assumption 1 implies $E[a_i \mid \theta_i] = \phi(\theta_i) + E[u_i \mid \theta_i] = \phi(\theta_i)$, and hence $u_i = a_i - E[a_i \mid \theta_i]$. Because $a_i = Y_{i0}$, using the same logic as above we have

$$\begin{aligned} E[u_i \mid F_i, \theta_i] &= E[a_i \mid F_i, \theta_i] - E[a_i \mid \theta_i] \\ &= E[a_i \mid \theta_i] - E[a_i \mid \theta_i] \\ &= 0. \end{aligned}$$

□

C Supplementary Results

This section provides additional results mentioned in the paper. First, we provide evidence that our approach is consistent with that of Kirkebøen et al. (2016). Second, we show results on the dynamics of the gender earnings gap through the lens of a decomposition between differences in returns and differences in match effects. Finally, we discuss a set of additional results that all point to qualitatively similar conclusions as the main results in the paper.

C.1 Relative Payoffs and Comparative Advantage:

Our findings complement and extend the evidence provided by Kirkebøen et al. (2016) on the importance of next-best alternatives and comparative advantage in the choice of college major.

Our approach resembles the fallback conditioning presented in Kirkebøen et al. (2016), as treatment effects vary according to the composition of a student's rank-ordered list, i.e., the payoff to a given field includes a match effect governed by the submitted rank-ordered list, including the fallback option and any subsequent fields a student ranks. This means that treatment effects for major j differ for individuals with different fallbacks; and even among individuals with similar fallbacks, additional differences in the composition of the rank-ordered list produce additional differences in treatment effects. Table C.1 shows the fallback-specific returns estimated by our model. Results are consistent with those of Kirkebøen et al. (2016) in that payoffs are highly heterogeneous depending on each student's fallback. For instance, the payoff to Science goes from positive for those with next best alternatives in Social Science, Teaching or Humanities to negative for those with next best alternatives in Business or Medicine.

C.2 Additional Results

Private Sector Earnings Data: Appendix Table C.3 reports estimates using the private sector data discussed in Section 2. The overarching results are qualitatively similar to our main estimates. In this case, however, the return to Medicine is relatively smaller. This stems from the fact that many medical doctors are employed in the public sector in Chile, so estimates using private sector data will invariably miss a significant portion of their labor market. For this reason, we use the pension system records as our primary data.

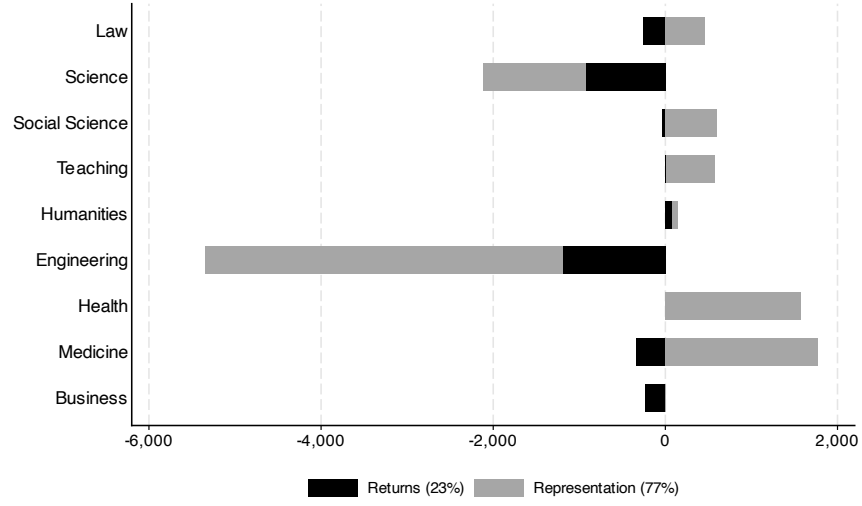
Total Earnings: Appendix Table C.2 reports estimates using the pension data but considering total yearly earnings instead of log earnings as the dependent variable. This allows us to include individuals not employed in the Chilean formal labor market in 2019. We find similar results when considering this outcome variable encompassing the intensive and extensive margins of the labor market. We find differences in returns across fields of study and gender. In line with our main results, men exhibit larger earnings in Science and Engineering, and women exhibit higher earnings in Teaching and Humanities, fields with relatively lower returns.

Alternative Non-Pecuniary Returns: The fertility results in the paper are on the extensive margin of having a child 13 years after college application. Appendix Table C.4 reports the estimates obtained when the outcome is the total number of children. This captures both extensive and intensive margin effects. Reassuringly, we find similar results when accounting for the intensive margin of fertility.

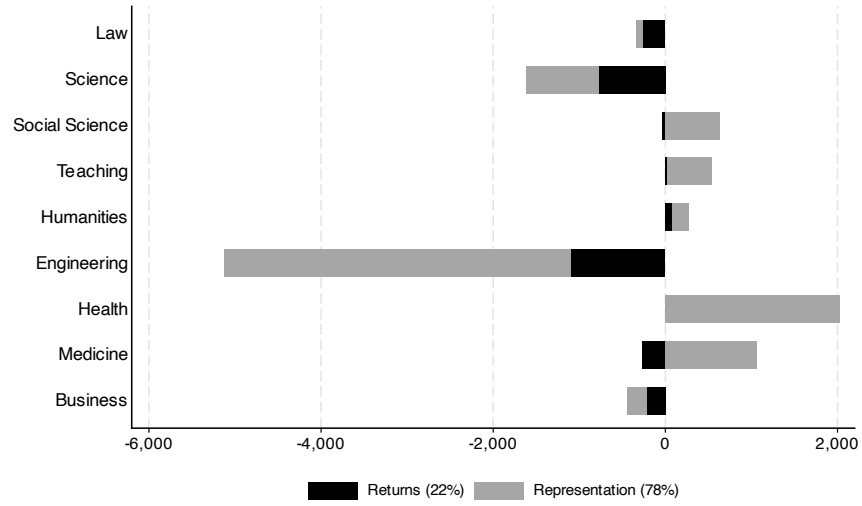
Fertility Correlations: We conjecture that majors impact fertility by allowing students to access jobs with different attributes and career opportunities. Our interpretation is consistent with the “new theories” underscoring that factors affecting the compatibility of women’s careers and families are key drivers of fertility (Doepke et al., 2023). We present evidence supporting our hypothesis in Figure 5, which reports the point estimates and confidence intervals obtained from separate regressions of fertility returns by field of study on job and field-specific attributes. For additional details, see Appendix D.

Counterfactual Preferences: Figure C.1 decomposes the annual earnings gap under the two joint preference-weight counterfactuals in Figure 6. The male-reference exercise assigns men’s estimated weights on both average earnings and family-formation effects to women and leaves men unchanged; the female-reference exercise does the reverse. Representation accounts for 77–78 percent of the absolute component contributions under these counterfactual shares.

Figure C.1: Counterfactual Earnings Gap: Sorting versus Returns



(a) Male Preferences



(b) Female Preferences

Notes: These figures use Equation (12) to decompose the annual gender earnings gap into differences in representation across fields and differences in earnings within fields. We consider annual earnings in 2019 for students who applied through the centralized admission system between 2004 and 2007, graduated between 2007 and 2019, and were employed in 2019. Panel (a) assigns men's estimated demand weights on both average earnings and family-formation effects to women, leaving men unchanged. Panel (b) assigns women's weights to men, leaving women unchanged. The counterfactual field shares use the same calibrated choice model and corrected all-36-cell estimates as Figure 6. Percentages in the legends report each component's share of the sum of absolute field-level contributions.

Table C.1: Estimates of Relative Payoffs

First-ranked field (j):		Second-ranked field (k):							
	Law	Science	Social Science	Teaching	Humanities	Engineering	Health	Medicine	Business
<i>Panel (a): Female</i>									
Law	–	0.18 (0.12)	0.30 (0.10)	0.14 (0.10)	0.48 (0.12)	–0.01 (0.10)	0.17 (0.10)	–0.13 (0.11)	–0.08 (0.10)
Science	–0.07 (0.11)	–	0.15 (0.08)	–0.01 (0.08)	0.34 (0.10)	–0.17 (0.09)	0.02 (0.08)	–0.30 (0.09)	–0.23 (0.08)
Social Science	–0.20 (0.09)	–0.08 (0.07)	–	–0.13 (0.04)	0.23 (0.07)	–0.28 (0.05)	–0.09 (0.04)	–0.42 (0.06)	–0.34 (0.04)
Teaching	–0.06 (0.08)	0.04 (0.06)	0.17 (0.03)	–	0.35 (0.06)	–0.16 (0.04)	0.03 (0.03)	–0.29 (0.05)	–0.22 (0.03)
Humanities	–0.33 (0.11)	–0.22 (0.10)	–0.11 (0.08)	–0.26 (0.08)	–	–0.42 (0.08)	–0.24 (0.08)	–0.48 (0.08)	–0.48 (0.08)
Engineering	0.10 (0.09)	0.21 (0.07)	0.33 (0.05)	0.16 (0.04)	0.52 (0.07)	–	0.20 (0.05)	–0.13 (0.06)	–0.05 (0.05)
Health	–0.04 (0.09)	0.08 (0.07)	0.20 (0.04)	0.04 (0.03)	0.39 (0.07)	–0.12 (0.04)	–	–0.26 (0.06)	–0.19 (0.04)
Medicine	0.20 (0.10)	0.30 (0.08)	0.41 (0.07)	0.25 (0.06)	–	0.10 (0.07)	0.29 (0.06)	–	0.04 (0.06)
Business	0.17 (0.09)	0.28 (0.07)	0.40 (0.04)	0.24 (0.04)	0.59 (0.07)	0.08 (0.05)	0.27 (0.04)	0.01 (0.08)	–
<i>Panel (b): Male</i>									
Law	–	0.13 (0.12)	0.44 (0.12)	0.31 (0.11)	0.71 (0.14)	0.07 (0.11)	0.27 (0.12)	–	0.03 (0.12)
Science	0.01 (0.10)	–	0.35 (0.06)	0.23 (0.05)	0.63 (0.09)	–0.02 (0.04)	0.17 (0.06)	–0.20 (0.07)	–0.05 (0.05)
Social Science	–0.27 (0.11)	–0.23 (0.07)	–	–0.05 (0.06)	0.35 (0.10)	–0.29 (0.06)	–0.10 (0.07)	–0.47 (0.08)	–0.33 (0.07)
Teaching	–0.17 (0.10)	–0.12 (0.04)	0.18 (0.06)	–	0.46 (0.09)	–0.19 (0.03)	–0.01 (0.05)	–0.35 (0.07)	–0.23 (0.04)
Humanities	–0.53 (0.14)	–0.49 (0.10)	–0.18 (0.11)	–0.30 (0.10)	–	–0.54 (0.10)	–0.40 (0.11)	–0.58 (0.10)	–0.58 (0.10)
Engineering	0.04 (0.10)	0.08 (0.04)	0.39 (0.05)	0.27 (0.03)	0.67 (0.08)	–	0.21 (0.05)	–0.17 (0.06)	–0.02 (0.03)
Health	–0.10 (0.11)	–0.05 (0.06)	0.25 (0.07)	0.15 (0.06)	0.57 (0.10)	–0.11 (0.06)	–	–0.29 (0.08)	–0.17 (0.06)
Medicine	0.21 (0.12)	0.24 (0.08)	0.55 (0.09)	0.43 (0.09)	0.86 (0.13)	0.19 (0.07)	0.37 (0.08)	–	0.15 (0.08)
Business	0.11 (0.10)	0.15 (0.04)	0.45 (0.06)	0.34 (0.04)	0.74 (0.09)	0.09 (0.03)	0.27 (0.05)	–0.10 (0.07)	–

Notes: In the spirit of Kirkeboen et al. (2016), this table presents the matrix of the pecuniary payoffs to field j as compared to k (for those who prefer j and have k as next-preferred field) implied by our model. The rows represent completed fields and the columns represent next-ranked fields. Equipped with model estimates of earning payoffs, we calculate the average fallback-specific return among those that list major j as their most-preferred field and who list major k as their next-preferred field. Each cell reports an estimate of these averages.

Table C.2: Gender Differences in Returns and Match Effects: Total Earnings

	Law (1)	Science (2)	Social Science (3)	Teaching (4)	Humanities (5)	Engineering (6)	Health (7)	Medicine (8)	Business (9)
Panel A: Female									
Returns	7,079 (1,185)	4,261 (961)	3,212 (336)	7,245 (148)	-2,402 (558)	10,040 (660)	7,442 (286)	17,444 (956)	12,056 (483)
Selection on Gains	1,418 (417)	197 (480)	539 (188)	-196 (115)	1,245 (235)	486 (362)	1,008 (172)	40 (415)	1,002 (234)
Panel B: Male									
Returns	6,042 (1,368)	9,170 (698)	2,507 (538)	6,451 (258)	-2,739 (671)	14,226 (374)	7,013 (604)	21,592 (1,200)	14,093 (542)
Selection on Gains	2,450 (482)	1,128 (413)	1,289 (253)	625 (155)	1,548 (253)	744 (325)	1,221 (264)	38 (478)	1,396 (284)
Est. Females = Est. Males									
Δ Returns (p-val)	0.566	0.000	0.262	0.005	0.696	0.000	0.511	0.007	0.005
Δ Match Effects (p-val)	0.105	0.141	0.017	0.000	0.379	0.596	0.499	0.997	0.285

Notes: This table presents the estimates obtained from our main regression model, presented in Equation (9). We consider the sum of earnings in 2019 (in USD) instead of the log of earnings to account for the extensive margin. We restrict the sample to students who applied and were accepted through the centralized admission system between 2004 and 2005. The sample includes 137,339 observations. The final two rows of the table report p-values from two additional statistical tests. The Δ Returns p-values correspond to the null hypothesis that the Female and Male returns are equivalent for a given column. The Δ Match Effects p-values correspond to the null hypothesis that the Female and Male match effects are equivalent for a given column. All specifications include quadratic polynomials of the college admission scores in mathematics and language, and control for level-effects as defined in the main text, cohort fixed effects and cell fixed effects, where a cell is defined by school type, macro-region, financial-aid relevant score ranges, and gender. For ease of interpretation, $\hat{\lambda}_{ij}$ is standardized within each cell. The R^2 of this regression is 0.18.

Table C.3: Gender Differences in Returns and Match Effects: Private Sector Data

	Law (1)	Science (2)	Social Science (3)	Teaching (4)	Humanities (5)	Engineering (6)	Health (7)	Medicine (8)	Business (9)
Panel A: Female									
Returns	0.391 (0.071)	0.376 (0.045)	0.222 (0.020)	0.284 (0.011)	-0.023 (0.047)	0.517 (0.022)	0.246 (0.023)	0.429 (0.054)	0.525 (0.018)
Selection on Gains	0.058 (0.023)	0.009 (0.022)	0.039 (0.011)	0.021 (0.008)	0.069 (0.019)	0.016 (0.013)	0.065 (0.013)	-0.013 (0.023)	0.019 (0.009)
Panel B: Male									
Returns	0.434 (0.092)	0.531 (0.023)	0.192 (0.036)	0.193 (0.021)	-0.124 (0.065)	0.614 (0.013)	0.278 (0.049)	0.428 (0.069)	0.580 (0.019)
Selection on Gains	0.105 (0.030)	0.036 (0.014)	0.060 (0.016)	0.066 (0.011)	0.087 (0.023)	0.004 (0.011)	0.066 (0.019)	0.032 (0.027)	0.052 (0.010)
Est. Females = Est. Males									
Δ Returns (p-val)	0.710	0.002	0.459	0.000	0.203	0.000	0.551	0.983	0.025
Δ Match Effects (p-val)	0.224	0.307	0.289	0.001	0.532	0.489	0.942	0.203	0.014

Notes: This table presents the estimates obtained from our main regression model, presented in Equation (9). We consider the earnings of workers in the private sector. The sample includes 78,336 observations of students who applied and were accepted through the centralized admission system between 2004 and 2007 and were employed in the private sector in 2019. The final two rows of the table report p-values from two additional statistical tests. The Δ Returns p-values correspond to the null hypothesis that the Female and Male returns are equivalent for a given column. The Δ Match Effects p-values correspond to the null hypothesis that the Female and Male match effects are equivalent for a given column. All specifications include quadratic polynomials of the college admission scores in mathematics and language, and control for level-effects as defined in the main text, cohort fixed effects and cell fixed effects, where a cell is defined by school type, macro-region, financial-aid relevant score ranges, and gender. For ease of interpretation, $\hat{\lambda}_{i,j}$ is standardized within each cell. The R^2 of this regression is 0.25.

Table C.4: Fertility: Number of children 13 years after college application

	Law (1)	Science (2)	Social Science (3)	Teaching (4)	Humanities (5)	Engineering (6)	Health (7)	Medicine (8)	Business (9)
Panel A: Female									
Returns	-0.184 (0.050)	-0.239 (0.040)	-0.072 (0.022)	0.119 (0.015)	-0.151 (0.040)	-0.063 (0.028)	0.087 (0.019)	-0.006 (0.033)	-0.005 (0.023)
Selection on Gains	0.060 (0.018)	0.001 (0.020)	-0.006 (0.012)	-0.020 (0.011)	0.022 (0.016)	-0.016 (0.016)	-0.010 (0.011)	0.018 (0.015)	0.012 (0.011)
Panel B: Male									
Returns	-0.241 (0.049)	-0.345 (0.023)	-0.277 (0.030)	-0.136 (0.022)	-0.436 (0.036)	-0.126 (0.015)	-0.142 (0.031)	-0.144 (0.037)	-0.122 (0.021)
Selection on Gains	0.058 (0.018)	0.043 (0.014)	0.012 (0.014)	-0.003 (0.012)	0.047 (0.014)	-0.008 (0.013)	0.020 (0.013)	0.046 (0.015)	-0.018 (0.011)
Est. Females = Est. Males									
Δ Returns (p-val)	0.415	0.018	0.000	0.000	0.000	0.040	0.000	0.004	0.000
Δ Match Effects (p-val)	0.929	0.075	0.338	0.294	0.239	0.687	0.091	0.177	0.059

Notes: This table presents the estimates obtained from our main regression model, presented in Equation (9). The sample includes 137,339 observations of students who applied and were accepted through the centralized admission system between 2004 and 2007. The final two rows of the table report p-values from two additional statistical tests. The Δ Returns p-values correspond to the null hypothesis that the Female and Male returns are equivalent for a given column. The Δ Match Effects p-values correspond to the null hypothesis that the Female and Male match effects are equivalent for a given column. All specifications include quadratic polynomials of the college admission scores in mathematics and language, and control for level-effects as defined in the main text, cohort fixed effects and cell fixed effects, where a cell is defined by school type, macro-region, financial-aid relevant score ranges, and gender. For ease of interpretation, $\hat{\lambda}_{ij}$ is standardized within each cell. The R^2 of this regression is 0.05.

D Child Penalties and Gender Norms

This section details how we leverage the administrative data to estimate the child penalties and gender norms measures discussed in Section 4 of the main body of the paper.

D.1 Child Penalties

Following Kleven et al. (2019b), we estimate child penalties with the model:

$$\tilde{Y}_{it} = \sum_{k \neq -2} \beta_k D_{it}^k + \mu_a + \gamma_t + \nu_{it}, \quad (18)$$

where $D_{it}^k \equiv \mathbb{1}[t = c + k]$ is a binary variable that indicates the time relative to the period in which the first child was born c . μ_a are age-fixed effects that account for the life cycle effects on the outcome variable, and γ_t are time-fixed effects that control for temporal trends at the 6-month frequency. We omit the time binary variable corresponding to $t = -2$ since, in our case, it represents the period of child conception. Our primary outcome variable is labor market earnings in period t relative to the period of child conception t^* , i.e., $\tilde{Y}_{it} = Y_{it}/Y_{it^*}$.

We estimate the impacts of children on women and men separately and define the child penalty at event time t as $\hat{\beta}_t^m - \hat{\beta}_t^w$, which measures the percentage by which women fall behind men due to children. Figure D.1 shows the effects of parenthood on earnings across fields of study. To use a symmetric window, and since we have earnings starting only in 2015, we focus on college graduates whose first child was born in 2016 or 2017. This figure also reports the estimated child penalty at time $t = 4$. We have a more limited window of coverage compared to existing studies (Kleven et al., 2019a) and can estimate child penalties only up to 4 years after the birth of a child, so our estimates are relatively short-run. Nonetheless, most evidence shows a somewhat immediate drop in mothers' earnings, which changes minimally over time.

Table D.1 complements previous evidence and reports the estimated child penalty before child conception, at time $t = -4$, and after it, at time $t = 4$; and Appendix Figure D.2 reports child penalty estimates for each field of study. There is a vast amount of heterogeneity, with Science and Social Science exhibiting the largest penalties (36% and 32%, respectively) and Business and Medicine exhibiting the lowest child penalties. The average penalty across majors is a 16% reduction in earnings for women relative to men after the birth of their first child, with a noise-adjusted standard deviation of 10%. Although the heterogeneity we document alludes to an empirical interaction between major choice, gender, earnings, and fertility, the evidence reported in Figure D.2 is not causal.³⁵

Another explanation is that some majors lead to careers in industries or occupations with more flexible work arrangements, which could help mitigate or amplify the child penalty. Additionally, as suggested by the evidence in Table 5, individuals who prioritize work-life balance and anticipate

³⁵An econometric concern is that we observe child penalties for a selected sample of parents who choose to have children earlier. Melentyeva and Riedel (2023) finds that age heterogeneity matters but that child penalties are largest for younger mothers as they are more likely to disconnect from the labor market in years with steep career growth.

having children may choose majors perceived as more family-friendly. However, certain industries—often aligned with specific majors—may enforce stronger gender norms or biases. In traditionally male-dominated fields like Engineering or Science, the child penalty might be more or less pronounced depending on implicit or explicit biases against women taking maternity leave or reducing work hours.

D.2 Gender Norms

We follow the pioneering work by Bertrand et al. (2015) and construct a measure of major-specific gender norms associated with the distribution of relative labor earnings within households. We focus on graduates who had their first child after they obtained their college degree and on their partners (i.e., a couple). For each couple of parents, we use their labor market earnings from the third and fourth quarter before childbirth and focus on couples where both members earn positive income and are between 20 and 40 years old. We define *Relative Earnings_i* as $\frac{Woman\ Earnings_i}{Woman\ Earnings_i + Man\ Earnings_i}$, where i indexes the couple, and *Woman Earnings_i* and *Male Earnings_i* are the total labor income of the will-be mothers and fathers, respectively.

Our proxy for the progressivity of gender norms is measured by the size of the cliff of *Relative Earnings_i* at 0.5. To gauge the magnitude of the drop at the point where the female starts to earn more than the male, we calculate the fraction of couples in each of twenty 0.05 relative income bins and project it on a discontinuity indicator while controlling for quadratic polynomials at each side of the 0.5 cutoff. Table D.2 presents the OLS estimates associated with the discontinuity indicators obtained from separate regressions for each of our nine fields of study. Panel (a) presents the results when we focus on female graduates (and their partners), and Panel (b) repeats the exercise but focuses on male graduates (and their partners).³⁶

Figure D.3 depicts, for each of the nine fields of study, the frequency distribution of relative earnings grouped in 20 bins, along with a lowess (locally weighted scatterplot smoothing) estimate of the distribution on each side of 0.5, the point where the woman starts to earn more than the man. In most fields of study, the distribution of relative earnings exhibits a sharp drop at 0.5, with the largest drops in Law, Science, Engineering, Medicine, and Business and the smallest in Social Science, Teaching, Humanities, and Health.

What drives the variation in the estimated cliffs at 0.5? Besides gender norms, a few other features could contribute to the variation in the estimated cliffs. First, majors offering students a menu of jobs with more flexible and less non-linear pay schedules allow for choices that generate a smoother within-household earnings distribution (Goldin, 2024). From that perspective, the estimated cliffs are not a product of norms but a consequence of differences in major-specific access to jobs. To partly gauge the extent of that, we corroborate our norm estimates in publicly available survey data, where we can further probe the sensitivity of the estimates to differences in labor hours.

³⁶Overall, we find similar variation in gender norms across fields of study, independently of whether we focus on male or female graduates (and their partners). Two exceptions are Humanities and Medicine, which show smaller and larger drops (at the point where the female starts to earn more than the male), respectively, when the graduate is male.

As shown by Figure D.4, we find broadly similar norm estimates among the subset of couples with full-time jobs whose hours worked per week are 40 or more.³⁷ Second, some majors have closer ties to the public sector, and variation in public sector access could contribute to variation in the cliffs. We do not find evidence of this as Health, Medicine, and Teaching all provide strong access to the public sector in Chile. Third, some scholars have noted that cliff estimates may be sensitive to bunching at 0.5, casting doubt on its interpretation as variation in gender norms, especially when using survey data (Binder and Lam, 2022; Hederos and Stenberg, 2022; Zinovyeva and Tverdostup, 2021). In our *administrative* records, however, few couples (7%) have equal earnings. Figure D.5 further investigates the potential impact of bunching on interpreting differences in cliffs as variations in gender norms. Panel (a) presents the cliffs when pooling data across fields of study, while Panel (b) shows the cliffs after removing cases where parents have equal earnings. Reassuringly, the cliffs are similar across panels.

We, therefore, argue that the variation in the estimated cliffs can be interpreted as variation in breadwinner norms across fields of study.

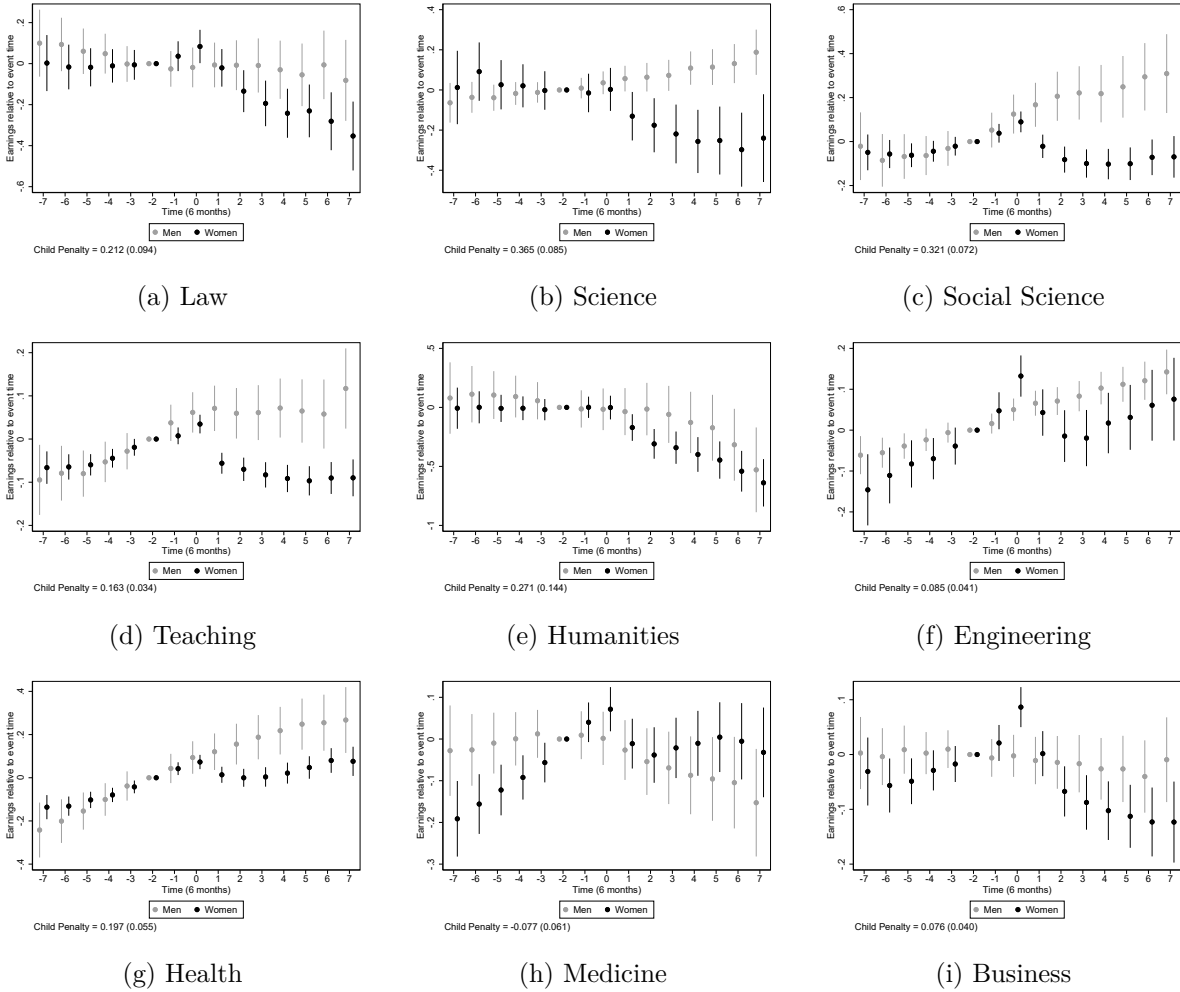
³⁷The patterns in the estimated cliffs using survey data remain broadly the same. The CASEN survey groups Health and Medicine into a single field of study, so we cannot estimate the cliffs separately for Health and Medicine.

Table D.1: Child Penalties

	Law (1)	Science (2)	Social Science (3)	Teaching (4)	Humanities (5)	Engineering (6)	Health (7)	Medicine (8)	Business (9)
Female - Male Gap Before ($t = -4$)	-0.060 (0.064)	0.038 (0.058)	0.020 (0.049)	0.008 (0.023)	-0.099 (0.098)	-0.046 (0.028)	0.021 (0.038)	-0.093 (0.042)	-0.032 (0.027)
Female - Male Gap After ($t = 4$)	-0.212 (0.094)	-0.365 (0.085)	-0.321 (0.072)	-0.163 (0.034)	-0.271 (0.144)	-0.085 (0.041)	-0.197 (0.055)	0.077 (0.061)	-0.076 (0.040)
Number of observations	24,155	24,436	60,385	131,242	19,859	95,012	71,754	27,935	80,003

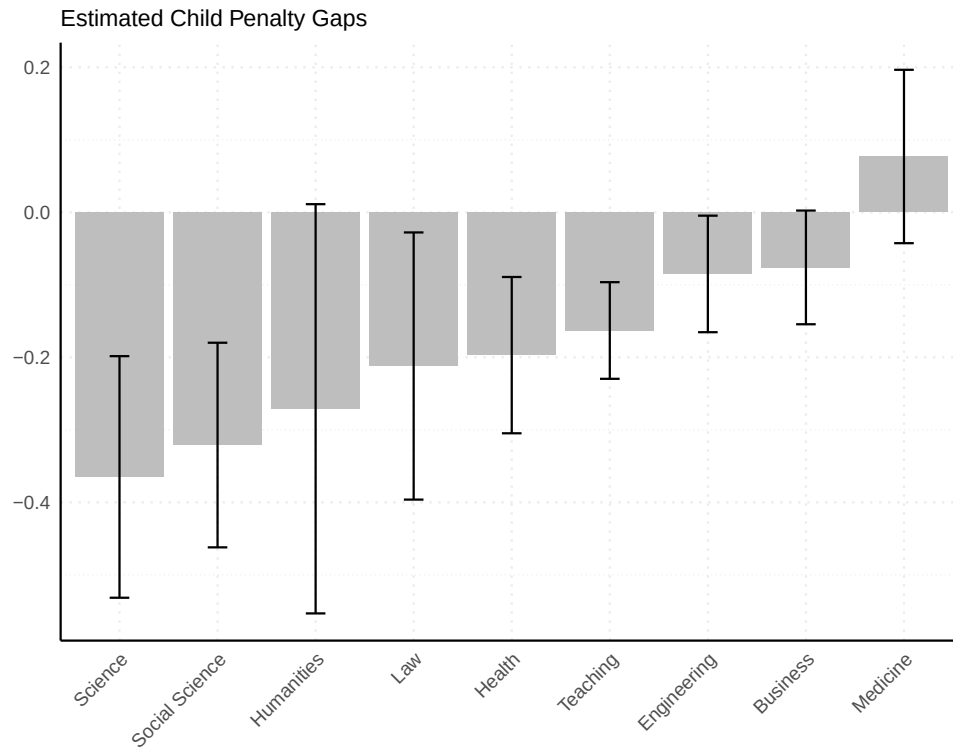
Notes: This table presents estimates obtained from estimating Equation (18). Specifically, it shows the estimated child penalty (and its standard error) before child conception, at time $t = -4$, and after it, at time $t = 4$.

Figure D.1: Child Penalty by Field of Study



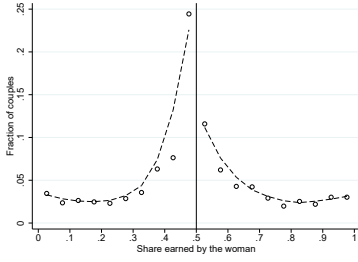
Notes: These figures show the percentage effects of parenthood on earnings across event time for men and women. Each panel corresponds to one of the nine fields of study considered in our analysis. We consider college graduates who had their first child between 2016 and 2017 and after their graduation. The figure also displays the average child penalty at the end of our time window (4 years after childbirth). Earnings are not conditional on employment status, and the effects include extensive and intensive margins.

Figure D.2: Child Penalty Gaps Across Fields of Study

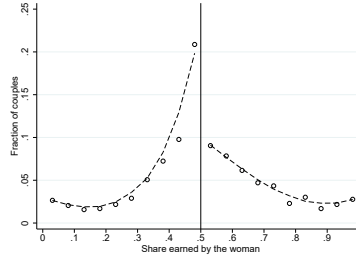


Notes: This figure shows the estimated child penalty gap four years after birth (and the corresponding confidence interval). We consider college graduates who had their first child between 2016 and 2017 and after graduation. Earnings are not conditional on employment status, and the effects include extensive and intensive margins. Estimation follows Kleven et al. (2019b) and for additional details, see Appendix D.

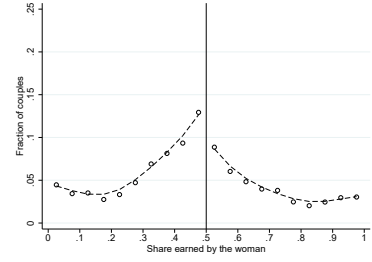
Figure D.3: Gender Norms by Field of Study



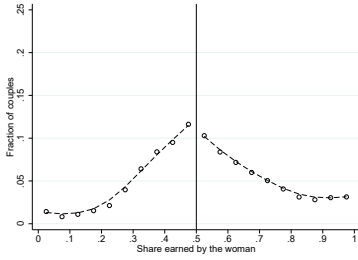
(a) Law



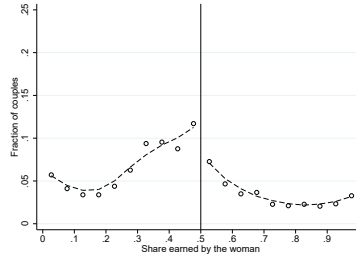
(b) Science



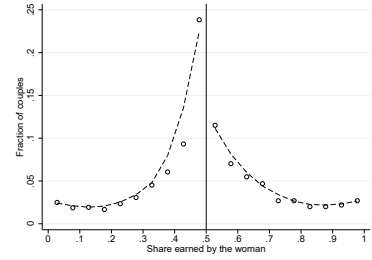
(c) Social Science



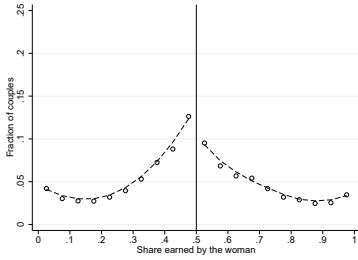
(d) Teaching



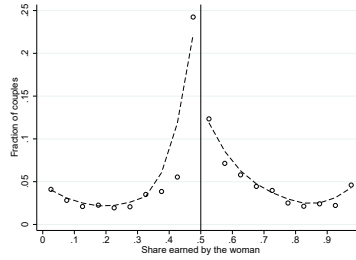
(e) Humanities



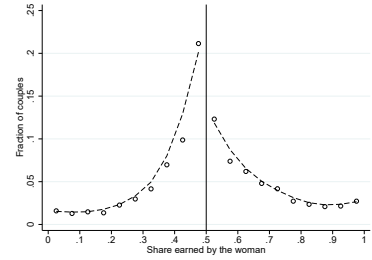
(f) Engineering



(g) Health



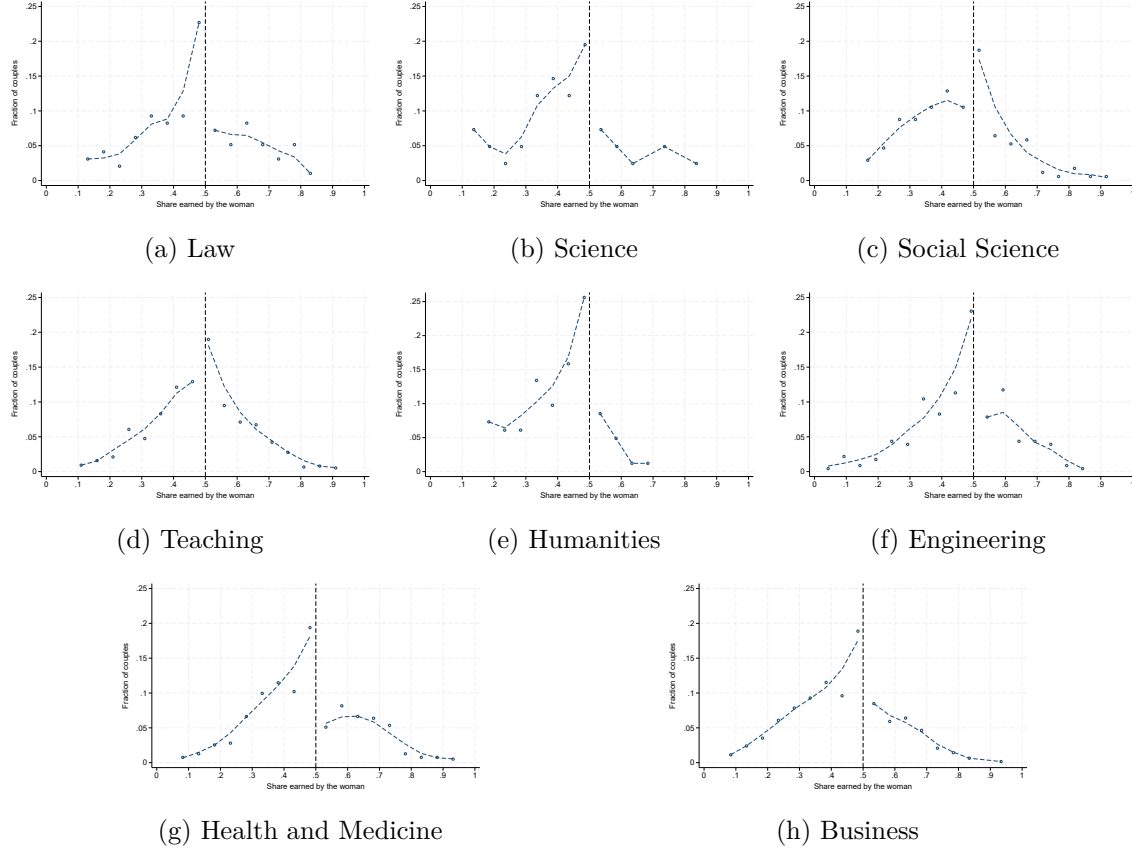
(h) Medicine



(i) Business

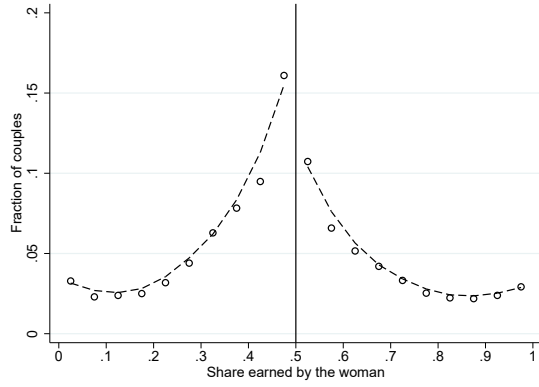
Notes: These figures report the relative income distribution for females who graduated from different fields of study. The sample includes couples of parents who both earn positive income. We use the observation from the third and fourth quarters before childbirth. Each dot is the fraction of couples in a 0.05 relative income bin. The vertical line indicates the relative income share=0.5. The dashed line is the lowess smoother applied to the distribution, allowing for a break at 0.5.

Figure D.4: Gender Norms by Field of Study: Full-time workers (CAsEN survey)

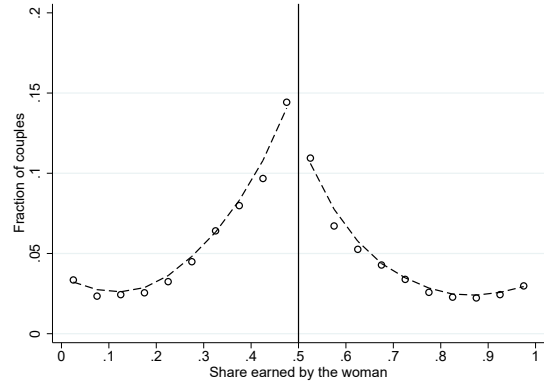


Notes: These figures report the distribution of relative income for couples in the household survey CAsEN in which the woman is a college graduate herself or her partner is, both partners earn positive income, and both work full-time (at least 40 hours a week). Each dot is the fraction of couples in a 0.05 relative income bin. The vertical line indicates the relative income share=0.5. The dashed line is the lowess smoother applied to the distribution, allowing for a break at 0.5. Since the survey records do not separate Medicine from Health, both fields are embedded into the Health figure.

Figure D.5: Gender Norms: Removing bunching at 0.5 (pooled)



(a) All



(b) No Bunching

Notes: These figures report the distribution of relative income. The sample includes couples of parents who both earn positive income. Panel (a) consider all parents, while Panel (b) removes cases in which parents report the same income. We use the observation from the third and fourth quarters before childbirth. Each dot is the fraction of couples in a 0.05 relative income bin. The vertical line indicates the relative income share=0.5. The dashed line is the lowest smoother applied to the distribution, allowing for a break at 0.5.

Table D.2: Gender Norms

	Law	Science	Social Science	Teaching	Humanities	Engineering	Health	Medicine	Business
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Panel (a): Female graduates (and their partners)									
Break	-0.104 (0.034)	-0.095 (0.019)	-0.048 (0.007)	-0.023 (0.006)	-0.054 (0.014)	-0.099 (0.028)	-0.036 (0.005)	-0.080 (0.039)	-0.080 (0.021)
Panel (b): Male graduates (and their partners)									
Break	-0.109 (0.019)	-0.046 (0.013)	-0.033 (0.011)	-0.019 (0.011)	-0.012 (0.012)	-0.078 (0.011)	-0.008 (0.010)	-0.143 (0.026)	-0.111 (0.019)

Notes: This table reports the estimated discontinuity in the distribution of relative earnings of individuals from different fields of study. The dependent variable is the fraction of couples. We calculate the fraction of couples in each of the twenty 0.05 relative income bins and project it on a discontinuity indicator while controlling for a quadratic polynomial at each side of the 0.5 cutoff. This table presents the OLS estimates of the discontinuity indicators obtained from separate regressions for each field of study. Panel (a) considers mothers who graduated from a field of study and their partners, and Panel (b) considers fathers who graduated from a field of study and their couples. We use the observations from the third and fourth quarters before childbirth and focus on couples where both members earn positive income.

E Counterfactual Details

This section reports additional details related to the counterfactual policies we analyze in Section 5. We discuss our implementation of the deferred acceptance algorithm, assess the within-field calibration slope of the linear control-function completion model, and provide additional details underpinning the main counterfactual results.

E.1 Deferred Acceptance Algorithm

We use the following algorithm to produce an outcome of the student-proposing Deferred Acceptance algorithm.

Step 1) Each student proposes to her first choice. Each program tentatively assigns seats to its proposers one at a time, following their priority order based on the weighted score. Students are rejected if no seats are available at the time of consideration.

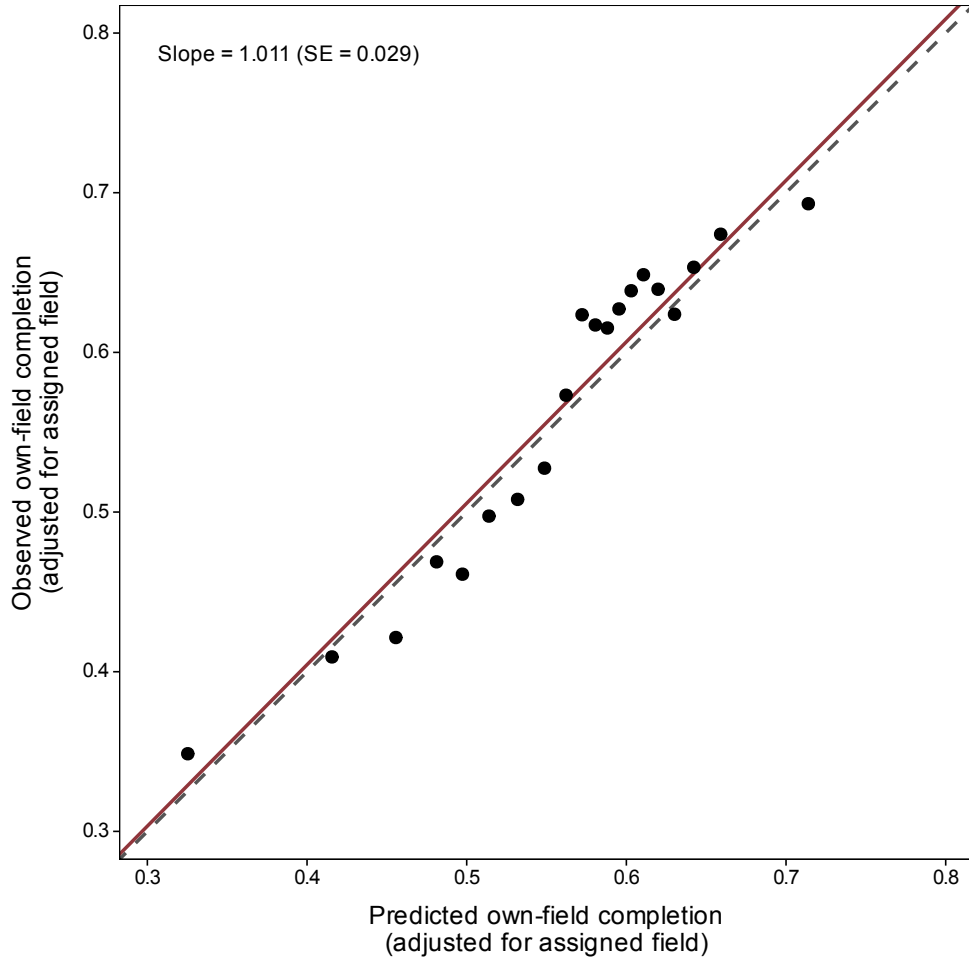
In general, in

Step k) Each student who was rejected in the previous step proposes to her next best choice. Each program considers the students it has tentatively assigned together with its new proposers and tentatively assigns its seats to these students one at a time following the program’s priority order based on the weighted score. The student is rejected if no seats are available when she is considered.

The algorithm terminates either when there are no new proposals or when all rejected students have exhausted their rank order list of preferences. When introducing quotas, we apply a precedence order by first processing the open seats and then the gender-specific seats. Following Dur et al. (2018), this order of precedence maximizes the impact of quotas.

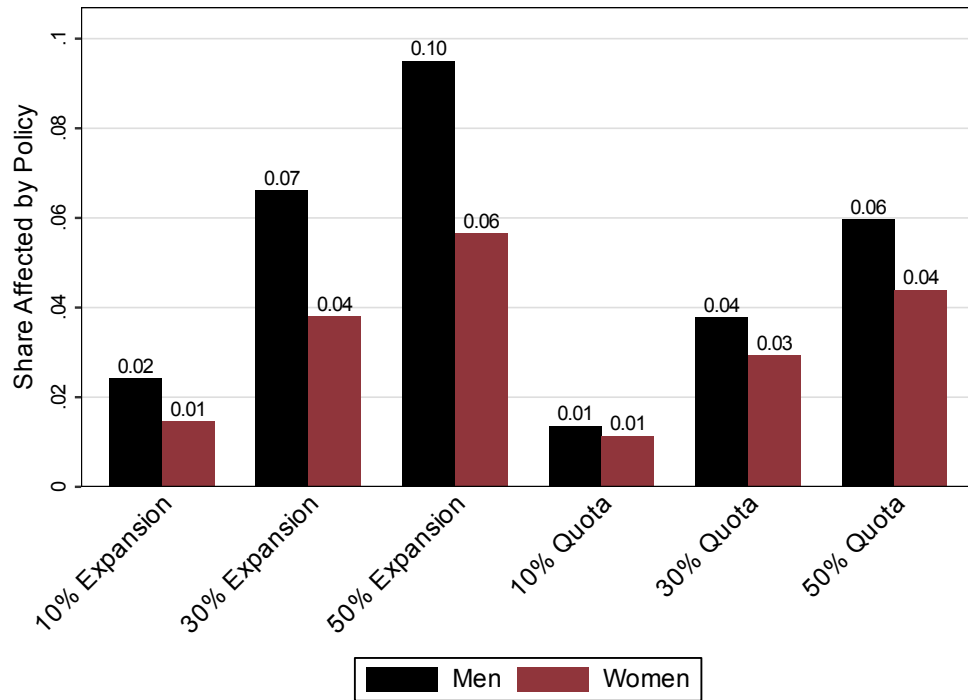
E.2 Causal Graduation Probabilities

Figure E.1: Predictive Accuracy of Graduation Probability Model



Notes: This figure pools 39,714 held-out applicants in 2007 assigned to one of the nine retained fields, excluding applicants with multiple retained-field completions. Dots show means in 20 approximately equal-frequency bins after subtracting assigned-field means from observed and predicted own-field completion. For display, the mean prediction $c = 0.557351$ is added horizontally and $\hat{\beta}c = 0.563476$ vertically; these coordinates are not raw completion rates. The solid line uses the slope from an individual-level regression with assigned-field fixed effects, with its intercept normalized to zero; the dashed line has unit slope. The annotation reports the slope and its HC1 robust standard error. The linear control-function model is estimated on 2008–2010 applicants, and held-out predictions use the 2008 cohort effect. Forecasts are not clipped. This is a pooled within-field slope check; it does not test field-specific level calibration or establish causal identification.

Figure E.2: Share Affected by Each Policy



Notes: This figure shows the share of male and female applicants whose broad-field assignment changes in each counterfactual scenario, including transitions to or from Unassigned. Shares use the policy-specific paired sample used in Figure 8, separately by gender; changes of program within the same field are not counted. Assignment shares are identical under the linear and bounded completion models.

Table E.1: Origin of Affected Students: 50% Engineering Seat Expansion

Fields	Women		Men	
	Percent among women affected in Engineering (1)	Percent among all women (2)	Percent among men affected in Engineering (3)	Percent among all men (4)
Panel (a): Effect in Engineering				
Law	0.17	0.01	0.09	0.01
Science	12.16	0.45	16.24	1.24
Social Science	1.51	0.06	0.84	0.06
Teaching	5.20	0.19	2.61	0.20
Humanities	5.54	0.21	2.84	0.22
Health	1.68	0.06	0.60	0.05
Medicine	0.00	0.00	0.00	0.00
Business	12.25	0.46	6.79	0.52
Unassigned	55.45	2.07	64.63	4.94
Reverse-direction moves	6.04	0.23	5.35	0.41
All affected Engineering	100.00	3.73	100.00	7.64
Panel (b): Effect in Other Fields				
All affected other fields	-	1.92	-	1.86

Notes: This table presents a breakdown of changes in broad-field assignment under a 50% seat expansion in Engineering relative to baseline. Panel (a) reports moves involving Engineering. Columns (1) and (3) report percentages among all women and men, respectively, who move into or out of Engineering; columns (2) and (4) report percentages of all women and men in the policy-specific paired sample used in Figure 8. Field rows give the baseline field of applicants moved into Engineering. The reverse-direction row records applicants moved out of Engineering. Panel (b) reports applicants whose field changes without entering or leaving Engineering. The two panels together recover the affected shares in Figure E.2. Percentages may not sum exactly because of rounding.

Table E.2: Origin and Destination of Affected Students: 50% Engineering Quota

Fields	Women		Men	
	Percent among women affected in Engineering (1)	Percent among all women (2)	Percent among men affected in Engineering (3)	Percent among all men (4)
Panel (a): Effect in Engineering				
Law	0.24	0.01	0.27	0.01
Science	14.98	0.58	26.29	1.37
Social Science	1.20	0.05	1.22	0.06
Teaching	4.78	0.19	3.33	0.17
Humanities	5.10	0.20	3.26	0.17
Health	1.75	0.07	2.72	0.14
Medicine	0.08	0.00	0.07	0.00
Business	13.78	0.54	10.94	0.57
Unassigned	57.45	2.24	49.18	2.57
Reverse-direction moves	0.64	0.02	2.72	0.14
All affected Engineering	100.00	3.90	100.00	5.23
Panel (b): Effect in Other Fields				
All affected other fields	-	0.49	-	0.74

Notes: This table presents a breakdown of changes in broad-field assignment under a 50% female quota in Engineering relative to baseline. Panel (a) reports moves involving Engineering. Columns (1) and (3) report percentages among all women and men, respectively, who move into or out of Engineering; columns (2) and (4) report percentages of all women and men in the policy-specific paired sample used in Figure 8. Field rows give the baseline field for women moved into Engineering and the destination field for men displaced from Engineering. The reverse-direction row records women moved out of Engineering and men moved into Engineering. Panel (b) reports applicants whose field changes without entering or leaving Engineering. The two panels together recover the affected shares in Figure E.2. Percentages may not sum exactly because of rounding.

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